

President University
Faculty Of Computer Science



DIGITAL FORENSIC INVESTIGATION REPORT

Operation X – Dark Web Human Trafficking & Murder Investigation

Prepared in Partial Fulfillment of the Digital Forensics Course Requirements

Course: Project Bootcamp - Company X Case 3

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1 EXECUTIVE SUMMARY

Operation X is a digital forensic investigation into a suspected international human trafficking network with links to a related murder case. The primary subject, Daniel Vega, became a person of interest after law enforcement intercepted dark web forum activity discussing the trafficking operation, and a missing person identified as A.C. was later found dead in connection with the same network.

The examination covered a ten-segment forensic disk image (Daniel_Laptop.E01) and a network traffic capture (Daniel_Network.pcap) belonging to the suspect. Additional files recovered from within these two sources, including a password-protected archive (encrypted.zip), two image files (HiddenMessage.bmp and safehouse location.bmp), and several PDF, spreadsheet, and email files, were examined using Autopsy, Wireshark, ExifTool, Binwalk, FTK Imager, NetworkMiner, John the Ripper, Hashcat, and pyzipper.

The investigation recovered documents identifying four trafficking victims and their intended destinations, forged identity documents, financial records indicating large offshore transactions, and email communications discussing shipments, safehouses, and operational activities. Network traffic analysis further revealed file transfers involving trafficking-related documents and communications with external systems. Additional examination identified the installation of anonymity and steganography tools, indicating deliberate attempts to conceal digital evidence.

Overall, the evidence recovered is highly consistent with the objectives of this investigation and supports the presence of organised trafficking activities, deliberate evidence concealment, and communications linking the suspect to the wider criminal network. Certain aspects, including encrypted Tor traffic contents and cryptocurrency transaction attribution, could not be independently verified within the scope of this examination and are therefore acknowledged as investigation limitations rather than confirmed findings.

2 INTRODUCTION

2.1 Background

Operation X is a digital forensic investigation involving a suspected human trafficking and murder case linked to dark web activities. Law enforcement agencies intercepted discussions on dark web forums regarding an international human trafficking network. During the investigation, a missing person identified as A.C. was believed to have become a victim of the trafficking organization and was later found dead. Investigators suspected that the case was connected to a larger organized crime network operating through encrypted communications and the Tor network.

Following the investigation, law enforcement conducted a raid and seized a laptop belonging to the suspect, Daniel [REDACTED]. A forensic disk image (Daniel_Laptop.E01) and a captured network traffic file (Daniel_Network.pcap) were submitted for digital forensic examination to identify digital evidence related to the investigation.

2.2 Objectives

This examination aims to:

1. Identify digital evidence related to the suspect's activities.
2. Analyze communications and transferred files associated with the investigation.
3. Reconstruct the sequence of events based on recovered digital evidence.
4. Produce a forensic report supporting the investigation.

2.3 Scope

This examination covers the following evidence items:

- DanielVe v. (R.E01–R.E10) – Split forensic disk image.
- Daniel_Network.pcap – Network traffic capture.
- HiddenMessage.bmp – Image recovered during network analysis.
- Safehouse location.bmp – Image recovered during network analysis.
- encrypted.zip – Password-protected archive recovered during network analysis.
- [Additional recovered files identified by other examination teams]

The following items or aspects were NOT covered in this examination:

- Physical crime scene investigation
- DNA examination
- Medical examination
- Live network monitoring
- Evidence not provided by law enforcement

3 CASE MANAGEMENT

3.1 Evidence Receipt from Law Enforcement

The digital evidence used in this examination was provided by the course instructor as part of the Operation X digital forensic case scenario. The examination team did not perform the original evidence acquisition or forensic imaging process. Instead, the team received the evidence in the form of a segmented forensic disk image (DanielVe v. R.E01–R.E10) and a network capture file (Daniel_Network.pcap) for examination.

At the time of receipt, all evidence files were accessible and showed no indication of corruption. Integrity verification was performed prior to examination, as described in Chapter 4.

3.2 Chain of Custody

The digital evidence was provided directly by the course instructor as part of the academic investigation scenario. No formal law enforcement chain-of-custody documentation was provided to the examination team prior to receipt. Throughout the examination process, the evidence remained under the control of the assigned examination team and was not modified. Evidence integrity was verified before analysis.

3.3 Team Roles & Responsibilities

Table 3.1 summarizes the division of responsibility across the examination team.

Team Member	Responsibility (Jira Epic)	Related Chapter(s)
Fathir Barhouti Awlya	Network & Steganography Analysis — EPIC 5 (KAN-46)	6.2, 6.4, 10 (log entries provided)

Team Member	Responsibility (Jira Epic)	Related Chapter(s)
Aurelio Sami Rabbani	Disk Image Acquisition & Hash Verification — EPIC 1 (KAN-38)	4, 6.1.1
Reva Stefvany br Ginting	Deleted File & Artifact Recovery — EPIC 2 (KAN-31)	6.1.2, 6.5
Melisa Anamaria Nainggolan	Hidden Data & Password Recovery — EPIC 3 (KAN-56)	6.3.1
Berliana Veni Jaya Kesuma	Encryption & File Cracking — EPIC 4 (KAN-42)	6.3.2
All Members	Report Compilation	All chapters

Table 3.1 — Team Roles & Responsibilities

4 EVIDENCE VERIFICATION & VALIDATION

Before examination, all digital evidence was verified to ensure data integrity. SHA-256 hash values were calculated for each evidence item using the sha256sum utility. No reference hash values were provided by law enforcement; therefore, integrity verification was limited to calculating and documenting the hash values of the received evidence prior to analysis. No indication of file corruption or modification was observed during the verification process.

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
E01	DanielVe v. R.E01	N/A	c89187bfd893d3 269141c017a496 5858be3cfad1717 5fe04fe410b045f 42a241	N/A	Autopsy

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
E02	DanielVe v. R.E02	N/A	5d821c5e20ef12f ddf772db10873a ccf1a279c495517 2dcb86734f7d6d 24a684	N/A	Autopsy
E03	DanielVe v. R.E03	N/A	07836542de8bdd 7b6fbca623e38a3 412ef43ee5c85f0 e70cdc906d4c04 b42508	N/A	Autopsy
E04	DanielVe v. R.E04	N/A	ca2f5141cd88ce9 1b96b68ded7110 47147e908184a5 0679e96e91ca8b 2712eb8	N/A	Autopsy
E05	DanielVe v. R.E05	N/A	c4e5f720f82b1d5 d9d090718c7083 3d201c8c28dc78 6c85df864d499d bc6c87f	N/A	Autopsy
E06	DanielVe v. R.E06	N/A	9704e188bbaef3a d023695296386b 71408d1a8c806b 085bb046ac7b07 869ee60	N/A	Autopsy

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
E07	DanielVe v. R.E07	N/A	b617a85c33c5d1 17cc2c148d267e c9f36827ca7fb52 a81080ea25d525 90e8fca	N/A	Autopsy
E08	DanielVe v. R.E08	N/A	47c4fe43739299 08769f8297bd5fd 6803b7db31d456 c37ef3b9301e0ec 770cba	N/A	Autopsy
E09	DanielVe v. R.E09	N/A	cc78c572c434c3 d7eb30897e9048 8fe007782e6c202 1c47efb715edb73 478bf6	N/A	Autopsy
E10	DanielVe v. R.E10	N/A	5b44ffee19c88a8 8fc3439cf1b6366 45684b107bbe76 ffdf3e7c9080976 5daed	N/A	Autopsy
PCAP	Daniel_Network.pcap	N/A	4345eb708d18ec c4f2c1fdcf64bb5 70243ddc42800a ce1fec0c41c4adb 0495c5	N/A	sha256sum

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
IMG-1	HiddenMessage.bmp	N/A	fed66c4038d2b9 4ce6b013993d82 fc79c878ba2016b cdb13ba48d50bb 6d3e95e	N/A	sha256sum
IMG-2	Safehouse location.bmp	N/A	b9efea7df0b1326 92865ad6cf8cf97 9246db1ce01d3c e33b5b27aa0c76f 19b4f	N/A	sha256sum
ZIP-1	encrypted.zip	N/A	77384ccb7efb109 c28f191e3bc9950 6f5ef9c98788ebb 47622682d391d0 f0cf4	N/A	sha256sum
DOC-1	INVOICE.pdf	N/A	704d65f7d2bae8c 7a0ffc03894d71e cd5a64cc081e5b 6dde0d9796cfa2a 6c056	N/A	sha256sum
DOC-2	Shipment List.pdf	N/A	64145ce8dca46b 1582359b2eac01 ecfa000c7894284 5ca9f344519edd2 c6b030	N/A	sha256sum

Evidence ID	Filename	Provided Hash	Recalculated Hash	Match	Tool
XLS-1	client_list.xlsx	N/A	5362d32b257708 69a0d9868b7e0c a8c4b9b08c00c7 9468f243135cb5 6d5a2313	N/A	sha256sum
TXT-1	contacts.txt	N/A	8cdd6b87208893 e17d0d5c8ddd12 d766e36e1143bb b701cb0dcb537b 4c64cc45	N/A	sha256sum
TXT-2	safehouse_location.txt	N/A	88576bbb42a152 f7e7e6847a50b8c 830c7502665098 fd7c398a03e4c66 531e63	N/A	sha256sum

Table 4.1 — Hash Verification (Network & Steganography evidence rows pre-filled with tool name; hash values pending actual computation)

5 METHODOLOGY

5.1 Forensic Framework

This examination follows the digital forensic methodology described in **NIST SP 800-86 – Guide to Integrating Forensic Techniques into Incident Response**. Since the examination team received forensic images and network captures that had already been acquired by law enforcement, the acquisition phase was not performed by the examination team. Therefore, the workflow implemented in this report consists of:

- Evidence Verification
- Examination
- Analysis
- Reporting

This approach ensures that the integrity of the provided evidence is verified before examination, the findings are analyzed systematically, and all results are documented in a structured forensic report.

5.2 Examination Workflow

The examination followed four sequential phases: Evidence Verification, Examination, Analysis, and Reporting.

5.3 Tools and Environment

Tool	Version	Purpose	Used By
Wireshark / TShark	4.7.2	Packet capture inspection and protocol analysis of Daniel_Network.pcap. Used to analyze HTTP, DNS, TLS, FTP, OCSP, and identify suspicious network activity.	Network & Steganography
GNU strings	GNU Binutils 2.47	Extracted printable strings from files to identify readable artifacts, URLs, filenames, and hidden text.	Network & Steganography
ExifTool	13.55	Examined image metadata to determine whether images contained hidden metadata or embedded information.	Network & Steganography
Binwalk	2.4.3	Detected embedded or appended file signatures to identify potential hidden data within image files.	Network & Steganography

Tool	Version	Purpose	Used By
Capinfos	4.7.2	Generated capture file statistics, timestamps, packet counts, and general summary information.	Network & Steganography
zsteg	Unknown (installed release)	Detected potential LSB steganography and hidden data patterns within BMP images.	Network & Steganography
file	5.48	Identified actual file types and validated file signatures (magic bytes).	Network & Steganography
sha256sum	GNU Coreutils 9.11	Compute SHA-256 hashes to verify evidence integrity.	Network & Steganography
xxd	2026-06-16	Raw hexadecimal inspection of recovered files.	Network & Steganography
John the Ripper	1.9.0-jumbo-1	Recover passwords and crack hashes for encrypted files or archives.	Encryption
Cryptii	v4.0.11	a web app for encoding, decoding, and encrypting text (Caesar cipher, Base64, Morse code, ROT13, etc.), all running in your browser with no data sent to a server.	Email & Document Recovery
7-Zip / p7zip-full	23.01	AE-compatible archive listing, dictionary testing, and extraction of encrypted.zip	Encryption & File Cracking
pyzipper (Python)	0.4.0	AESZipFile dictionary attack — recovered password for encrypted.zip	Encryption & File Cracking
zipinfo / unzip	ZipInfo 3.00 / UnZip 6.00	Identified encrypted.zip as WinZip AE (compression method 99); initial (unsuccessful) dictionary attack	Encryption & File Cracking

Tool	Version	Purpose	Used By
Capinfos	4.7.2	Capture file statistics and summary information	Network & Steganography
Exiftool	13.55	Image metadata inspection	Network & Steganography
zsteg	Unknown (installed release)	Detect potential LSB steganography within BMP images	Network & Steganography
zip2john	John the Ripper 1.9.0-jumbo-1	Convert encrypted ZIP archive into crackable hash	Encryption

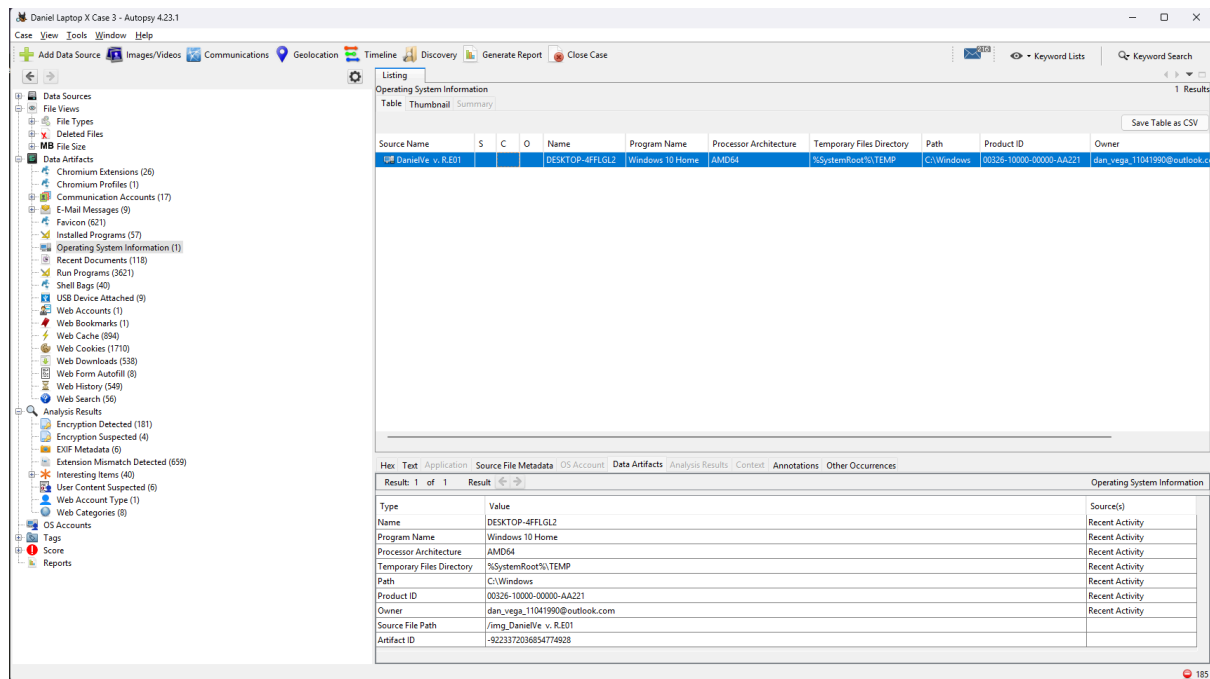
Table 5.1 — Tools and Environment

6 FINDINGS

This chapter presents the factual findings obtained during the examination of the digital evidence. Each finding is reported objectively based on the evidence recovered, without interpretation or correlation. Cross-evidence analysis is presented separately in Chapter 8 (Analysis).

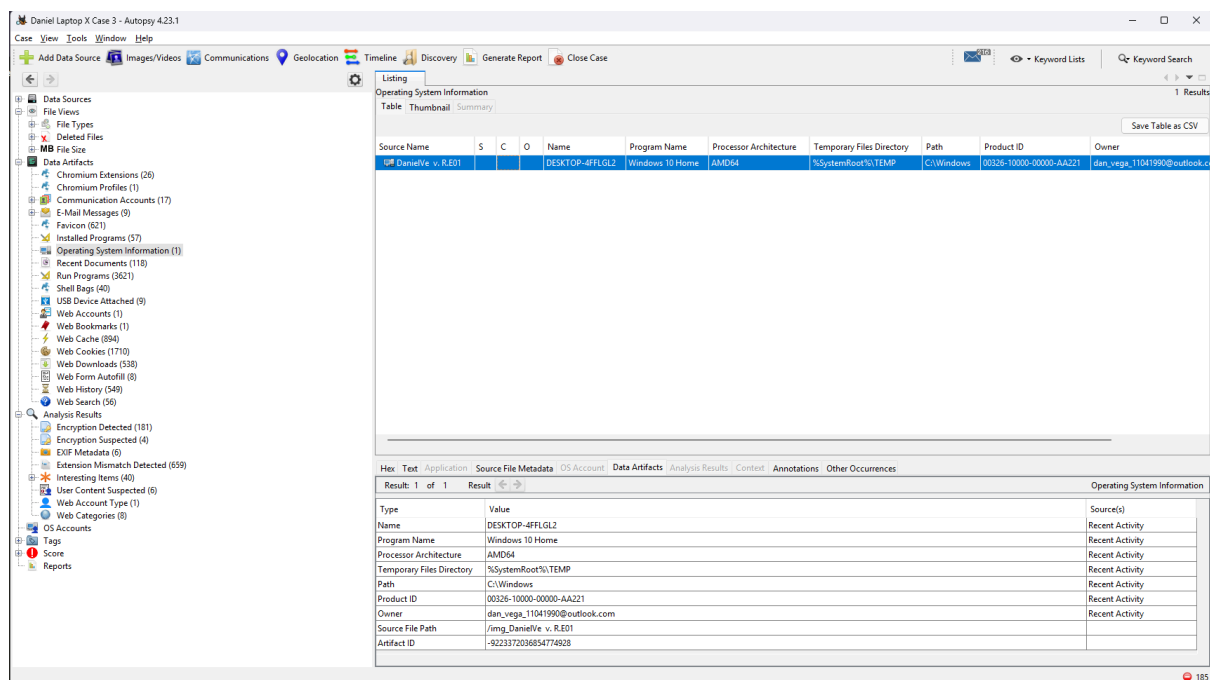
6.1 Disk Image Findings

This section presents the findings obtained from the forensic examination of the Daniel_Laptop.E01 disk image. The examination focused on identifying system artifacts, recovered files, browser activity, and file system metadata relevant to the investigation. All findings reported in this section are presented objectively without interpretation.



6.1.1 Disk Image Acquisition and Hash Verification

The forensic disk image (DanielVe v. R.E01–R.E10) was successfully loaded into Autopsy for examination. The image was recognized as a valid EnCase Evidence File (EWF) and the file system was parsed successfully. System artifacts including operating system information, user accounts, browser history, registry entries, installed applications, and file system metadata were identified and made available for further examination.



Prior to examination, integrity verification was performed as documented in Table 4.1. SHA-256 hash values were calculated for the received evidence. Since no reference hash values were provided by law enforcement, the calculated hash values were recorded as baseline integrity values. No indication of file corruption or modification was observed.

6.1.2 Timeline Analysis

Timeline reconstruction was performed using Autopsy 4.22.1's Timeline feature on Daniel_Laptop.E01, filtering file-system and web-activity events by filename to isolate artifacts relevant to the case from the general noise of installed-software files (over 1,000,000 total timeline events were present unfiltered, the majority originating from bundled Tor Browser localization files).

The following chronological sequence was established for key artifacts:

- **2025-03-07 17:33:06–17:38:21 WIB** — tor-browser-windows-x86_64-portable-14.0.7.exe created, modified, and changed (Downloads), consistent with download and installation.
- **2025-03-11 00:22:19–05:47:09 WIB** — nightclub.jpg created, changed, and accessed (Downloads).
- **2025-03-11 01:06:26 & 01:16:32 WIB** — transaction logs.xlsx modified and accessed, located within usb_backup_files.zip (OneDrive/Documents/Hidden).
- **2025-03-11 01:14:50 & 01:16:32 WIB** — identity docs.pdf modified and accessed at the same time as transaction logs.xlsx, within the same archive.
- **2025-03-13 00:55:01–00:55:14 WIB** — Web History shows a search for "OpenStego" on Bing, followed by visits to openstego.com and the project's GitHub releases page, immediately followed by a Web Download of Setup-OpenStego-0.8.6.exe.
- **2025-03-13 00:55:19–00:56:38 WIB** — OpenStego installed (registry entries, shortcuts, and Prefetch execution recorded) and its library accessed shortly after installation.
- **2025-03-13 01:11:24 WIB** — Shipment List[48].pdf and Everything is in place[47].eml both recorded as attachment-save events (ACBM) at the identical timestamp, indicating the shipment manifest was saved as an email attachment.

- **2025-03-13 11:32:27–11:32:28 WIB —**
tor-browser-windows-x86_64-portable-14.0.7.exe and Setup-OpenStego-0.8.6.exe were both accessed within one second of each other, indicating the two tools were used together in the same session.
- **2025-03-13 19:18–19:56 WIB and 2025-03-14 00:23–12:14 WIB —** Repeated access to OpenStego program files and Prefetch entries across the following day, indicating continued use rather than a single isolated installation.

Email content: The email Everything is in place[47].eml (sent 2025-03-11 09:23:40, from dan_vega_11041990@outlook.com) was found to contain body text encoded with the same Caesar-cipher (shift of 3) pattern identified elsewhere in the case evidence. Decoded, the message reads: *"...planned. No issues. Payment confirmed. Awaiting final instructions. Will proceed as discussed."*

The findings presented in this section are reported objectively. Interpretation of their significance is discussed in Chapter 8 (Analysis).

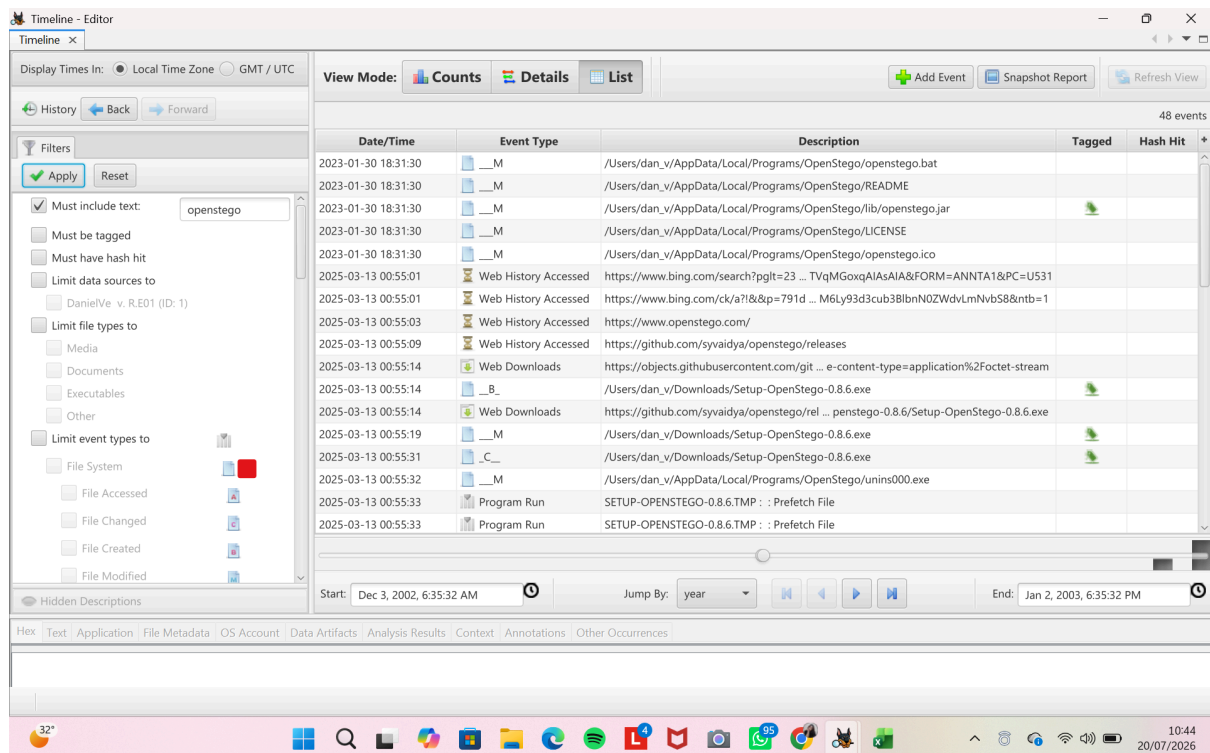


Figure 6.1.1 — Autopsy timeline / artifact overview

6.1.3 Deleted File and Artifact Recovery

The forensic examination identified deleted files and recoverable artifacts from the Daniel_Laptop.E01 disk image using Autopsy. Deleted files were identified through the Deleted Files directory, while additional recoverable artifacts were identified from the Recent Documents and File Types categories. These findings indicate that previously deleted or recently accessed files remained recoverable from the forensic image.

The forensic examination recovered several documents and related artifacts from the disk image, including PDF documents, spreadsheet files, email messages, and Windows shortcut (.lnk) files. The recovered documents include Shipment List.pdf, identity docs.pdf, INVOICE.pdf, and transaction logs.xlsx. In addition, multiple email artifacts associated with the account dan_vega_11041990@outlook.com were successfully recovered. The identified email subjects included Everything is in Place, Safehouse Ready, Urgent: Cleanup Needed, and File Transfer – Urgent, indicating previous communication activities stored within the forensic image.

Metadata analysis was performed on the recovered Windows shortcut (.lnk) artifacts, including victim list.txt.lnk, HiddenMessage.bmp.lnk, encrypted.zip.lnk, Dark Web.lnk, and Emails.lnk. The examination compared the creation, modification, and access timestamps of each artifact. The metadata showed consistent timestamp information across the recovered artifacts. The encrypted.zip.lnk and Dark Web.lnk artifacts remained in use after their creation, while no internal timestamp inconsistencies or indications of metadata manipulation were identified based on the available forensic metadata.

Email 1.eml	Safehouse Ready	handler@darkops.xyz	12/03/2025 15:48:31
Email 2.eml	Urgent: Cleanup Needed	fixer@darkops.xyz	12/03/2025 15:50:07
Email 3.eml	Everything is in place	boss@onionmarket.xyz	12/03/2025 15:51:47
Email 4.eml	File Transfer - Urgent	handler@darkops.xyz	12/03/2025 15:54:14

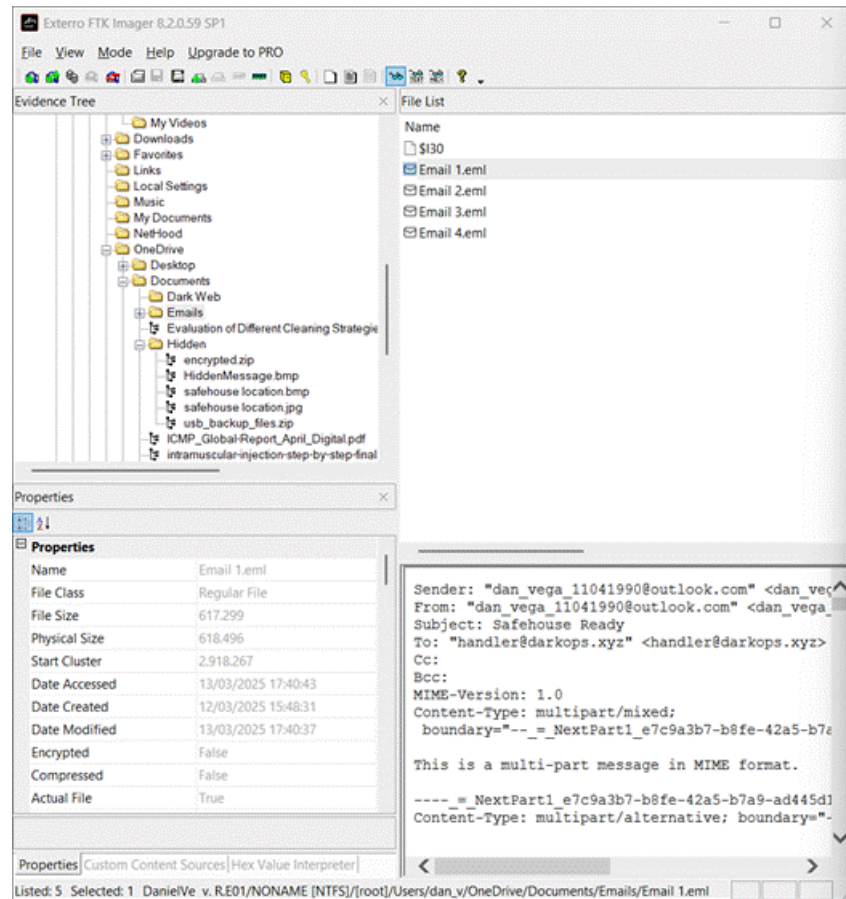


Figure 6.1.3a — Email 1.eml ("Safehouse Ready") properties in FTK Imager.

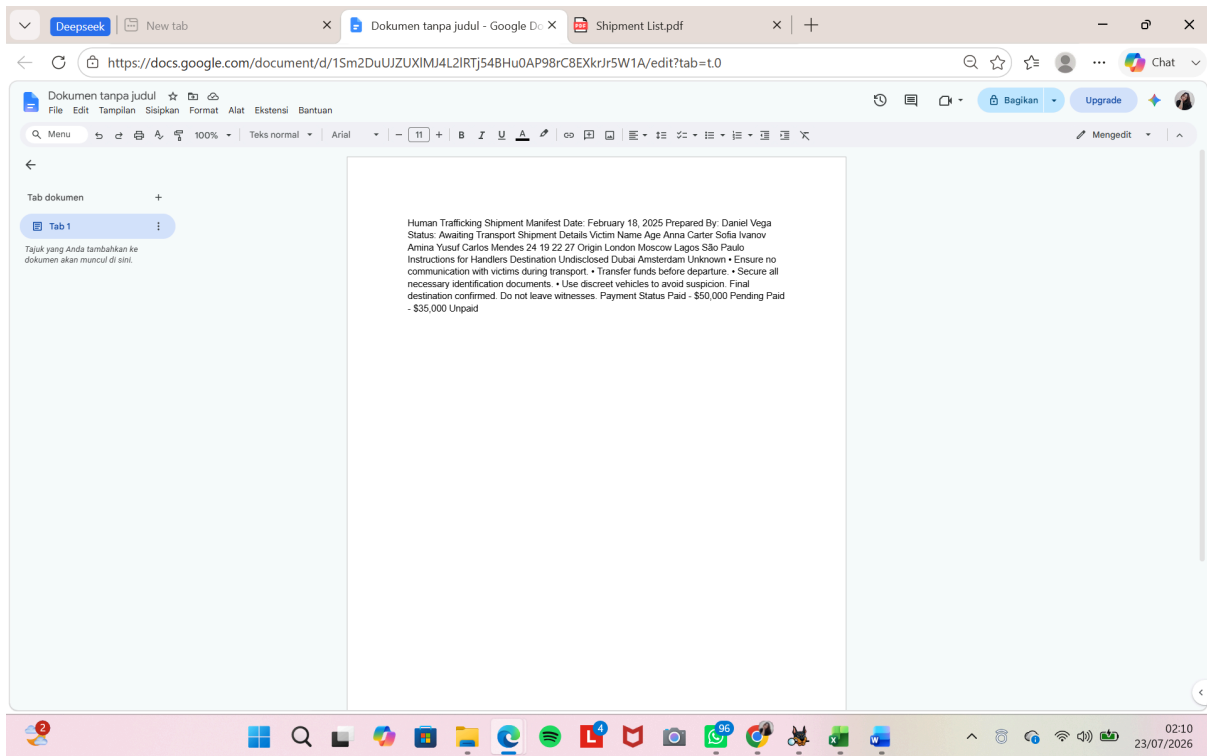


Figure 6.1.2 — Recuva / file carving results

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing

/img_DanielVe v. R.E01/Users/dan_v/AppData/Local/Packages/microsoft.windowscommunicationsapps_8wekyb3d8bbwe/LocalState/Files/S0/7/Attachments/Safehouse Ready[51].eml 1 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
safehouse.location.bmp		2		0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	450038	Allocated	Allocated	unknown	/img_DanielV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Tags Menu



Attachments

- Everything
- Safehouse
- Urgent_C

Figure 6.1.3a — Email 1.eml ("Safehouse Ready")

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

Default

Table Thumbnail Summary

9 Results

Save Table as CSV

Source Name	S	C	O	E-Mail From	E-Mail To	Subject
Safehouse Ready[51].eml				dan_vega_11041990@outlook.com;	handler@darkops.xyz;	Safehouse Ready
Everything is in place[47].eml				dan_vega_11041990@outlook.com;	boss@onionmarket.xyz;	Everything is in pl
Urgent_Cleanup Needed[50].eml				dan_vega_11041990@outlook.com;	fixer@darkops.xyz;	Urgent: Cleanup
msg_43.txt				; >;	webmaster@python.org;	Banned file: auto
msg_25.txt				MAILER-DAEMON@zinfandel.lacita.com;	linuxuser-admin@www.linux.org.uk;	Returned mail: To
Email 2.eml				dan_vega_11041990@outlook.com;	fixer@darkops.xyz;	Urgent: Cleanup
Email 1.msg_25.txt				dan_vega_11041990@outlook.com;	handler@darkops.xyz;	Safehouse Ready
Email 3.eml				dan_vega_11041990@outlook.com;	boss@onionmarket.xyz;	Everything is in pl
Email 4.eml				dan_vega_11041990@outlook.com;	handler@darkops.xyz;	File Transfer - Urg

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 3 of 3 Result

E-Mail Messages

From: dan_vega_11041990@outlook.com; 2025-03-11 10:10:14 WIB

To: fixer@darkops.xyz;

CC:

Subject: Urgent: Cleanup Needed

Headers Text HTML RTF Attachments (0) Accounts

Download Images

Xqiruhvhq frpsoldwlrq. Qhgh lpphgdwh dvvlwdqfh zlwklvsvrdo.
Orfdwlrq frpsurplvhg. Pxxw dfw idvw.

Lqvwuxfwlrq
Wdujhw holplqdwhg. Lpphgdwh glvsrdo uhtxlgh.
Hqvxuh qr iruhqvlh hylghqfh uhpdlqv.
Wudqvsruw uhpdlqv wr ghvjlqdwgh vdihkrxvh.
Frqwdfw "Udyhq" iru ilqdo duudqjhpqvw.

Figure 6.1.3b — Email 2.eml ("Urgent: Cleanup Needed")

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing

/img_DanielVe v. R.E01/Users/dan_v/OneDrive/Documents/Emails/Email 3.eml 1 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
Shipment List.pdf			2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	44615	Allocated	Allocated	unknown	/img_DanielVe v. R

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 3 of 3 Result

E-Mail Messages

From: dan_vega_11041990@outlook.com;
To: boss@onionmarket.xyz;
CC:
Subject: Everything is in place

Headers Text HTML RTF Attachments (1) Accounts

Download Images

Vklsphqw zl00 duulyh dv sodqqhg. Qr lvvxhv. Sdbphqw frqiluphg.
Dzdlwlqj ilqdo lqvwxwflrqv. Zl00 surfhhg dv glvfxvvhg.

Figure 6.1.3c — Email 3.eml ("Everything is in Place")

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

+ Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing

/img_DanielVe v. R.E01/Users/dan_v/OneDrive/Documents/Emails/Email 4.eml 0 Results

Table Thumbnail Summary

Save Table as CSV

Source Name

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 3 of 3 Result

E-Mail Messages

From: dan_vega_11041990@outlook.com;
To: handler@darkops.xyz;
CC:
Subject: File Transfer - Urgent

Headers Text HTML RTF Attachments (0) Accounts

Download Images

Secure server is live.

Connect to: <http://192.168.40.135:8080>

Download the necessary files ASAP.

Do not reply. Delete after use.

sions (26)
es (1)
Accounts (17)
(9)
ult]]

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Keyword Lists Keyword Search

Listing

Default 9 Results

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	E-Mail From	E-Mail To	St
Safehouse Ready[51].eml	!			dan_vega_11041990@outlook.com;	handler@darkops.xyz;	Sa
Everything is in place[47].eml	!			dan_vega_11041990@outlook.com;	boss@onionmarket.xyz;	Ev
Urgent_ Cleanup Needed[50].eml	!			dan_vega_11041990@outlook.com;	fixer@darkops.xyz;	Ur
msg_43.txt	!			; >;	webmaster@python.org;	Ba
msg_25.txt	!			MAILER-DAEMON@zinfandel.lacita.com;	linuxuser-admin@www.linux.org.uk;	Re
Email 2.eml	!			dan_vega_11041990@outlook.com;	fixer@darkops.xyz;	Ur
Email 1.eml	!			dan_vega_11041990@outlook.com;	handler@darkops.xyz;	Sa
Email 3.eml	!			dan_vega_11041990@outlook.com;	boss@onionmarket.xyz;	Ev
Email 4.eml	!			dan_vega_11041990@outlook.com;	handler@darkops.xyz;	Fil

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 3 of 3 Result

E-Mail Messages

From: MAILER-DAEMON@zinfandel.lacita.com; 2001-04-07 00:23:06 WIB

To: linuxuser-admin@www.linux.org.uk;

CC:

Subject: Returned mail: Too many hops 19 (17 max): from <linuxuser-admin@www.linux.org.uk> via [199.164.235.226], to <scoffman@wellpart...

Headers Text HTML RTF Attachments (0) Accounts

Original Text

This is a MIME-encapsulated message

--JAB03225.986577786/zinfandel.lacita.com

The original message was received at Fri, 6 Apr 2001 09:23:03 -0800 (GMT-0800) from [199.164.235.226]

----- The following addresses have delivery notifications -----

<scoffman@wellpartner.com> (unrecoverable error)

----- Transcript of session follows -----

554 Too many hops 19 (17 max): from <linuxuser-admin@www.linux.org.uk> via [199.164.235.226], to <scoffman@wellpartner.com>

--JAB03225.986577786/zinfandel.lacita.com

Content-Type: message/delivery-status

Disposition: MTAs: dan.zinfandel.lacita.com

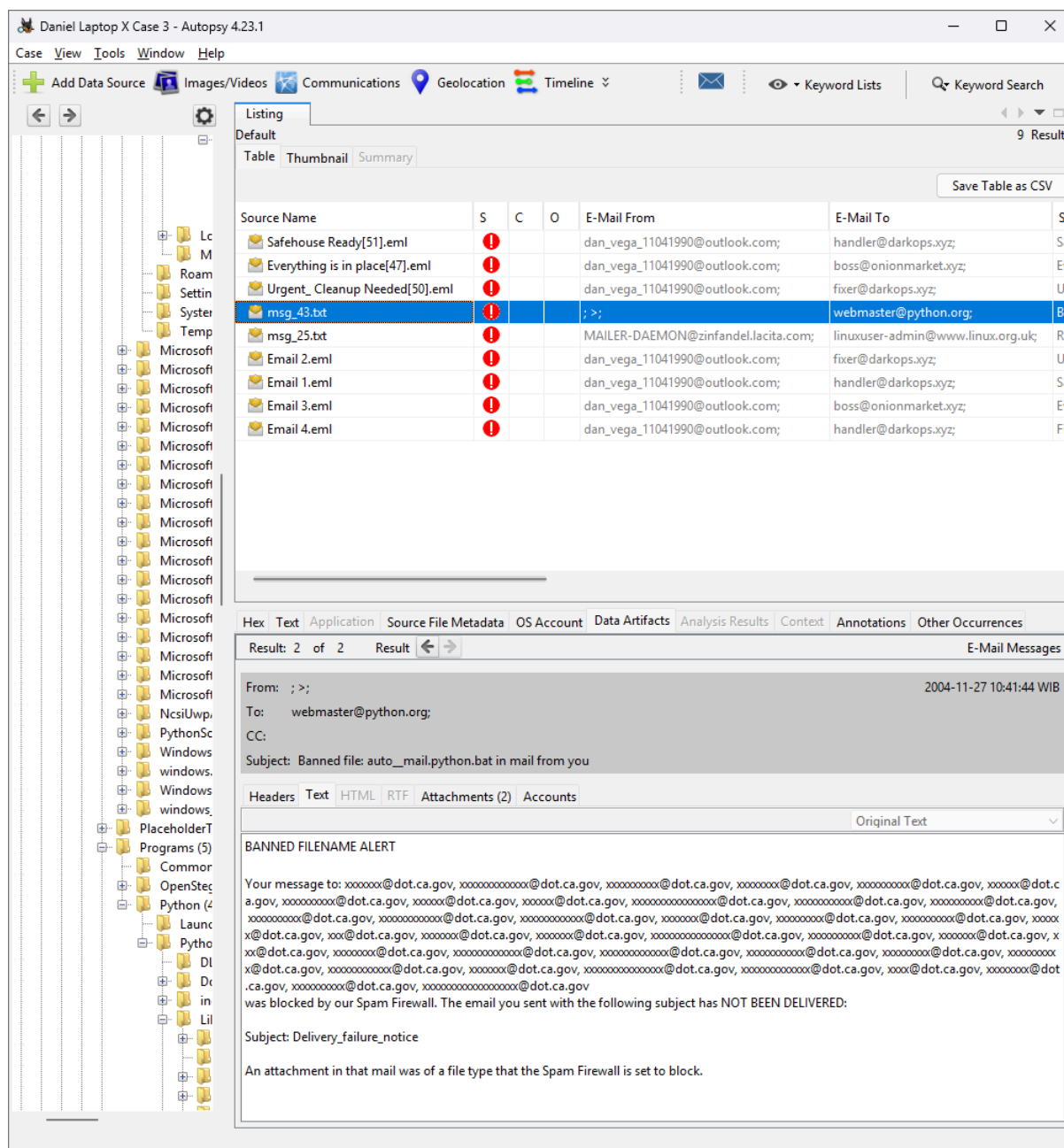


Figure 6.1.3d — Email 4.eml, msg 25.txt & msg 43.txt ("File Transfer – Urgent")

cryptii.com/pipes/caesar-cipher/

cryptii

Protect your code and reputation. See why top mobile app developers trust Guardsquare for security.

hubs | Ads by EthicalAds

VIEW
Plaintext

If he had anything confidential to say, he wrote it in cipher, that is, by so changing the order of the letters of the alphabet, that not a word could be made out.

ENCODE **DECODE**
Caesar cipher

SHIFT

— 7 a→h +

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

Maintain case

FOREIGN CHARS

Include Ignore

→ Encoded 163 chars

VIEW
Ciphertext

Pm ol ohk hufaop
jvumpkluaphs av
pa pu jpwoly, ao
johunpun aol vyk
slaalyz vm aol h
uva h dvyk jvbsk

cryptii

Protect your code and reputation. See why top mobile app developers trust Guardsquare for security.

hubs | Ads by EthicalAds

VIEW
Ciphertext

Vklsphqw zloo duulyh dv
sodqqhg. Qr lvvxhv. Sdbphqw
frqiluphg.
Dzdlwlqj ilqdo lqvwuxfwlrqv.
Zloo surfhhg dv glvfxvvhg.

ENCODE **DECODE**
Caesar cipher

SHIFT

— 3 a→d +

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

Maintain case

FOREIGN CHARS

Include Ignore

→ Decoded 121 chars

VIEW
Plaintext

Shipment will arrive as
planned. No issues. Payment
confirmed.
Awaiting final instructions.
Will proceed as discussed.

Protect your code and reputation. See why top mobile app developers trust GuardSquare for security.

VIEW

Ciphertext

Xqiruhvhhq frpsolfdwlrq. Qhhg lpphglwdh dvvlvwdqfh zlwklvsvrdo.

Orfdwlrq frpsurplvhg. Pxxw dfwidvw.

Lqvwuxfwlrqv

Wdujhw holplqdwhg. Lpphglwdh glsvrdo uhtxluhg.

Hqvuh qr iruhqvlf hylghqfh uhpdlqv.

Wudqvsruw uhpdlqv wr ghvljqdwhg vdihkrxvh.

Frqwdfw "Udyhq" iru ilqdo duudqjhphqvv.

ENCODE DECODE

Caesar cipher

SHIFT

—

3 a→d

+

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

Maintain case

FOREIGN CHARS

Include Ignore

→ Decoded 300 chars

VIEW

Plaintext

Unforeseen complication. Need immediate assistance with disposal.

Location compromised. Must act fast.

Instructions

Target eliminated. Immediate disposal required.

Ensure no forensic evidence remains.

Transport remains to designated safehouse.

Contact "Raven" for final arrangements.

Secure your mobile apps against attackers. See how GuardSquare provides industry-leading protection.

VIEW

Plaintext

Unforeseen complication. Need immediate assistance with disposal.

Location compromised. Must act fast.

Instructions

Target eliminated. Immediate disposal required.

Ensure no forensic evidence remains.

Transport remains to designated safehouse.

Contact "Raven" for final arrangements.

ENCODE DECODE

Caesar cipher

SHIFT

—

3 a→d

+

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

Maintain case

FOREIGN CHARS

Include Ignore

← Decoded 294 chars

VIEW

Ciphertext

Xqiruhvhhq frpsolfdwlrq. Qhhg lpphglwdh dvvlvwdqfh zlwklvsvrdo.

Orfdwlrq frpsurplvhg. Pxxw dfwidvw.

Lqvwuxfwlrqv

Wdujhw holplqdwhg. Lpphglwdh glsvrdo uhtxluhg.

Hqvuh qr iruhqvlf hylghqfh uhpdlqv.

Wudqvsruw uhpdlqv wr ghvljqdwhg vdihkrxvh.

Frqwdfw "Udyhq" iru ilqdo duudqjhphqvv.

The screenshot displays the cryptii website's Caesar cipher tool. The interface is divided into three main sections: Ciphertext, Caesar cipher settings, and Plaintext.

Top Section (First Decoding):

- Ciphertext:** Orfdwlrq lv vhfuxh. Qr ixuwkhu lvvxhv. Dzdlwlqj ilqdo frqilupdwlrq.
- Caesar cipher:** SHIFT 3 a→d, ALPHABET abcdefghijklmnopqrstuvwxyz, CASE STRATEGY Maintain case, FOREIGN CHARS Include Ignore. Decoded 69 chars.
- Plaintext:** Location is secure. No further issues. Awaiting final confirmation.

Bottom Section (Second Decoding):

- Ciphertext:** Vklspqh w zloo duulyh dv sodqqhg. Qr lvvxhv. Sdbphqw frqiluphg. Dzdlwlqj ilqdo lqvwxflrqv. Zloo surfhgh dv glvfvvhg.
- Caesar cipher:** SHIFT 3 a→d, ALPHABET abcdefghijklmnopqrstuvwxyz, CASE STRATEGY Maintain case, FOREIGN CHARS Include Ignore. Decoded 121 chars.
- Plaintext:** Shipment will arrive as planned. No issues. Payment confirmed. Awaiting final instructions. Will proceed as discussed.

Below the tool interface, there is a section titled "Caesar cipher: Encode and decode online" with a brief description and a list of other tools: Decimal to text, URL encode, Enigma decoder, Text to binary, and Commercial Enigma.

Figure 6.1.3e — Decoded Ciphertext in each Email sources

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source
 Images/Videos
 Communications
 Geolocation
 Timeline
 Discovery
 Keyword Lists
 Keyword Search

Listing

Installed Programs

Table Thumbnail Summary

Save Table as CSV

57 Results

Source Name	S	C	O	Program Name	Date/Time	Data Source
SOFTWARE	!		1	AddressBook	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	AddressBook	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	Connection Manager	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	Connection Manager	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	DXM_Runtime	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	DXM_Runtime	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	DirectDrawEx	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	DirectDrawEx	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE	!		1	Exterro FTK Imager v.4.7.3.81	2025-03-14 14:40:20 WIB	DanielVe v. R.E01
SOFTWARE			1	Fontcore	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	Fontcore	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IE40	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IE40	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IE4Data	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IE4Data	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IESBAKEX	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IESBAKEX	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	IEData	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	IEData	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE			1	MPlayer2	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	MPlayer2	2019-12-07 09:52:45 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Edge Update v.1.3.195.45	2025-03-08 14:17:45 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Edge WebView2 Runtime v.133.0.3065.92	2025-03-05 17:30:17 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Edge v.134.0.3124.66	2025-03-13 14:41:43 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Update Health Tools v.3.74.0.0	2025-03-06 14:08:38 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2015-2022 Redistributable (x64) - ...	2025-03-14 14:27:00 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2015-2022 Redistributable (x86) - ...	2025-03-05 18:20:23 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X64 Additional Runtime - 1...	2025-03-14 14:26:58 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X64 Minimum Runtime - 1...	2025-03-14 14:26:56 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X86 Additional Runtime - 1...	2025-03-05 18:20:23 WIB	DanielVe v. R.E01
SOFTWARE			1	Microsoft Visual C++ 2022 X86 Minimum Runtime - 1...	2025-03-05 18:20:23 WIB	DanielVe v. R.E01
SOFTWARE			1	MobileOptionPack	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE			1	MobileOptionPack	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE	!		1	Npcap v.1.79	2025-03-14 14:28:39 WIB	DanielVe v. R.E01
SOFTWARE	!		1	Python 3.13.2 Add to Path (64-bit) v.3.13.2150.0	2025-03-13 03:46:51 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Core Interpreter (64-bit) v.3.13.2150.0	2025-03-13 03:46:24 WIB	DanielVe v. R.E01
SOFTWARE			1	Python 3.13.2 Development Libraries (64-bit) v.3.13.21...	2025-03-13 03:46:25 WIB	DanielVe v. R.E01

SOFTWARE		1	Python 3.13.2 Documentation (64-bit) v.3.13.2150.0	2025-03-13 03:46:36 WIB	DanielVe v. R.E01
SOFTWARE		1	Python 3.13.2 Executables (64-bit) v.3.13.2150.0	2025-03-13 03:46:24 WIB	DanielVe v. R.E01
SOFTWARE		1	Python 3.13.2 Standard Library (64-bit) v.3.13.2150.0	2025-03-13 03:46:27 WIB	DanielVe v. R.E01
SOFTWARE		1	Python 3.13.2 Tcl/Tk Support (64-bit) v.3.13.2150.0	2025-03-13 03:46:39 WIB	DanielVe v. R.E01
SOFTWARE		1	Python 3.13.2 Test Suite (64-bit) v.3.13.2150.0	2025-03-13 03:46:34 WIB	DanielVe v. R.E01
SOFTWARE		1	Python 3.13.2 pip Bootstrap (64-bit) v.3.13.2150.0	2025-03-13 03:46:40 WIB	DanielVe v. R.E01
SOFTWARE		1	Python Launcher v.3.13.2150.0	2025-03-13 03:46:40 WIB	DanielVe v. R.E01
SOFTWARE	▽	1	Quick Stego 1.2	2025-03-10 10:48:06 WIB	DanielVe v. R.E01
SOFTWARE		1	SchedulingAgent	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE		1	SchedulingAgent	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE		1	SnippingTool-eeFeb196-db28-4d99-816d-fa9a63b4a377	2025-03-06 06:22:23 WIB	DanielVe v. R.E01
SOFTWARE	!	1	USBPcap 1.5.4.0 v.1.5.4.0	2025-03-14 14:31:42 WIB	DanielVe v. R.E01
SOFTWARE		1	Update for x64-based Windows Systems (KB5001716) v...	2025-03-08 15:16:20 WIB	DanielVe v. R.E01
SOFTWARE		1	VMware Tools v.12.4.5.23787635	2025-03-05 18:20:55 WIB	DanielVe v. R.E01
SOFTWARE		1	WIC	2019-12-07 09:17:28 WIB	DanielVe v. R.E01
SOFTWARE		1	WIC	2019-12-07 09:17:27 WIB	DanielVe v. R.E01
SOFTWARE		1	WinRAR 7.10 (64-bit) v.7.10.0	2025-03-10 17:35:47 WIB	DanielVe v. R.E01
SOFTWARE	!	1	Wireshark 4.4.5 x64 v.4.4.5	2025-03-14 14:32:14 WIB	DanielVe v. R.E01
SOFTWARE	!	1	mspaint-b330ad9e-f80b-4c96-9949-4b4228be9a6e	2025-03-06 06:22:23 WIB	DanielVe v. R.E01
SOFTWARE		1	mstsc-4b0a31aa-df6a-4307-9b47-d5cc50009643	2025-03-06 06:22:23 WIB	DanielVe v. R.E01

Figure 6.1.4b — Installed applications identified during forensic.

[illegible]

Daniel Laptop X Case 3 - Autopsy 43.31
Case View Tools Window Help

Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case

Data Sources
Datafile v. RE01_1 Host
File Views
File Types
Deleted Files
MB File Size
Data Artifacts
Chromium Extensions (28)
Communication Profiles (1)
Chromium Accounts (17)
E-Mail Messages (9)
Recent Documents (118)
Installed Programs (57)
Operating System Information (1)
Recent Files (118)
Run Programs (362)
Shell Bags (40)
USB Device Attached (9)
Web Bookmarks (1)
Web Cache (94)
Web Cookies (1710)
Web Downloads (538)
Web Form Autofill (8)
Web History (546)
Web Search (56)
Analysis Results
Encryption Detected (181)
Encryption Suspected (4)
EXIF Metadata (8)
Extension Manually Detected (65)
Interesting Items (46)
Interesting Programs (1)
Privacy Programs (3)
Remote Monitoring Managem
User Content Suspected (8)
Web Account Type (1)
Web Categories (6)
OS Accounts
Tags
Score
Reports

Listing
Run Programs
Table
Thumbnail
Summary

Source Name
S
C
Program Name
O
Username
Date/Time
Comment
Data Source
Bytes Sent
Bytes Received

SYSTEM
Windows\ImmersionControlPanel_cw\sn1h2ywpv
2025-03-07 19:20:40 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Microsoft.Windows.SecHealthUI_cw\sn1h2ywpv
2025-03-07 17:36:50 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\MobilityManagementUser.exe
2025-03-13 19:19:51 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Microsoft.Windows.CloudExperienceHost_cw\sn1h2ywpv
2025-03-14 12:05:12 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program Files (x86)\QuickStep\quickstep.exe
2025-03-13 00:57:44 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Microsoft.Windows.Apprep.ChwApp_cw\sn1h2ywpv
2025-03-10 17:54:20 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\SystemReporterComputerMain.exe
2025-03-07 17:55:01 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Microsoft.Windows.Photos_cw\sn1h2ywpv
2025-03-14 09:45:01 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\ncryptapi.exe
2025-03-13 11:35:01 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\SystemSettingAdminFlows.exe
2025-03-13 10:46:02 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\VMware\VMware Tools\vmtoolsd.exe
2025-03-13 19:19:25 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\Windows NT\Accessories\wordpad.exe
2025-03-10 22:43:17 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\OpenWith.exe
2025-03-10 04:03:45 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
S02WP\ServerPrint32\S3PSreadinst_3bth\wpqrg...
2025-03-13 09:19:42 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\Downloads\comms-v66-710.exe
2025-03-11 00:35:46 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\WinRAR\UnrarUI.exe
2025-03-11 00:37:47 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\WinRAR\WinRAR.exe
2025-03-13 00:57:12 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Microsoft.OutlookForWindows_Bvexy63d3b3bwe
2025-03-13 11:35:51 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\AppData\Local\Microsoft\WPS Office12...
2025-03-11 06:06:54 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\AppData\Local\Microsoft\WPS Office12...
2025-03-11 06:03:32 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Microsoft.Cryptui_Bvexy63d3b3bwe
2025-03-12 11:08:05 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File (x86)\Microsoft\EdgeWebView\Applia...
2025-03-13 11:25:20 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\AppData\Local\Microsoft\WPS Office12...
2025-03-13 22:26:07 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
38833F2B8A1D\UnigramViewer_gfbc92yppwpe
2025-03-14 12:02:55 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\cryptui.exe
2025-03-13 00:56:10 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\AppData\Local\Microsoft\WPS Office12...
2025-03-11 00:57:08 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\AppData\Local\Microsoft\WPS Office12...
2025-03-13 22:02:06 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\Downloads\Winetrack-4.2.5-64.exe
2025-03-13 21:36:24 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\Winetrack\tracp-1.72m
2025-03-14 21:31:31 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\Ncpap\MPFInstall.exe
2025-03-14 21:26:36 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\Winetrack\USB\CapSetup-1.5.4.0.exe
2025-03-14 21:31:44 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\undoc32.exe
2025-03-13 19:23:56 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\msiexec.exe
2025-03-14 21:40:22 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\AccessData\FTK Imager\FTK Imager.exe
2025-03-13 21:39:32 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Program File\Winetrack\Winetrack.exe
2025-03-13 21:35:05 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Windows\System32\mpimapi.exe
2025-03-14 09:45:01 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01
SYSTEM
Users\den_v\OneDrive\Desktop\Tor Browser\Browser...
2025-03-14 10:40:33 WIB
Recently Executed according to Background Activity...
Danielle/v. v.R01

3621 Results
Save Table as CSV

Figure 6.1.4c — Executed program artifacts recovered from the forensic image.

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing Recent Documents 118 Results

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Path	Date Accessed	Data Source
3 headed serpent Stego.bmp.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Steg Images for pr...	0000-00-00 00:00:00	DanielVe v. R.E01
3 headed serpent Stego.Ink	!			C:\Users\dan_v\Pictures\Steg Images for practice\3 he...	2025-03-10 18:00:21 WIB	DanielVe v. R.E01
3 headed serpent.jpg.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\3 h...	0000-00-00 00:00:00	DanielVe v. R.E01
3 headed serpent.Ink	!			C:\Users\dan_v\Pictures\Saved Pictures\3 headed serp...	2025-03-10 17:57:21 WIB	DanielVe v. R.E01
Camera Roll.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Camera Roll	0000-00-00 00:00:00	DanielVe v. R.E01
DanielVega_Network.pcap.Ink	!			C:\Users\Public\Downloads\DanielVega_Network.pcap	0000-00-00 00:00:00	DanielVe v. R.E01
Dark Web.Ink	!			C:\Users\dan_v\OneDrive\Documents\Dark Web	0000-00-00 00:00:00	DanielVe v. R.E01
Desktop.Ink	!			C:\Users\dan_v\Desktop	0000-00-00 00:00:00	DanielVe v. R.E01
Desktop.Ink	!			C:\Users\dan_v\OneDrive\Desktop	0000-00-00 00:00:00	DanielVe v. R.E01
Documents.Ink	!			C:\Users\dan_v\OneDrive\Documents	0000-00-00 00:00:00	DanielVe v. R.E01
Downloads.Ink	!			C:\Users\dan_v\Downloads	0000-00-00 00:00:00	DanielVe v. R.E01
Downloads.Ink	!			C:\Users\Public\Downloads	0000-00-00 00:00:00	DanielVe v. R.E01
E.Ink	!			C:\Users\dan_v\Pictures\E	2025-03-10 18:06:44 WIB	DanielVe v. R.E01
Emails.Ink	!			C:\Users\dan_v\OneDrive\Documents\Emails	0000-00-00 00:00:00	DanielVe v. R.E01
Employment Contract.docx.Ink	!			C:\Users\dan_v\Documents\Work\Employment Contr...	0000-00-00 00:00:00	DanielVe v. R.E01
Employment Contract.docx.Ink	!			C:\Users\dan_v\OneDrive\Documents\Work\Employ...	0000-00-00 00:00:00	DanielVe v. R.E01
Employment Contract.Ink	!			C:\Users\dan_v\Documents\Work\Employment Contr...	2023-01-05 22:39:20 WIB	DanielVe v. R.E01
Evaluation of Different Cleaning Strategies for Rem...	!			C:\Users\dan_v\Documents\Evaluation of Different CL...	0000-00-00 00:00:00	DanielVe v. R.E01
Evaluation of Different Cleaning Strategies for Rem...	!			C:\Users\dan_v\OneDrive\Documents\Evaluation of D...	0000-00-00 00:00:00	DanielVe v. R.E01
Evil Serpent 1.Ink	!			C:\Users\dan_v\Pictures\Exiftool Images for practice\E...	2025-03-10 18:03:05 WIB	DanielVe v. R.E01
Evil Serpent 1.png.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\Evil...	0000-00-00 00:00:00	DanielVe v. R.E01
Evil Serpent 1.png.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Exiftool Images for ...	0000-00-00 00:00:00	DanielVe v. R.E01
Evil Serpent.jpg.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\Evil...	0000-00-00 00:00:00	DanielVe v. R.E01
Evil Serpent.jpg.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Exiftool Images for ...	0000-00-00 00:00:00	DanielVe v. R.E01
Evil Serpent.Ink	!			C:\Users\dan_v\Pictures\Exiftool Images for practice\E...	2025-03-10 18:02:35 WIB	DanielVe v. R.E01
Exiftool Images for practice.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Exiftool Images for ...	0000-00-00 00:00:00	DanielVe v. R.E01
Exiftool Images for practice.Ink	!			C:\Users\dan_v\Pictures\Exiftool Images for practice	2025-03-10 18:15:49 WIB	DanielVe v. R.E01
Hidden.Ink	!			C:\Users\dan_v\OneDrive\Documents\Hidden	0000-00-00 00:00:00	DanielVe v. R.E01
HiddenMessage.bmp.Ink	!			C:\Users\dan_v\OneDrive\Documents\Hidden\Hidde...	0000-00-00 00:00:00	DanielVe v. R.E01
HiddenMessage.bmp.Ink	!			C:\Users\dan_v\Documents\Hidden\HiddenMessage....	2025-03-11 05:49:30 WIB	DanielVe v. R.E01
Human Decomposition Stages.jpg.Ink	!			C:\Users\dan_v\OneDrive\Pictures\Human Decompos...	0000-00-00 00:00:00	DanielVe v. R.E01
Human Decomposition Stages.jpg.Ink	!			C:\Users\dan_v\Pictures\Human Decomposition Stag...	2025-03-13 13:50:59 WIB	DanielVe v. R.E01
ICMP_Global-Report_April_Digital.pdf.Ink	!			C:\Users\dan_v\OneDrive\Documents\ICMP_Global-R...	0000-00-00 00:00:00	DanielVe v. R.E01
ICMP_Global-Report_April_Digital.pdf.Ink	!			C:\Users\dan_v\Documents\ICMP_Global-Report_Apri...	0000-00-00 00:00:00	DanielVe v. R.E01
Intresting Night Club.bmp.Ink	!			C:\Users\dan_v\Documents\Hidden\Intresting Night ...	2025-03-11 00:25:10 WIB	DanielVe v. R.E01
KingsoftData.Ink	!			C:\Users\dan_v\OneDrive\Documents\KingsoftData	0000-00-00 00:00:00	DanielVe v. R.E01
Meeting Notes.txt.Ink	!			C:\Users\dan_v\Documents\Work\Meeting Notes.txt	0000-00-00 00:00:00	DanielVe v. R.E01

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

+ Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing 118 Results

Recent Documents Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Path	Date Accessed	Data Source
Meeting Notes.txt.Ink				C:\Users\dan_v\OneDrive\Documents\Work\Meeting ...	0000-00-00 00:00:00	DanielVe v. R.E01
Music.Ink				C:\Users\dan_v\Music	0000-00-00 00:00:00	DanielVe v. R.E01
NTUSER.DAT				C:\Users\dan_v\OneDrive\Documents\Hidden\safeho...		DanielVe v. R.E01
NTUSER.DAT				Computer\HKEY_CURRENT_CONFIG\Software		DanielVe v. R.E01
New Text Document.txt.Ink				C:\Users\dan_v\Documents\Dark Web\New Text Docu...	2025-03-11 00:37:59 WIB	DanielVe v. R.E01
New folder.Ink				C:\Users\dan_v\OneDrive\Documents\New folder	0000-00-00 00:00:00	DanielVe v. R.E01
No preferred path found.Ink				No preferred path found	0000-00-00 00:00:00	DanielVe v. R.E01
Notes.Ink				C:\Users\dan_v\Pictures\Exiftool Images for practice...	2025-03-10 18:17:44 WIB	DanielVe v. R.E01
Notes.txt.Ink				C:\Users\dan_v\Pictures\Exiftool Images for practice...	0000-00-00 00:00:00	DanielVe v. R.E01
Run Programs (				C:\Users\dan_v\OneDrive\Pictures\Exiftool Images for ...	0000-00-00 00:00:00	DanielVe v. R.E01
Shell Bags (40)				C:\Users\dan_v\Pictures	0000-00-00 00:00:00	DanielVe v. R.E01
USB Device Atta				C:\Users\dan_v\OneDrive\Pictures	0000-00-00 00:00:00	DanielVe v. R.E01
Web Accounts (				C:\Users\dan_v\OneDrive\Pictures	0000-00-00 00:00:00	DanielVe v. R.E01
Web Bookmarks				C:\Users\dan_v\OneDrive\Pictures\Profile Picture.jpg	0000-00-00 00:00:00	DanielVe v. R.E01
Web Cache (894				C:\Users\dan_v\Pictures\Profile Picture.jpg	2025-03-11 05:35:44 WIB	DanielVe v. R.E01
Web Cookies (1				C:\Users\dan_v\Documents\Work\Project Proposal.pdf	2023-01-05 22:40:05 WIB	DanielVe v. R.E01
Web Downloads				C:\Users\dan_v\Documents\Work\Project Proposal.pdf	0000-00-00 00:00:00	DanielVe v. R.E01
Web Form Auto				C:\Users\dan_v\OneDrive\Documents\Work\Project P...	0000-00-00 00:00:00	DanielVe v. R.E01
Web History (34				C:\Users\Public\Downloads	2025-03-13 11:36:26 WIB	DanielVe v. R.E01
Web Search (56)				C:\Users\dan_v\Downloads\QS12Setup	0000-00-00 00:00:00	DanielVe v. R.E01
Analysis Results				C:\Users\dan_v\Downloads\QS12Setup.zip	2025-03-10 17:46:44 WIB	DanielVe v. R.E01
Encryption Dete				C:\Users\dan_v\Downloads\QS12Setup.zip	0000-00-00 00:00:00	DanielVe v. R.E01
Encryption Susp				C:\Users\dan_v\OneDrive\Documents\SIA_Passport_Fr...	0000-00-00 00:00:00	DanielVe v. R.E01
EXIF Metadata (t				C:\Users\dan_v\Documents\SIA_Passport_Fraud_Repo...	0000-00-00 00:00:00	DanielVe v. R.E01
Extension Mism:				C:\Users\dan_v\Pictures\Saved Pictures	0000-00-00 00:00:00	DanielVe v. R.E01
Interesting Item:				C:\Users\dan_v\OneDrive\Pictures\Saved Pictures	0000-00-00 00:00:00	DanielVe v. R.E01
Encryption P				C:\Users\dan_v\Documents\Hidden\Shipment List.pdf	0000-00-00 00:00:00	DanielVe v. R.E01
Privacy Prog				C:\Users\dan_v\OneDrive\Documents\Hidden\Shipm...	0000-00-00 00:00:00	DanielVe v. R.E01
Remote Mor				C:\Users\dan_v\OneDrive\Pictures\Steg Images for pr...	0000-00-00 00:00:00	DanielVe v. R.E01
User Content Su				C:\Users\dan_v\Pictures\Steg Images for practice	2025-03-10 17:59:22 WIB	DanielVe v. R.E01
Web Account Tj				C:\Windows\System32	0000-00-00 00:00:00	DanielVe v. R.E01
Web Categories				No preferred path found	2025-03-06 01:23:19 WIB	DanielVe v. R.E01
OS Accounts				C:\Users\dan_v\OneDrive\Pictures\Steg Images for pr...	0000-00-00 00:00:00	DanielVe v. R.E01
Tags				C:\Users\dan_v\Pictures\Steg Images for practice\Thr...	2025-03-10 18:01:34 WIB	DanielVe v. R.E01
Score				C:\Users\dan_v\OneDrive\Pictures\Saved Pictures\Thr...	0000-00-00 00:00:00	DanielVe v. R.E01
Reports				C:\Users\dan_v\Pictures\Saved Pictures\Three-Heade...	2025-03-10 17:58:19 WIB	DanielVe v. R.E01
				C:\Users\dan_v\OneDrive\Desktop\Tor Browser	0000-00-00 00:00:00	DanielVe v. R.E01
				C:\Users\dan_v\Videos	0000-00-00 00:00:00	DanielVe v. R.E01

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing Recent Documents 118 Results

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Path	Date Accessed	Data Source
Weekly Tasks.Ink				C:\Users\dan_v\Documents\Work\Weekly Tasks.xlsx	2025-03-10 23:01:17 WIB	DanielVe v. R.E01
Weekly Tasks.xlsx.Ink				C:\Users\dan_v\Documents\Work\Weekly Tasks.xlsx	0000-00-00 00:00:00	DanielVe v. R.E01
Weekly Tasks.xlsx.Ink				C:\Users\dan_v\OneDrive\Documents\Work\Weekly T...	0000-00-00 00:00:00	DanielVe v. R.E01
Work.Ink				C:\Users\dan_v\OneDrive\Documents\Work	0000-00-00 00:00:00	DanielVe v. R.E01
Work.Ink				C:\Users\dan_v\Documents\Work	2025-03-10 18:29:07 WIB	DanielVe v. R.E01
client_list.Ink				C:\Users\dan_v\OneDrive\Documents\Hidden\client_...	0000-00-00 00:00:00	DanielVe v. R.E01
client_list.xlsx.Ink				C:\Users\dan_v\Documents\Hidden\client_list.xlsx	2025-03-13 01:09:47 WIB	DanielVe v. R.E01
contacts.txt.Ink				C:\Users\dan_v\Documents\Hidden\contacts.txt	0000-00-00 00:00:00	DanielVe v. R.E01
encrypted.zip.Ink				C:\Users\dan_v\Documents\Hidden\encrypted.zip	0000-00-00 00:00:00	DanielVe v. R.E01
encrypted.zip.Ink				C:\Users\dan_v\OneDrive\Documents\Hidden\encryp...	0000-00-00 00:00:00	DanielVe v. R.E01
exiftool-13.24_64.Ink				C:\Users\dan_v\Downloads\exiftool-13.24_64	0000-00-00 00:00:00	DanielVe v. R.E01
exiftool-13.24_64.Ink				C:\Users\dan_v\Downloads\exiftool-13.24_64\exiftool-...	0000-00-00 00:00:00	DanielVe v. R.E01
exiftool-13.24_64.Ink				C:\Users\dan_v\Downloads\exiftool-13.24_64.zip	2025-03-10 17:53:08 WIB	DanielVe v. R.E01
exiftool-13.24_64.zip.Ink				C:\Users\dan_v\Downloads\exiftool-13.24_64.zip	0000-00-00 00:00:00	DanielVe v. R.E01
fake reviews.docx.Ink				C:\Users\dan_v\Documents\Dark Web\fake reviews.d...	0000-00-00 00:00:00	DanielVe v. R.E01
fake reviews.docx.Ink				C:\Users\dan_v\OneDrive\Documents\Dark Web\fake ...	0000-00-00 00:00:00	DanielVe v. R.E01
fake.docx.Ink				C:\Users\dan_v\Documents\Dark Web\fake.docx	2025-03-11 07:55:37 WIB	DanielVe v. R.E01
intramuscular-injection-step-by-step-final.pdf.Ink				C:\Users\dan_v\OneDrive\Documents\intramuscular-...	0000-00-00 00:00:00	DanielVe v. R.E01
intramuscular-injection-step-by-step-final.pdf.Ink				C:\Users\dan_v\Documents\intramuscular-injection-s...	0000-00-00 00:00:00	DanielVe v. R.E01
login details.txt.Ink				C:\Users\dan_v\Documents\Dark Web\login details.txt	0000-00-00 00:00:00	DanielVe v. R.E01
login details.txt.Ink				C:\Users\dan_v\OneDrive\Documents\Dark Web\logi...	0000-00-00 00:00:00	DanielVe v. R.E01
meeting notes.pdf.Ink				C:\Users\dan_v\Documents\Work\meeting notes.pdf	0000-00-00 00:00:00	DanielVe v. R.E01
meeting notes.pdf.Ink				C:\Users\dan_v\OneDrive\Documents\Work\meeting ...	0000-00-00 00:00:00	DanielVe v. R.E01
microsoft-edgehttps-www.bing.com-spotlightspc				No preferred path found	2025-03-14 00:22:48 WIB	DanielVe v. R.E01
ms-gamingoverlay--kgclcheck-.Ink				No preferred path found	2025-03-06 01:23:19 WIB	DanielVe v. R.E01
ms-photospareprocess-viewer.Ink				No preferred path found	2025-03-10 18:05:36 WIB	DanielVe v. R.E01
ms-windowsbackup---ReferrallId=WindowsBackup				No preferred path found	2025-03-10 17:12:57 WIB	DanielVe v. R.E01
nightclub.bmp.Ink				C:\Users\dan_v\Documents\Hidden\nightclub.bmp	2025-03-11 05:46:14 WIB	DanielVe v. R.E01
nightclub.jpg.Ink				C:\Users\dan_v\Downloads\nightclub.jpg	0000-00-00 00:00:00	DanielVe v. R.E01
onion links.txt.Ink				C:\Users\dan_v\OneDrive\Documents\Dark Web\onion...	0000-00-00 00:00:00	DanielVe v. R.E01
outlookcal.Ink				No preferred path found	2025-03-07 17:29:13 WIB	DanielVe v. R.E01
safehouse location.bmp.Ink				C:\Users\dan_v\OneDrive\Documents\Hidden\safeho...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse location.jpg.Ink				C:\Users\dan_v\OneDrive\Documents\Hidden\safeho...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse location.webp.Ink				C:\Users\dan_v\Documents\Hidden\safehouse locatio...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse location.webp.Ink				C:\Users\dan_v\OneDrive\Documents\Hidden\safeho...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse locations.rtf.Ink				C:\Users\dan_v\Documents\Hidden\safehouse locatio...	0000-00-00 00:00:00	DanielVe v. R.E01
safehouse locations.txt.Ink				C:\Users\dan_v\Documents\Hidden\safehouse locatio...	0000-00-00 00:00:00	DanielVe v. R.E01
sysdm.cpl.Ink				C:\Windows\system32\sysdm.cpl	0000-00-00 00:00:00	DanielVe v. R.E01
system32.Ink				C:\Windows\system32	2025-03-10 17:54:30 WIB	DanielVe v. R.E01
usb_backup_files (2).zip.Ink				E:\usb_backup_files (2).zip	0000-00-00 00:00:00	DanielVe v. R.E01
usb_backup_files.zip.Ink				C:\Users\dan_v\Documents\Hidden\usb_backup_files....	0000-00-00 00:00:00	DanielVe v. R.E01
usb_backup_files.zip.Ink				C:\Users\dan_v\OneDrive\Documents\Hidden\usb_ba...	0000-00-00 00:00:00	DanielVe v. R.E01
victim list.txt.Ink				C:\Users\dan_v\Desktop\victim list.txt	0000-00-00 00:00:00	DanielVe v. R.E01
windowsdefender---.Ink				No preferred path found	2025-03-07 17:33:42 WIB	DanielVe v. R.E01

Figure 6.1.4d — Recently accessed document artifacts.

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

+ Add Data Source
 Images/Videos
 Communications
 Geolocation
 Timeline
 Discovery
 Generate Report
 Close Case
 Keyword Lists
 Keyword Search

Listing

USB Device Attached

9 Results

Save Table as CSV

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
SYSTEM			1	2025-03-13 19:18:34 WIB		ROOT_HUB	58:2891968b&0	DanielVe v. R.E01
SYSTEM			1	2025-03-13 19:18:34 WIB		ROOT_HUB20	58:36a4b5d6&0	DanielVe v. R.E01
SYSTEM			1	2025-03-13 19:18:35 WIB		ROOT_HUB30	58:11106705&0&0	DanielVe v. R.E01
SYSTEM			1	2025-03-13 19:18:35 WIB	VMware, Inc.	Virtual USB Hub	68:39d724fe&0&7	DanielVe v. R.E01
SYSTEM			1	2025-03-13 19:18:35 WIB	VMware, Inc.	Virtual USB Hub	68:39d724fe&0&8	DanielVe v. R.E01
SYSTEM			1	2025-03-13 19:18:35 WIB	VMware, Inc.	Virtual Mouse	68:39d724fe&0&5	DanielVe v. R.E01
SYSTEM			1	2025-03-13 19:18:35 WIB	VMware, Inc.	Virtual Mouse	78:bcbfcc2&0&0000	DanielVe v. R.E01
SYSTEM			1	2025-03-13 19:18:35 WIB	VMware, Inc.	Virtual Mouse	78:bcbfcc2&0&0001	DanielVe v. R.E01
SYSTEM			1	2025-03-11 01:20:05 WIB		VID_346D&PID_5678	7399821011223036534	DanielVe v. R.E01

Data Sources
 DanielVe v. R.E01
 DanielVe v. R.E01
 File Views
 File Types
 Deleted Files
 MB File Size
 Data Artifacts
 Chromium Ext
 Chromium Profi
 Communication
 E-Mail Messages
 Favicon (621)
 Installed Progr
 Operating Syste
 Recent Docume
 Run Programs
 Shell Bags (40)
 USB Device Atta
 Web Accounts
 Web Bookmarks
 Web Cache (894)
 Web Cookies (17)
 Web Downloads
 Web Form Auto
 Web History (54)
 Web Search (56)
 Analysis Results
 Encryption Dete
 Encryption Susp
 EXIF Metadata (1)
 Extension Mism
 Interesting Item
 Encryption P
 Privacy Prog
 Remote Mor
 User Content Su
 Web Account T
 Web Categories
 OS Accounts
 Tags
 Score
 Reports

Figure 6.1.4e — USB device connection artifacts.

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source
 Images/Videos
 Communications
 Geolocation
 Timeline
 Discovery
 Generate Report
 Close Case
 Keyword Lists
 Keyword Search

Listing

Encryption Programs 1 Results

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Category	File Path
Tor Browser		!		File	Likely Notable		Encryption Programs		Tor Browser	/img_DanielVe v. R.E01/Users/dan_v/OneDrive/Desko... 2

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
 - Chromium Extens
 - Chromium Profile
 - Communication /
 - E-Mail Messages (
 - Favicon (621)
 - Installed Program
 - Operating System
 - Recent Document
 - Run Programs (36)
 - Shell Bags (40)
 - USB Device Attac
 - Web Accounts (1)
 - Web Bookmarks (
 - Web Cache (894)
 - Web Cookies (171)
 - Web Downloads (
 - Web Form Autofil
 - Web History (549)
 - Web Search (56)
- Analysis Results
 - Encryption Detect
 - Encryption Suspe
 - EXIF Metadata (6)
 - Extension Mismat
 - Interesting Items (
 - Encryption Pr
 - Privacy Progra
 - Remote Monit
 - User Content Susq
 - Web Account Typ
 - Web Categories (8
- OS Accounts
- Tags
- Score
- Reports

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

[Add Data Source](#)
[Images/Videos](#)
[Communications](#)
[Geolocation](#)
[Timeline](#)
[Discovery](#)
[Generate Report](#)
[Close Case](#)
[Keyword Lists](#)
[Keyword Search](#)

Listing

Encryption Suspected 4 Results

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Comment
mpenginedb.db			1	File	Likely Notable			Suspected encryption due to high entropy (7.992922).	Suspected encryption due to high entropy (7.992922).
thumbcache_768.db			1	File	Likely Notable			Suspected encryption due to high entropy (7.578199).	Suspected encryption due to high entropy (7.578199).
store.db			1	File	Likely Notable			Suspected encryption due to high entropy (7.999752).	Suspected encryption due to high entropy (7.999752).
cache.db-slack				File	Likely Notable			Suspected encryption due to high entropy (7.998194).	Suspected encryption due to high entropy (7.998194).

File Views

- File Types
- Deleted Files
- MB File Size
- Data Artifacts
 - Chromium Extens
 - Chromium Profile
 - Communication /
 - E-Mail Messages (
 - Favicon (621)
 - Installed Program
 - Operating System
 - Recent Document
 - Run Programs (36
 - Shell Bags (40)
 - USB Device Attach
 - Web Accounts (1)
 - Web Bookmarks (
 - Web Cache (894)
 - Web Cookies (171
 - Web Downloads (
 - Web Form Autofil
 - Web History (549)
 - Web Search (56)
- Analysis Results
 - Encryption Detect
 - Encryption Suspe
 - EXIF Metadata (6)
 - Extension Mismat
 - Interesting Items (
 - Encryption Prc
 - Privacy Progra
 - Remote Monit
 - User Content Susq
 - Web Account Typ
 - Web Categories (6
 - OS Accounts
 - Tags
 - Score

Figure 6.1.4f — Encryption-related artifacts identified during examination.

[illegible]

Figure 6.1.4g — Remote monitoring software identified from system artifacts.

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source
 Images/Videos
 Communications
 Geolocation
 Timeline
 Discovery
 Generate Report
 Close Case
 Keyword Lists
 Keyword Search

Listing

User Content Suspected

Table Thumbnail Summary

6 Results

Save Table as CSV

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Comment	File Path
WelcomeScan.jpg			2	File	Unknown				EXIF metadata data exists for this file.	/img_DanielVe v. R.E01/ProgramData
bg1a_thumb.png			1	File	Unknown				EXIF metadata data exists for this file.	/img_DanielVe v. R.E01/Program File
WelcomeScan.jpg			2	File	Unknown				EXIF metadata data exists for this file.	/img_DanielVe v. R.E01/Windows/Wi
WelcomeScan.jpg			2	File	Unknown				EXIF metadata data exists for this file.	/img_DanielVe v. R.E01/Windows/Wi
f000d6b				File	Unknown				EXIF metadata data exists for this file.	/img_DanielVe v. R.E01/\$OrphanFiles
f0483176.jpg			0	File	Unknown				EXIF metadata data exists for this file.	/img_DanielVe v. R.E01/\$CarvedFiles

Figure 6.1.4h — User content flagged during forensic.

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case Keyword Lists Keyword Search

Listing

/img_DanielVe v. R.E01/Windows/System32/config

59 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flag
COMPONENTS\53b39e62-18c4-11ea-a811-000d3aa				2025-03-13 19:19:32 WIB	2025-03-13 19:19:32 WIB	2025-03-13 19:19:32 WIB	2025-03-13 19:19:32 WIB	65536	Una
COMPONENTS\53b39e63-18c4-11ea-a811-000d3aa	0			2025-03-13 19:18:49 WIB	2025-03-13 19:18:49 WIB	2025-03-13 19:18:49 WIB	2025-03-06 13:23:44 WIB	65536	Allo
COMPONENTS\53b39e63-18c4-11ea-a811-000d3aa	0			2025-03-08 21:21:45 WIB	2025-03-08 21:21:45 WIB	2025-03-08 21:47:19 WIB	2025-03-06 13:23:44 WIB	524288	Allo
COMPONENTS\53b39e63-18c4-11ea-a811-000d3aa	0			2025-03-13 19:18:49 WIB	2025-03-13 19:18:49 WIB	2025-03-13 19:21:36 WIB	2025-03-06 13:23:44 WIB	524288	Allo
DEFAULT			0	2025-03-13 19:18:24 WIB	2025-03-06 13:20:44 WIB	2025-03-13 19:18:24 WIB	2019-12-07 16:03:44 WIB	524288	Allo
DEFAULT.LOG1			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	114688	Allo
DEFAULT.LOG2			0	2019-12-07 16:03:44 WIB	2025-03-06 13:22:11 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	147456	Allo
DRIVERS			0	2025-03-13 19:22:59 WIB	2025-03-06 13:21:34 WIB	2025-03-13 21:52:01 WIB	2019-12-07 16:03:44 WIB	4374528	Allo
DRIVERS.LOG1			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	1048576	Allo
DRIVERS.LOG2			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:43 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	208896	Allo
DRIVERS\53b39e6f-18c4-11ea-a811-000d3aa4692b\.				2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	1048576	Una
DRIVERS\53b39e6f-18c4-11ea-a811-000d3aa4692b\.				2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	1048576	Una
DRIVERS\53b39e6f-18c4-11ea-a811-000d3aa4692b\.				2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	1048576	Una
DRIVERS\53b39e6f-18c4-11ea-a811-000d3aa4692b\.				2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	2025-03-13 19:17:18 WIB	65536	Una
DRIVERS\53b39e70-18c4-11ea-a811-000d3aa4692b\.	0			2025-03-13 19:18:24 WIB	2025-03-13 19:18:24 WIB	2025-03-13 19:18:24 WIB	2025-03-06 13:22:49 WIB	65536	Allo
DRIVERS\53b39e70-18c4-11ea-a811-000d3aa4692b\.	0			2019-12-07 16:03:44 WIB	2025-03-13 21:52:01 WIB	2025-03-13 21:52:01 WIB	2025-03-06 13:22:49 WIB	524288	Allo
DRIVERS\53b39e70-18c4-11ea-a811-000d3aa4692b\.	2			2025-03-06 13:22:49 WIB	2025-03-06 13:22:49 WIB	2025-03-06 13:22:49 WIB	2025-03-06 13:22:49 WIB	524288	Allo
ELAM			0	2025-03-06 21:15:40 WIB	2025-03-06 13:21:34 WIB	2025-03-08 21:47:19 WIB	2019-12-07 16:03:44 WIB	32768	Allo
ELAM.LOG1			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:42 WIB	2025-03-08 21:47:19 WIB	2019-12-07 16:03:44 WIB	32768	Allo
ELAM.LOG2				2019-12-07 16:03:44 WIB	2025-03-06 13:22:11 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	0	Allo
ELAM\53b39eac-18c4-11ea-a811-000d3aa4692b\.	0			2025-03-06 13:23:06 WIB	2025-03-06 13:23:06 WIB	2025-03-06 13:23:06 WIB	2025-03-06 13:23:06 WIB	65536	Allo
ELAM\53b39eac-18c4-11ea-a811-000d3aa4692b\.	0			2025-03-06 13:23:06 WIB	2025-03-06 13:23:06 WIB	2025-03-06 21:15:40 WIB	2025-03-06 13:23:06 WIB	524288	Allo
ELAM\53b39eac-18c4-11ea-a811-000d3aa4692b\.	2			2025-03-06 13:23:06 WIB	2025-03-06 13:23:06 WIB	2025-03-06 13:23:06 WIB	2025-03-06 13:23:06 WIB	524288	Allo
SAM			0	2025-03-13 19:18:24 WIB	2025-03-06 13:21:54 WIB	2025-03-13 19:18:24 WIB	2019-12-07 16:03:44 WIB	131072	Allo
SAM.LOG1			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	65536	Allo
SAM.LOG2			0	2019-12-07 16:03:44 WIB	2025-03-06 13:22:11 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	82944	Allo
SECURITY			0	2025-03-13 19:18:24 WIB	2025-03-06 13:21:54 WIB	2025-03-13 19:18:24 WIB	2019-12-07 16:03:44 WIB	32768	Allo
SECURITY.LOG1			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	65536	Allo
SECURITY.LOG2			0	2019-12-07 16:03:44 WIB	2025-03-06 13:22:11 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	57344	Allo
SOFTWARE			0	2025-03-13 19:18:24 WIB	2025-03-06 13:21:55 WIB	2025-03-13 19:18:24 WIB	2019-12-07 16:03:44 WIB	79429632	Allo
SOFTWARE.LOG1			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	4194304	Allo
SOFTWARE.LOG2			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:43 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	8519680	Allo
SYSTEM			0	2025-03-13 19:18:24 WIB	2025-03-06 13:21:57 WIB	2025-03-13 19:18:24 WIB	2019-12-07 16:03:44 WIB	12845056	Allo
SYSTEM.LOG1			0	2019-12-07 16:03:44 WIB	2025-03-06 13:21:42 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	1572864	Allo
SYSTEM.LOG2			0	2019-12-07 16:03:44 WIB	2025-03-06 13:22:11 WIB	2019-12-07 16:03:44 WIB	2019-12-07 16:03:44 WIB	1409024	Allo
COMPONENTS.LOG1				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Una

Figure 6.1.4i — Windows System.

Daniel Laptop X Case 3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Keyword Lists Keyword Search

Listing Web Search 56 Results

Table Thumbnail Summary

Save Table as CSV

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History	!			bing.com	how to browse dark web	Microsoft Edge	2025-03-10 17:26:49 WIB	DanielVe v. R.E01
History				bing.com	how to browse dark web	Microsoft Edge	2025-03-10 17:26:49 WIB	DanielVe v. R.E01
History				bing.com	how to browse dark web	Microsoft Edge	2025-03-10 17:26:49 WIB	DanielVe v. R.E01
History	!			bing.com	tor browser	Microsoft Edge	2025-03-10 17:27:23 WIB	DanielVe v. R.E01
History				bing.com	tor browser	Microsoft Edge	2025-03-10 17:27:23 WIB	DanielVe v. R.E01
History	!			bing.com	image steganography tool	Microsoft Edge	2025-03-10 17:45:17 WIB	DanielVe v. R.E01
History				bing.com	image steganography tool	Microsoft Edge	2025-03-10 17:45:17 WIB	DanielVe v. R.E01
History				bing.com	image steganography tool	Microsoft Edge	2025-03-10 17:45:17 WIB	DanielVe v. R.E01
History	!			bing.com	quickstego download	Microsoft Edge	2025-03-10 17:46:10 WIB	DanielVe v. R.E01
History				bing.com	quickstego download	Microsoft Edge	2025-03-10 17:46:10 WIB	DanielVe v. R.E01
History	!			bing.com	3 serpent	Microsoft Edge	2025-03-10 17:56:34 WIB	DanielVe v. R.E01
History	!			bing.com	3 serpent	Microsoft Edge	2025-03-10 17:58:24 WIB	DanielVe v. R.E01
History				bing.com	3 serpent	Microsoft Edge	2025-03-10 17:58:24 WIB	DanielVe v. R.E01
History				bing.com	3 serpent	Microsoft Edge	2025-03-10 17:58:24 WIB	DanielVe v. R.E01
History	!			bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:02:19 WIB	DanielVe v. R.E01
History	!			bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History				bing.com	Evil Serpent	Microsoft Edge	2025-03-10 18:05:59 WIB	DanielVe v. R.E01
History	!			bing.com	how to delete my digital footprint	Microsoft Edge	2025-03-10 18:27:29 WIB	DanielVe v. R.E01
History				bing.com	how to delete my digital footprint	Microsoft Edge	2025-03-10 18:27:29 WIB	DanielVe v. R.E01
History	!			bing.com	Travel to Bangkok	Microsoft Edge	2025-03-10 22:47:54 WIB	DanielVe v. R.E01
History	!			bing.com	night club	Microsoft Edge	2025-03-11 00:21:30 WIB	DanielVe v. R.E01
History				bing.com	night club	Microsoft Edge	2025-03-11 00:21:30 WIB	DanielVe v. R.E01
History				bing.com	night club	Microsoft Edge	2025-03-11 00:21:30 WIB	DanielVe v. R.E01
History	!			bing.com	night club	Microsoft Edge	2025-03-11 00:21:33 WIB	DanielVe v. R.E01
History				bing.com	nightclub	Microsoft Edge	2025-03-11 00:22:00 WIB	DanielVe v. R.E01
History				bing.com	nightclub	Microsoft Edge	2025-03-11 00:22:00 WIB	DanielVe v. R.E01
History	!			bing.com	nightclub	Microsoft Edge	2025-03-11 00:22:00 WIB	DanielVe v. R.E01
History	!			bing.com	winrar	Microsoft Edge	2025-03-11 00:34:56 WIB	DanielVe v. R.E01
History				bing.com	winrar	Microsoft Edge	2025-03-11 00:34:56 WIB	DanielVe v. R.E01
History	!			bing.com	child labour country ranking	Microsoft Edge	2025-03-11 05:26:07 WIB	DanielVe v. R.E01
History				bing.com	child labour country ranking	Microsoft Edge	2025-03-11 05:26:07 WIB	DanielVe v. R.E01
History				bing.com	child labour country ranking	Microsoft Edge	2025-03-11 05:26:07 WIB	DanielVe v. R.E01

History	!		bing.com	methods of falsifying passports	Microsoft Edge	2025-03-11 05:40:15 WIB	DanielVe v. R.E01
History			bing.com	methods of falsifying passports	Microsoft Edge	2025-03-11 05:40:15 WIB	DanielVe v. R.E01
History			bing.com	methods of falsifying passports	Microsoft Edge	2025-03-11 05:40:15 WIB	DanielVe v. R.E01
History	!		bing.com	how to wash dna off	Microsoft Edge	2025-03-12 11:50:51 WIB	DanielVe v. R.E01
History			bing.com	how to wash dna off	Microsoft Edge	2025-03-12 11:50:51 WIB	DanielVe v. R.E01
History			bing.com	how to wash dna off	Microsoft Edge	2025-03-12 11:50:51 WIB	DanielVe v. R.E01
History	!		bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History			bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History			bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History			bing.com	alternative ways to be buried	Microsoft Edge	2025-03-13 00:44:21 WIB	DanielVe v. R.E01
History	!		bing.com	bangkok thailand	Microsoft Edge	2025-03-13 00:50:07 WIB	DanielVe v. R.E01
History			bing.com	bangkok thailand	Microsoft Edge	2025-03-13 00:54:25 WIB	DanielVe v. R.E01
History	!		bing.com	bangkok thailand	Microsoft Edge	2025-03-13 00:54:25 WIB	DanielVe v. R.E01
History	!		bing.com	stages of decomposition human chart	Microsoft Edge	2025-03-13 13:50:25 WIB	DanielVe v. R.E01
History	!		bing.com	stages of decomposition human chart	Microsoft Edge	2025-03-13 13:50:25 WIB	DanielVe v. R.E01
History			bing.com	stages of decomposition human chart	Microsoft Edge	2025-03-13 13:50:25 WIB	DanielVe v. R.E01
History	!		bing.com	\ftk imager	Microsoft Edge	2025-03-14 21:24:18 WIB	DanielVe v. R.E01
History			bing.com	\ftk imager	Microsoft Edge	2025-03-14 21:24:18 WIB	DanielVe v. R.E01

Figure 6.1.5a — Browser search history recovered.

Source Name	S	C	O	Path	URL	Date Accessed
windows_exiftool.txt:Zone.Identifier				/Users/dan_v/Downloads/exiftool-13.24_64/exiftool-13.24_64/exiftool.fil...		
README.txt:Zone.Identifier				/Users/dan_v/Downloads/exiftool-13.24_64/exiftool-13.24_64/README.txt		
Profile Picture.jpg:Zone.Identifier			1	/Users/dan_v/OneDrive/Pictures/Profile Picture.jpg	https://www.dcode.f...	
3 headed serpent.jpg:Zone.Identifier			1	/Users/dan_v/OneDrive/Pictures/Saved Pictures/3 headed serpent.jpg	https://th.bing.com/...	
SIA_Passport_Fraud_Report_Short_EN.pdf:Zone.Identifier			1	/Users/dan_v/OneDrive/Documents/SIA_Passport_Fraud_Report_Short_E...	https://secureidentit...	
Evil Serpent.jpg:Zone.Identifier			1	/Users/dan_v/OneDrive/Pictures/Saved Pictures/Evil Serpent.jpg	https://th.bing.com/...	
Evil Serpent 1.png:Zone.Identifier			1	/Users/dan_v/OneDrive/Pictures/Saved Pictures/Evil Serpent 1.png	https://i.pinimg.com/...	
Human Decomposition Stages.jpg:Zone.Identifier			1	/Users/dan_v/OneDrive/Pictures/Human Decomposition Stages.jpg	https://images.squar...	
Three-Headed serpent.jpg:Zone.Identifier			1	/Users/dan_v/OneDrive/Pictures/Saved Pictures/Three-Headed serpent.j...	https://c8.alamy.co...	
intramuscular-injection-step-by-step-final.Zon			1	/Users/dan_v/OneDrive/Documents/intramuscular-injection-step-by-ste...	https://www.ljmu.ac...	
Evaluation of Different Cleaning Strategies for Rem			1	/Users/dan_v/OneDrive/Documents/Evaluation of Different Cleaning Str...	https://pmc.ncbi.nlm...	
safehouse location.jpg:Zone.Identifier			1	/Users/dan_v/OneDrive/Documents/Hidden/safehouse location.jpg	https://th.bing.com/...	
History			0	C:\Users\dan_v\Downloads\tor-browser-windows-x86_64-portable-14.0...	https://www.torproj...	2025-03-07 17:33:06
History			0	C:\Users\dan_v\Downloads\tor-browser-windows-x86_64-portable-14.0...	https://dist.torprojec...	2025-03-07 17:33:06
History			0	C:\Users\dan_v\Downloads\Q512Setup.zip	http://www.quickcry...	2025-03-10 17:46:36
History			0	C:\Users\dan_v\Downloads\exiftool-13.24_64.zip	https://www.exiftool...	2025-03-10 17:52:52
History			1	C:\Users\dan_v\Pictures\Saved Pictures\3 headed serpent.jpg	https://th.bing.com/...	2025-03-10 17:56:53
History			1	C:\Users\dan_v\Pictures\Saved Pictures\Three-Headed serpent.jpg	https://c8.alamy.co...	2025-03-10 17:58:05
History			1	C:\Users\dan_v\Pictures\Saved Pictures\Evil Serpent.jpg	https://th.bing.com/...	2025-03-10 18:02:26
History			1	C:\Users\dan_v\Pictures\Saved Pictures\Evil Serpent 1.png	https://i.pinimg.com/...	2025-03-10 18:02:51
History			1	C:\Users\dan_v\Downloads\nightclub.jpg	https://th.bing.com/...	2025-03-11 00:22:05
History			1	C:\Users\dan_v\Downloads\winrar-x64-710.exe	https://www.win-rar...	2025-03-11 00:35:05
History			1	C:\Users\dan_v\Documents\ICMP_Global-Report_April_Digital.pdf	https://www.icmp.in...	2025-03-11 05:27:53
History			1	C:\Users\dan_v\Pictures\Profile Picture.jpg	https://www.dcode.f...	2025-03-11 05:35:27
History			1	C:\Users\dan_v\Documents\SIA_Passport_Fraud_Report_Short_EN.pdf	https://secureidentit...	2025-03-11 05:39:19
History			1	C:\Users\dan_v\Documents\intramuscular-injection-step-by-step-final...	https://www.ljmu.ac...	2025-03-12 11:26:06
History			0	C:\Users\dan_v\Documents\Hidden\safeshouse location.webp	https://thumbs.drea...	2025-03-13 00:53:51
History			1	C:\Users\dan_v\Downloads\Setup-OpenStego-0.8.6.exe	https://github.com/...	2025-03-13 00:55:14
History			0	C:\Users\dan_v\Downloads\Setup-OpenStego-0.8.6.exe	https://objects.githu...	2025-03-13 00:55:14
History			1	C:\Users\dan_v\Documents\Hidden\safeshouse location.jpg	https://th.bing.com/...	2025-03-13 00:56:29
History			1	C:\Users\dan_v\Documents\Evaluation of Different Cleaning Strategies f...	https://pmc.ncbi.nlm...	2025-03-13 13:47:58
History			1	C:\Users\dan_v\Pictures\Human Decomposition Stages.jpg	https://images.squar...	2025-03-13 13:50:39
History			1	C:\Users\dan_v\Pictures\Human Decomposition Stages.jpg	https://images.squar...	2025-03-13 13:50:43
History			1	C:\Users\dan_v\Downloads\python-3.13.2-amd64.exe	https://www.python...	2025-03-14 10:45:10
History			1	C:\Users\dan_v\Downloads\Wireshark-4.4.5-x64.exe	https://2.na.dl.wires...	2025-03-14 21:24:07
History			0	C:\Users\dan_v\Downloads\Exterro_FTK_Imager(x64)-4.7.3.81.exe	https://d1kpmuw7...	2025-03-14 21:25:23

Figure 6.1.5b — Browser download history recovered from the forensic image.

6.2 Network Findings

Protocol hierarchy analysis was performed using Wireshark to identify the communication protocols present within the captured network traffic. The capture consisted of 29,211 frames, with IPv4 accounting for approximately 99.5% of all packets and IPv6 accounting for approximately 0.3%.

The dominant application-layer protocols identified during the examination included QUIC, TLS, DNS, HTTP, and FTP/FTP Data.

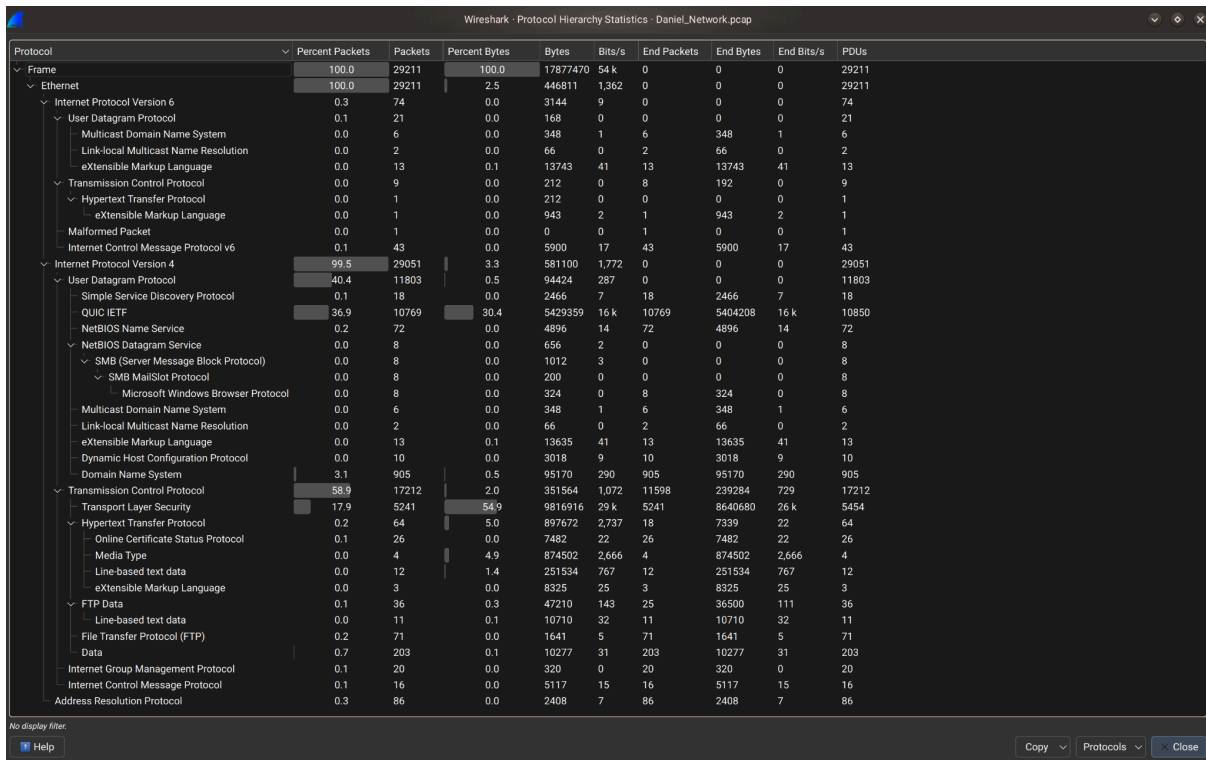


Figure 6.2.1 — Wireshark Protocol Hierarchy Statistics showing the distribution of network protocols observed within *Daniel_Network.pcap*. IPv4 traffic represented the majority of captured packets, with significant TLS, QUIC, DNS, HTTP, and FTP communications.

6.2.1 HTTP Analysis

HTTP traffic was examined in Wireshark using the HTTP display filter. Multiple HTTP request and response exchanges were identified between the suspect host (192.168.40.129) and the HTTP server (192.168.40.135). Analysis of the captured traffic confirmed the retrieval of several evidence files, including *HiddenMessage.bmp*, *safehouse location.bmp*, *Shipment List.pdf*, *client_list*, and *encrypted.zip* through HTTP GET requests. Exported HTTP objects were preserved for further examination.

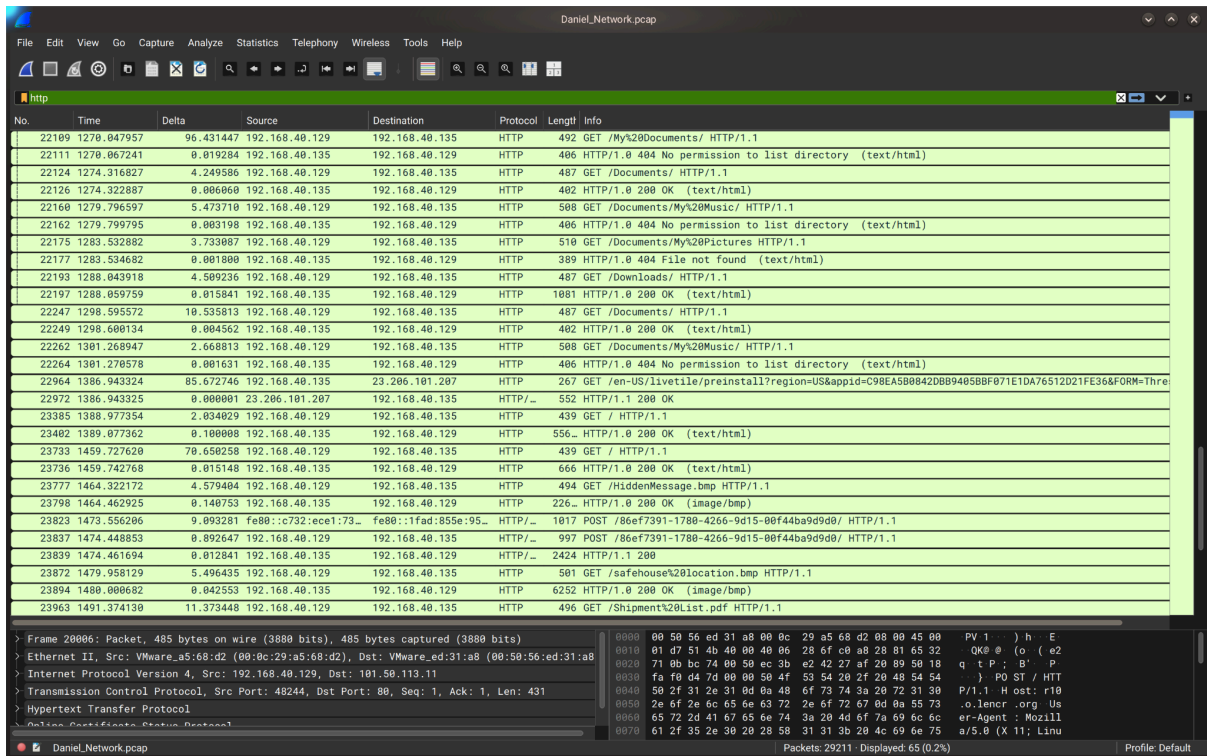


Figure 6.2.2 — HTTP traffic filter and requests.

6.2.2 FTP Analysis

FTP analysis identified multiple authentication attempts from host 192.168.40.135 to 192.168.40.129. An initial login attempt using the credentials "ftpuser" and password "123456" failed with a 530 Login incorrect response. Subsequent authentication attempts using the password "anonymous" were accepted by the FTP server (230 Login successful). Following successful authentication, the suspect retrieved INVOICE.pdf using the FTP RETR command.

Unlike the FTP activity, other evidence files, including HiddenMessage.bmp, safehouse location.bmp, Shipment List.pdf, client_list, and encrypted.zip, were transferred separately through HTTP GET requests from a Python SimpleHTTP server running on port 8080. This indicates that HTTP and FTP were used for different purposes during the communication.

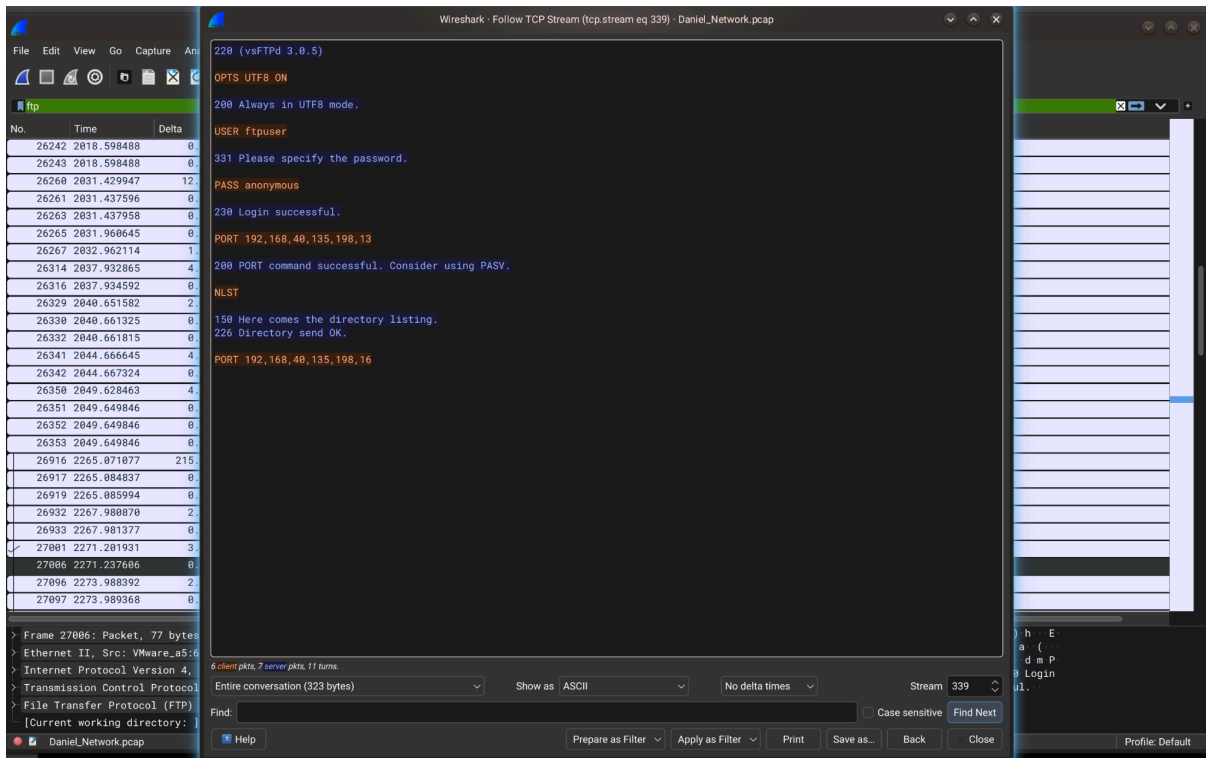


Figure 6.2.3 — FTP session — TCP Stream Follow.

6.2.3 DNS and TLS Analysis

DNS traffic was examined to identify resolved domains and communication endpoints associated with the captured network activity. The analysis identified multiple DNS queries to external services, including Microsoft, Mozilla, Google, Bing, Office 365, and OneDrive infrastructure. In addition, several DNS queries referenced .onion domains, including darkmarketxyz.onion, humantradeabcxyz.onion, shadowdealersxyz.onion, redphoenixtrades.onion, and deepwebroutesxyz.onion, indicating attempts to resolve hidden-service domain names.

TLS sessions were also identified throughout the network capture. Examination of the Server Name Indication (SNI) values showed encrypted communications primarily involving Microsoft, Mozilla, Google, and related cloud services. Because TLS encrypts application-layer content, the exchanged data could not be examined further. No direct TLS connections, Server Name Indication (SNI), or standard Tor service ports (9001, 9030, 9050, and 9150) associated with the identified .onion domains were observed within the captured traffic.

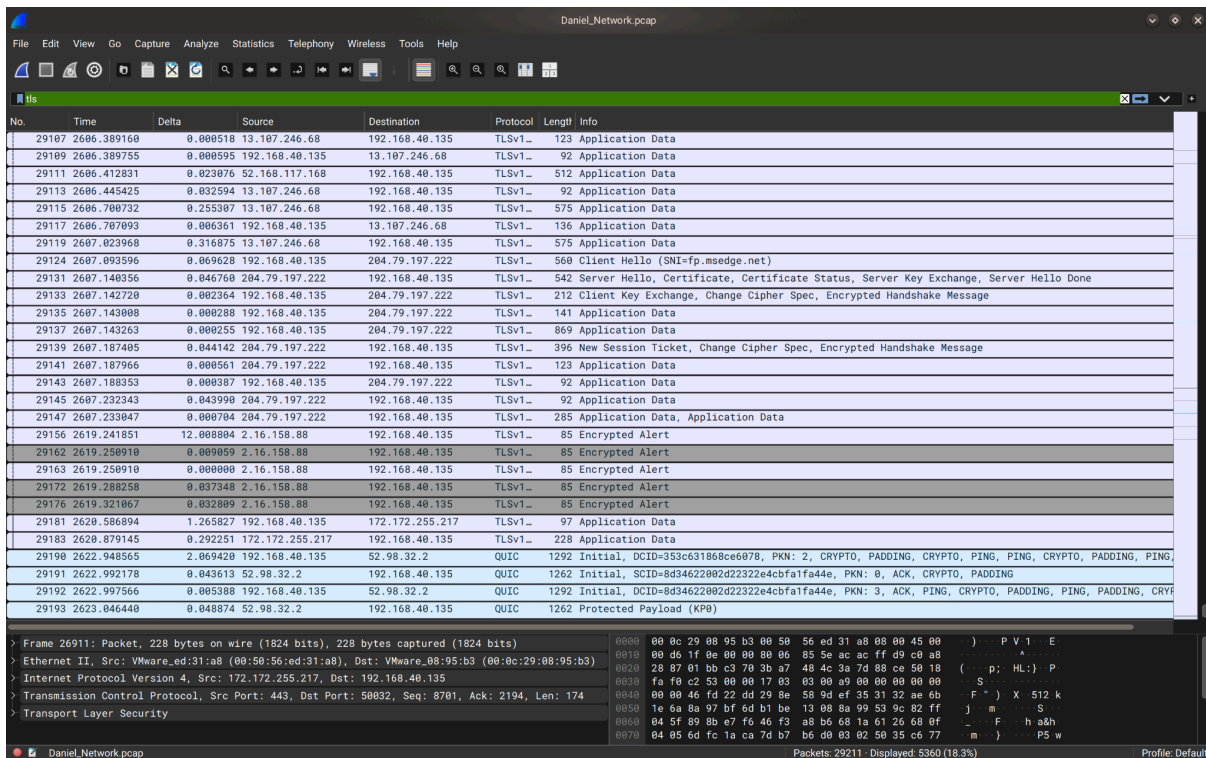
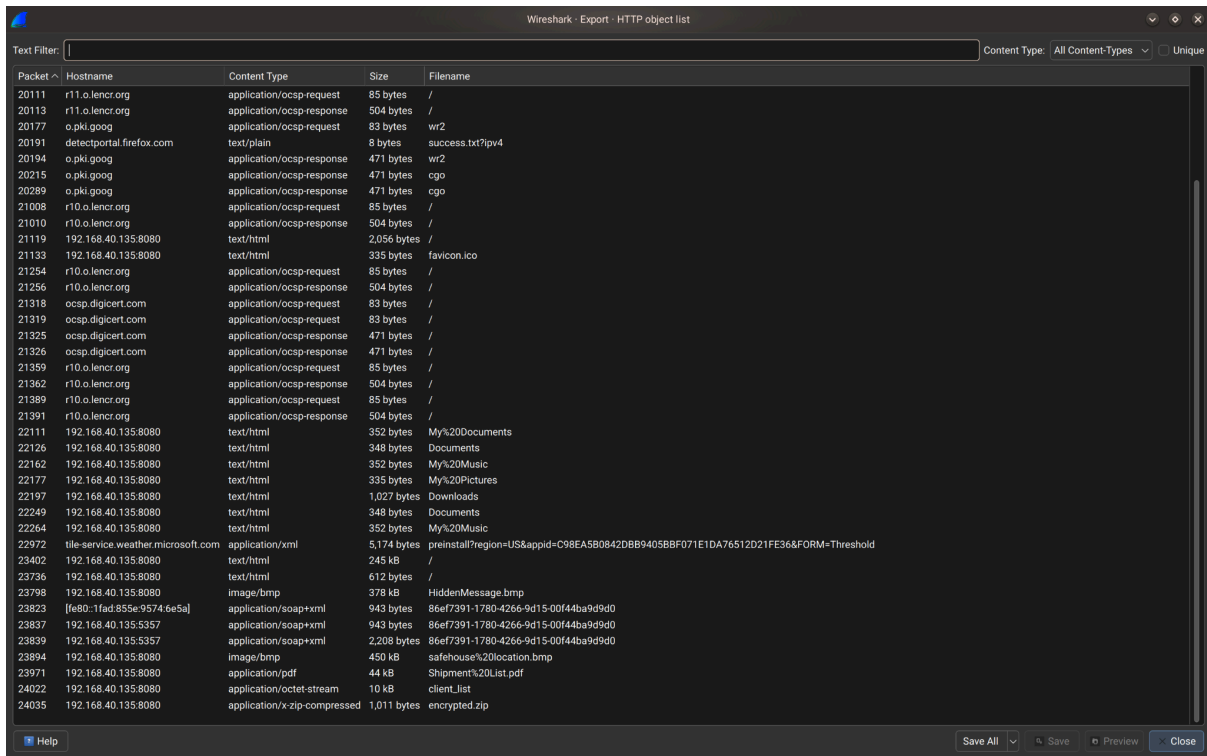


Figure 6.2.4 — DNS queries identifying external services and .onion domains together with TLS Server Name Indication (SNI) values observed within encrypted network sessions.

6.2.4 Exported Network Objects

Wireshark's Export Objects feature was used to recover files transmitted during HTTP and FTP sessions. The exported artifacts included **HiddenMessage.bmp**, **safehouse location.bmp**, **encrypted.zip**, and additional recovered files. These recovered objects were preserved and examined during the subsequent phases of the forensic investigation.



Wireshark - Export - HTTP object list

Text Filter: Content Type: All Content-Types Unique

Packet	Hostname	Content Type	Size	Filename
20111	r11.o.lencr.org	application/ocsp-request	85 bytes	/
20113	r11.o.lencr.org	application/ocsp-response	504 bytes	/
20177	o.pki.goog	application/ocsp-request	83 bytes	wr2
20191	detectportal.firefox.com	text/plain	8 bytes	success.txt?ipv4
20194	o.pki.goog	application/ocsp-response	471 bytes	wr2
20215	o.pki.goog	application/ocsp-response	471 bytes	cgo
20289	o.pki.goog	application/ocsp-response	471 bytes	cgo
21008	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21010	r10.o.lencr.org	application/ocsp-response	504 bytes	/
21119	192.168.40.135:8080	text/html	2,056 bytes	/
21133	192.168.40.135:8080	text/html	335 bytes	favicon.ico
21254	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21256	r10.o.lencr.org	application/ocsp-response	504 bytes	/
21318	ocsp.digicert.com	application/ocsp-request	83 bytes	/
21319	ocsp.digicert.com	application/ocsp-request	83 bytes	/
21325	ocsp.digicert.com	application/ocsp-response	471 bytes	/
21326	ocsp.digicert.com	application/ocsp-response	471 bytes	/
21359	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21362	r10.o.lencr.org	application/ocsp-response	504 bytes	/
21389	r10.o.lencr.org	application/ocsp-request	85 bytes	/
21391	r10.o.lencr.org	application/ocsp-response	504 bytes	/
22111	192.168.40.135:8080	text/html	352 bytes	My%20Documents
22126	192.168.40.135:8080	text/html	348 bytes	Documents
22162	192.168.40.135:8080	text/html	352 bytes	My%20Music
22177	192.168.40.135:8080	text/html	335 bytes	My%20Pictures
22197	192.168.40.135:8080	text/html	1,027 bytes	Downloads
22249	192.168.40.135:8080	text/html	348 bytes	Documents
22264	192.168.40.135:8080	text/html	352 bytes	My%20Music
22972	tile-service.weather.microsoft.com	application/xml	5,174 bytes	preinstall?region=US&appid=C98EA5B0842DBB9405BBF071E1DA76512D21FE36&FORM=Threshold
23402	192.168.40.135:8080	text/html	245 kB	/
23736	192.168.40.135:8080	text/html	612 bytes	/
23798	192.168.40.135:8080	image/bmp	378 kB	HiddenMessage.bmp
23823	[fe80::1fad:85e:9574:6e5a]	application/soap+xml	943 bytes	86ef7391-1780-4266-9d15-00f44ba9d9d0
23837	192.168.40.135:5357	application/soap+xml	943 bytes	86ef7391-1780-4266-9d15-00f44ba9d9d0
23839	192.168.40.135:5357	application/soap+xml	2,208 bytes	86ef7391-1780-4266-9d15-00f44ba9d9d0
23894	192.168.40.135:8080	image/bmp	450 kB	safehouse%20location.bmp
23971	192.168.40.135:8080	application/pdf	44 kB	Shipment%20List.pdf
24022	192.168.40.135:8080	application/octet-stream	10 kB	client_list
24035	192.168.40.135:8080	application/x-zip-compressed	1,011 bytes	encrypted.zip

Help Save All Save Preview Close

Figure 6.2.5 — Exported HTTP/FTP objects

6.2.5 Cryptocurrency Transaction Tracing

The network capture and recovered files were examined for cryptocurrency-related artifacts, including wallet addresses, transaction identifiers (TXIDs), exchange references, and textual references to cryptocurrency payments.

No cryptocurrency wallet addresses or transaction identifiers were identified within the examined evidence.

However, textual references indicating that payments were intended to be conducted using cryptocurrency were identified within **client_list**.

No evidence linking these references to a specific blockchain transaction or cryptocurrency wallet was recovered during the examination.

Based on the available evidence, cryptocurrency was identified only as the intended payment method. No blockchain artifacts were recovered that would permit transaction tracing.

Timestamp	Protocol	Source	Destination	Description
1143.611	OCSP	192.168.40.129	101.50.113.11	OCSP request to verify a digital certificate before establishing a secure connection.
1464.322	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving HiddenMessage.bmp
1479.958	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving safehouse location.bmp
1491.374	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving Shipment List.pdf
1505.190	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving client_list
35.127	DNS	192.168.40.135	192.168.40.2	DNS query for humantradeabcxyz.onion
164.181	DNS	192.168.40.135	192.168.40.2	DNS query for shadowdealersxyz.onion
361.680	DNS	192.168.40.135	192.168.40.2	DNS query for deepwebroutesxyz.onion
576.047	DNS	192.168.40.135	192.168.40.2	DNS query for darkmarketxyz.onion
812.114	DNS	192.168.40.135	192.168.40.2	DNS query for redphoenixtrades.onion

Timestamp	Protocol	Source	Destination	Description
0.358	TLS	192.168.40.135	20.44.10.123	TLS Client Hello initiating an encrypted connection (SNI: browser.events.data.msn.com)
1508.917	HTTP	192.168.40.129	192.168.40.135	HTTP GET request retrieving encrypted.zip
2002.443	FTP	192.168.40.135	192.168.40.129	FTP authentication attempt using credentials " ftpuser " / " 123456 " (530 Login incorrect).
2271.238	FTP	192.168.40.129	192.168.40.135	Successful FTP login (230 Login successful) after authentication using the password " anonymous ".
2569.615	FTP	192.168.40.135	192.168.40.129	FTP RETR command used to download INVOICE.pdf .

Table 6.2.1 — Network Traffic Log

Analysis of this evidence is discussed in Chapter 8.

6.3 Encrypted File Findings

This section presents the findings obtained from the forensic examination of encrypted and password-protected files recovered during the investigation. The examination focused on identifying protected files, evaluating the encryption methods used, and documenting the outcome of password recovery and decryption attempts. All findings are presented objectively without interpretation.

6.3.1 Hidden Data and Password Recovery

This subsection presents the findings obtained from the examination of encrypted and password-protected files recovered during the investigation. The examination focused on identifying protected files, verifying their file signatures, and attempting password recovery using forensic password-cracking techniques.

File Signature and Header Verification

File signature (magic number) verification was performed on the files recovered after password recovery from encrypted.zip and usb_backup_files.zip (cross-reference Section 6.3.2). Each file's actual content was visually and structurally confirmed to match its stated extension.

File	Stated Extension	Verified Content / Signature	Match
contacts.txt	.txt	ASCII plain text — dark-web contact list	Yes
safehouse_locations.txt	.txt	ASCII plain text — safehouse address list	Yes
identity docs.pdf	.pdf	PDF document (%PDF header) — forged passport record	Yes
transaction logs.xlsx	.xlsx	Microsoft Excel 2007+ (PK\x03\x04 / OOXML) — transaction ledger	Yes
messages.txt.txt	.txt	ASCII plain text — chat log transcript	Yes

Result: no file extension mismatches were identified among the files recovered from within encrypted.zip and usb_backup_files.zip. This complements the single mismatch identified separately at the container level (client_list, Section 6.3.2, Table 6.3.2) and is reported here as a negative finding for this specific set of files.

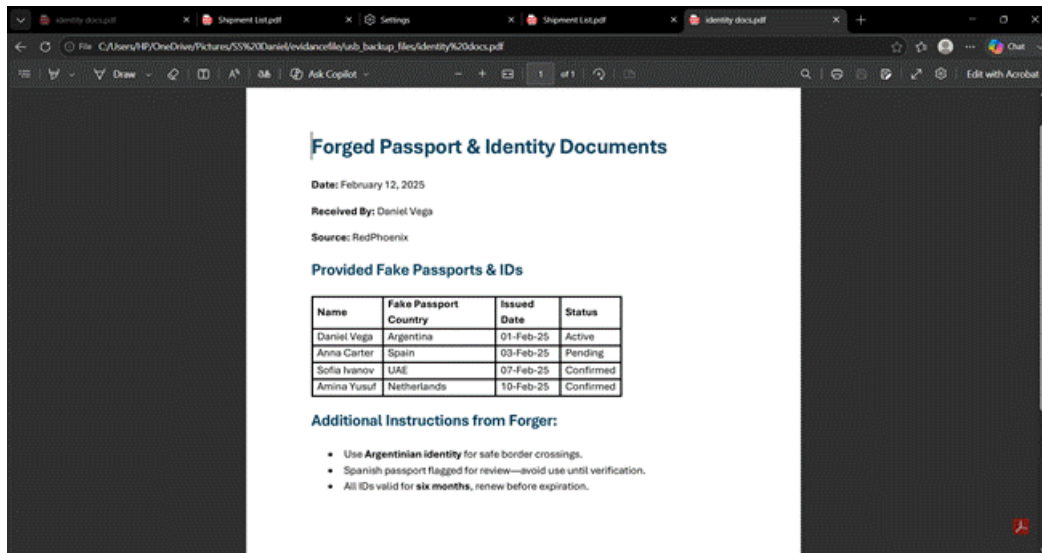


Figure 6.3.1a — Content of *identity docs.pdf*, confirming genuine PDF structure and content (Forged Passport & Identity Documents record).

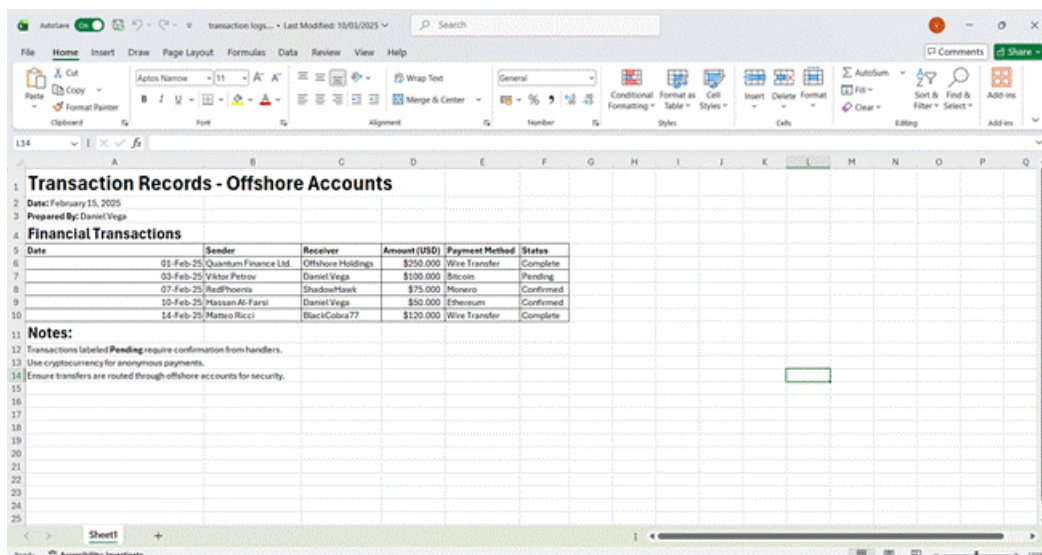
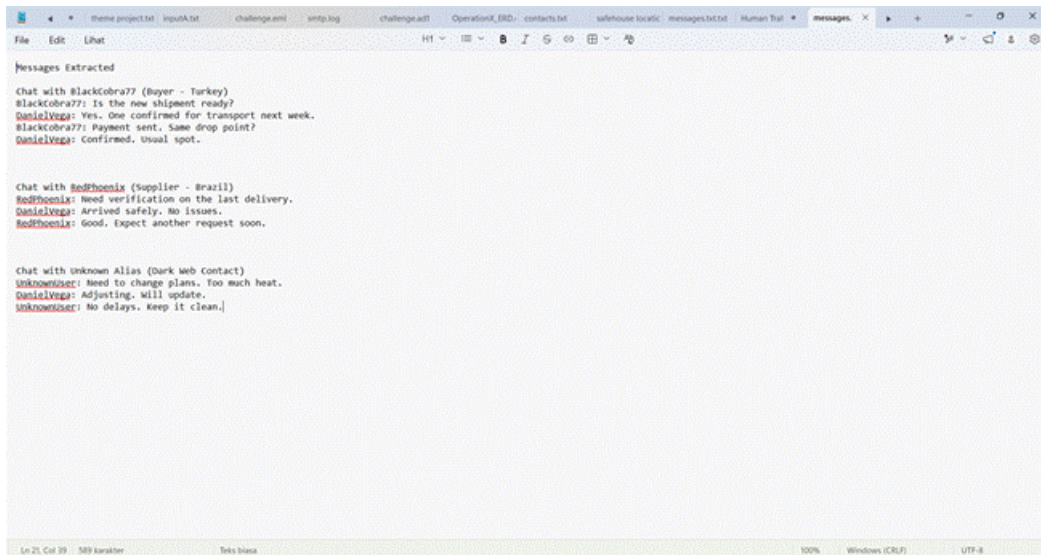


Figure 6.3.1b — Content of *transaction logs.xlsx*, confirming genuine Excel/OOXML structure (Transaction Records — Offshore Accounts).



```

Messages Extracted

Chat with BlackCobra77 (Buyer - Turkey)
BlackCobra77: Is the new shipment ready?
DanielVega: Yes. One confirmed for transport next week.
BlackCobra77: Payment sent. Same drop point?
DanielVega: Confirmed. Usual spot.

Chat with RedPhoenix (Supplier - Brazil)
RedPhoenix: Need verification on the last delivery.
DanielVega: Arrived safely. No issues.
RedPhoenix: Good. Expect another request soon.

Chat with Unknown Alias (Dark web contact)
UnknownUser: Need to change plans. Too much heat.
DanielVega: Adjusting. Will update.
UnknownUser: No delays. Keep it clean.

```

Figure 6.3.1c — Content of messages.txt.txt, confirming plain-text structure (extracted chat log).

Identify of Password-Protected Files

All files transferred and recovered under this scope were checked to identify which required a password before their contents could be accessed.

File	Password-Protected?	Verification Method
encrypted.zip	Yes — AES encrypted	Password prompt on manual open attempt; confirmed as WinZip AE (method 99) via zipinfo -v (Section 6.3.2)
usb_backup_files.zip	No	Opened directly without a password prompt; consistent with standard Deflate compression (Table 6.3.2)

Result: encrypted.zip was confirmed as the only password-protected archive within this analyst's scope. Both encrypted.zip and HiddenMessage.bmp were located within the OneDrive/Documents/Hidden folder of Daniel_Laptop.E01, alongside safehouse location.bmp/.jpg and usb_backup_files.zip.

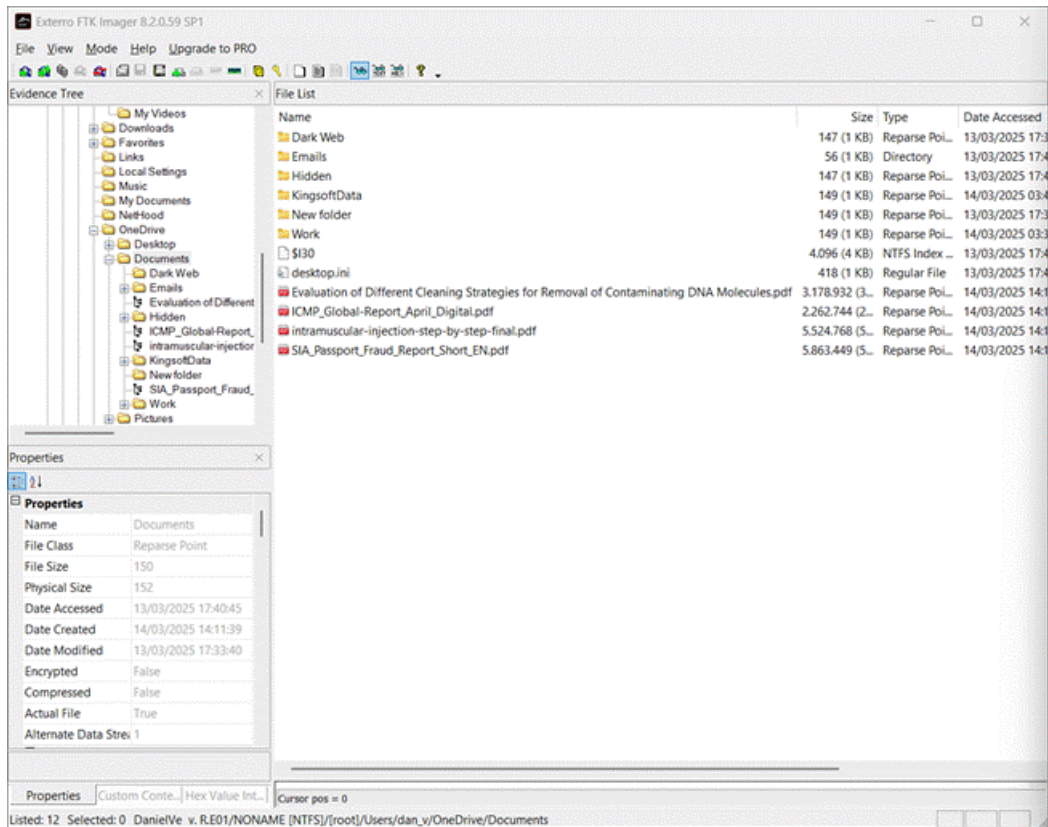


Figure 6.3.1d — FTK Imager Evidence Tree showing the Hidden folder contents, including *encrypted.zip* and *usb_backup_files.zip*.

Password Recovery

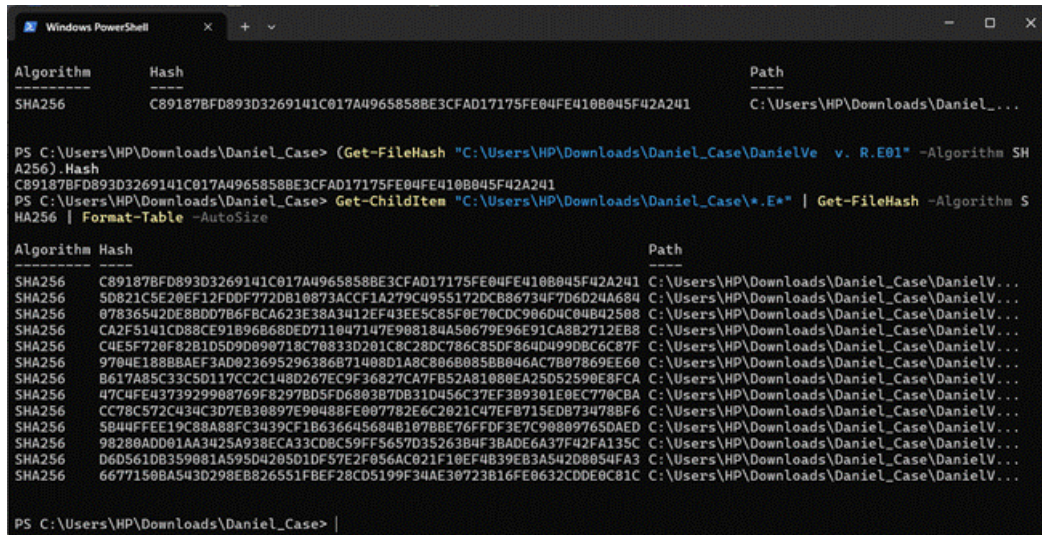
A password hash was extracted from *encrypted.zip* using the *zip2john* utility (John the Ripper suite), and a dictionary attack was performed using the *rockyou.txt* wordlist:

- `zip2john encrypted.zip > zip_hash.txt`
- `john --wordlist=rockyou.txt zip_hash.txt`
- `john --show zip_hash.txt`

Result: the password for *encrypted.zip* was recovered as 123456. The recovered password is consistent with the password obtained independently using an alternative tool chain described in Section 6.3.2. Once decrypted, *encrypted.zip* was confirmed to contain *contacts.txt* and *safehouse_locations.txt*, consistent with the extraction reported in Section 6.3.2.

Password Recovery Session

The following screenshots corroborate the SHA-256 hash values already recorded in Table 4.1 for the disk image segments (E01–E13).



```

Windows PowerShell
Algorithm      Hash
-----
SHA256         C89187BFD893D3269141C017A4965858BE3CFAD17175FE04FE410B045F42A241 C:\Users\HP\Downloads\Daniel_...

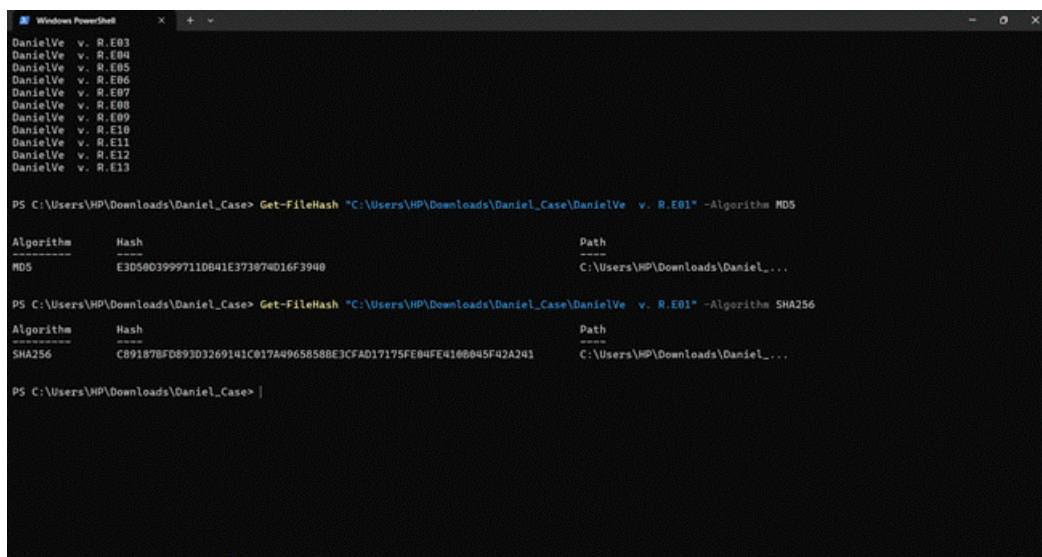
PS C:\Users\HP\Downloads\Daniel_Case> (Get-FileHash "C:\Users\HP\Downloads\Daniel_Case\DanielVe v. R.E01" -Algorithm SH
A256).Hash
C89187BFD893D3269141C017A4965858BE3CFAD17175FE04FE410B045F42A241
PS C:\Users\HP\Downloads\Daniel_Case> Get-Childitem "C:\Users\HP\Downloads\Daniel_Case\*.E*" | Get-FileHash -Algorithm S
HA256 | Format-Table -AutoSize

Algorithm Hash
-----
SHA256      C89187BFD893D3269141C017A4965858BE3CFAD17175FE04FE410B045F42A241 C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      5D821C5E20EF12FDDF772DB10873ACCF1A279C4955172DCB86734F7D6D24A684 C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      07836542DE8BD0786FBCA623E38A3412EF43EE5C85F0E70CDC906D4C04B42508 C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      CA2F5141CD88CE91B96B68DE711047147E908184A50679E9691CA8B2712E88 C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      C4E5F720F82B105D9D090718C78833D201C8C28DC786C85DF864D4990BC6C87F C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      9704E188BBAEF3AD023695296386871408D1A8C806B085B8046AC7B07869EE60 C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      B617A85C33C5D117C2C148D267EC9F36827CA7FB52A81080EA25D52590E8FCA C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      47C4FE4373929908769F8297BD5FD6803B7DB31D456C37EF3B9301E0EC770CBA C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      CC78C572C434C3D7E3B0897E90488FE087782E6C2021C47EFB715EDB73478BF6 C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      5B44FFEE19C8A88FC3439CF18636645684B107B8E76FFDF3E7C908097650AED C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      98280ADD01AA3425A938ECA33CDBCC59FF5657D35263B4F3BADE6A37F42FA135C C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      D6D561DB359081A595D4205D1DF57E2F056AC021F10EF4B39EB3A542D8054FA3 C:\Users\HP\Downloads\Daniel_Case\DanielV...
SHA256      6677150BA543D298EB826551FBEF28C05199F34AE30723B16FE0632CDE0C81C C:\Users\HP\Downloads\Daniel_Case\DanielV...

PS C:\Users\HP\Downloads\Daniel_Case>

```

Figure 6.3.1e — *Get-FileHash* (SHA-256) output for *DanielVe v. R.E01*, matching Table 4.1.



```

Windows PowerShell
DanielVe v. R.E03
DanielVe v. R.E04
DanielVe v. R.E05
DanielVe v. R.E06
DanielVe v. R.E07
DanielVe v. R.E08
DanielVe v. R.E09
DanielVe v. R.E10
DanielVe v. R.E11
DanielVe v. R.E12
DanielVe v. R.E13

PS C:\Users\HP\Downloads\Daniel_Case> Get-FileHash "C:\Users\HP\Downloads\Daniel_Case\DanielVe v. R.E01" -Algorithm MD5

Algorithm      Hash
-----
MD5             E3D56039997110B41E373074D16F3940 C:\Users\HP\Downloads\Daniel_...

PS C:\Users\HP\Downloads\Daniel_Case> Get-FileHash "C:\Users\HP\Downloads\Daniel_Case\DanielVe v. R.E01" -Algorithm SHA256

Algorithm      Hash
-----
SHA256         C89187BFD893D3269141C017A4965858BE3CFAD17175FE04FE410B045F42A241 C:\Users\HP\Downloads\Daniel_...

PS C:\Users\HP\Downloads\Daniel_Case>

```

Figure 6.3.1f — *Get-FileHash* (MD5 and SHA-256) verification session for *DanielVe v. R.E01–R.E13*.

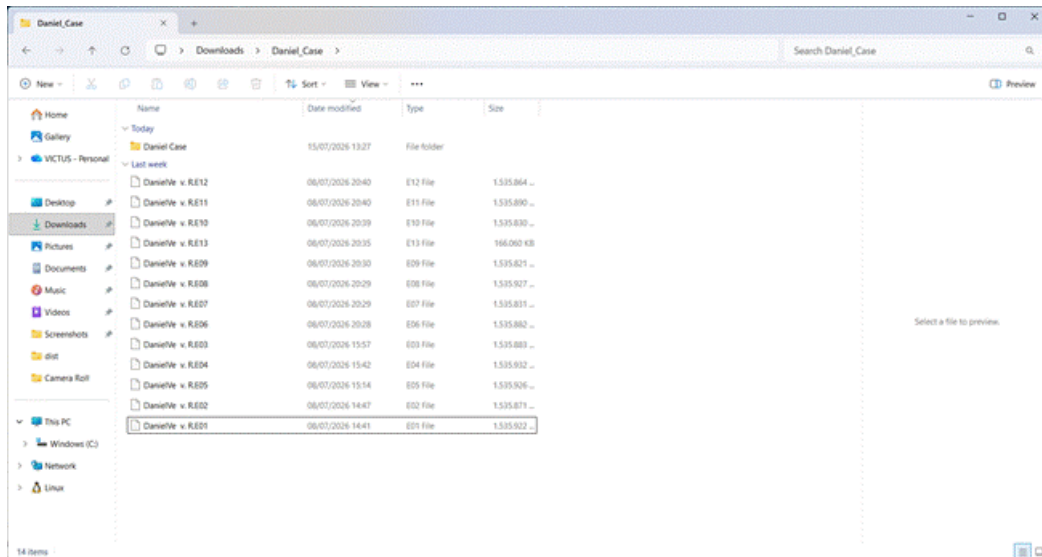


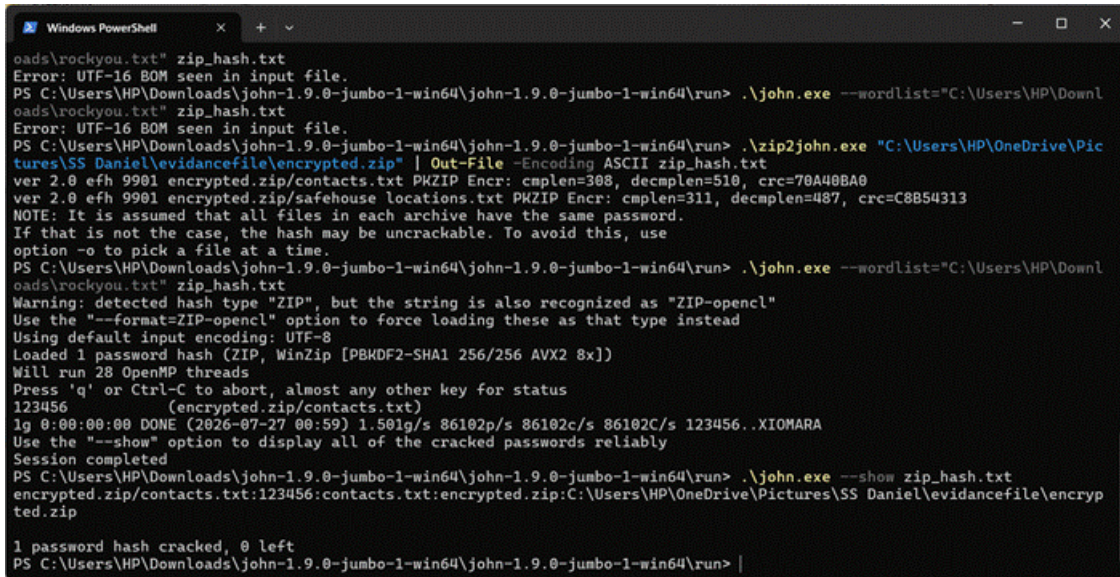
Figure 6.3.1g — File Explorer listing of DanielVe v. R.E01–R.E13 disk image segments.

```

Windows PowerShell
False
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> Test-Path "C:\Users\HP\Downloads\rockyou.txt"
True
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\john.exe --wordlist="C:\Users\HP\Downloads\rockyou.txt" zip_hash.txt
Error: UTF-16 BOM seen in input file.
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\john.exe --wordlist="C:\Users\HP\Downloads\rockyou.txt" zip_hash.txt
Error: UTF-16 BOM seen in input file.
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\zip2john.exe "C:\Users\HP\OneDrive\Pictures\SS Daniel\evadancefile\encrypted.zip" | Out-File -Encoding ASCII zip_hash.txt
ver 2.0 efh 9901 encrypted.zip/contacts.txt PKZIP Encr: cmplen=308, decmplen=510, crc=70A40BA0
ver 2.0 efh 9901 encrypted.zip/safehouse locations.txt PKZIP Encr: cmplen=311, decmplen=487, crc=C0B54313
NOTE: It is assumed that all files in each archive have the same password.
If that is not the case, the hash may be uncrackable. To avoid this, use
option -o to pick a file at a time.
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\john.exe --wordlist="C:\Users\HP\Downloads\rockyou.txt" zip_hash.txt
Warning: detected hash type "ZIP", but the string is also recognized as "ZIP-opencl"
Use the "--format=ZIP-opencl" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Will run 28 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
123456 (encrypted.zip/contacts.txt)
lg 0:00:00:00 DONE (2026-07-27 00:59) 1.501g/s 86102p/s 86102c/s 86102C/s 123456..XIOMARA
Use the "--show" option to display all of the cracked passwords reliably
Session completed
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> |

```

Figure 6.3.1h — zip2john hash extraction from encrypted.zip, producing zip_hash.txt (both \$zip2\$ and \$pkzip2\$ hash formats), followed by locating the rockyou.txt wordlist.



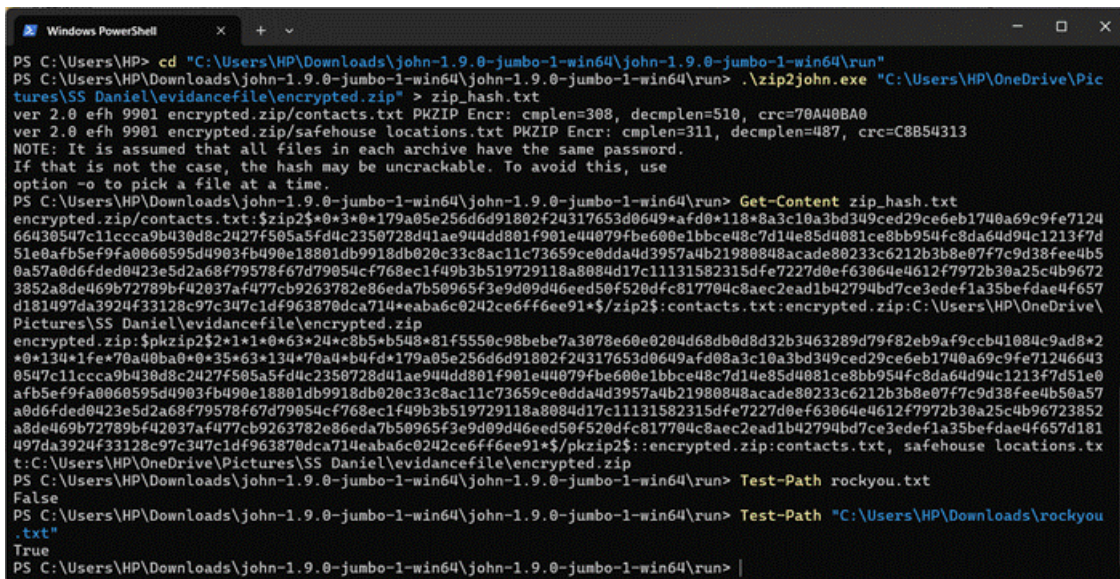
```

Windows PowerShell
oads\rockyou.txt" zip_hash.txt
Error: UTF-16 BOM seen in input file.
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\john.exe --wordlist="C:\Users\HP\Downl
oads\rockyou.txt" zip_hash.txt
Error: UTF-16 BOM seen in input file.
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\zip2john.exe "C:\Users\HP\OneDrive\Pic
tures\SS Daniel\evidencefile\encrypted.zip" | Out-File -Encoding ASCII zip_hash.txt
ver 2.0 efh 9901 encrypted.zip/contacts.txt PKZIP Encr: cmplen=308, decmplen=510, crc=70A40BA0
ver 2.0 efh 9901 encrypted.zip/safehouse locations.txt PKZIP Encr: cmplen=311, decmplen=487, crc=C8B54313
NOTE: It is assumed that all files in each archive have the same password.
If that is not the case, the hash may be uncrackable. To avoid this, use
option -o to pick a file at a time.
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\john.exe --wordlist="C:\Users\HP\Downl
oads\rockyou.txt" zip_hash.txt
Warning: detected hash type "ZIP", but the string is also recognized as "ZIP-openc1"
Use the "--format=ZIP-openc1" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Will run 28 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
123456
(encrypted.zip/contacts.txt)
lg 0:00:00:00 DONE (2026-07-27 00:59) 1.501g/s 86102p/s 86102c/s 86102C/s 123456..XIOMARA
Use the "--show" option to display all of the cracked passwords reliably
Session completed
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\john.exe --show zip_hash.txt
encrypted.zip/contacts.txt:123456:contacts.txt:encrypted.zip:C:\Users\HP\OneDrive\Pictures\SS Daniel\evidencefile\encryp
ted.zip

1 password hash cracked, 0 left
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> |

```

Figure 6.3.1i — John the Ripper dictionary attack (--wordlist=rockyou.txt) against zip_hash.txt, successfully recovering the password 123456 for encrypted.zip/contacts.txt in under 1 second.



```

Windows PowerShell
PS C:\Users\HP> cd "C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run"
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> .\zip2john.exe "C:\Users\HP\OneDrive\Pic
tures\SS Daniel\evidencefile\encrypted.zip" > zip_hash.txt
ver 2.0 efh 9901 encrypted.zip/contacts.txt PKZIP Encr: cmplen=308, decmplen=510, crc=70A40BA0
ver 2.0 efh 9901 encrypted.zip/safehouse locations.txt PKZIP Encr: cmplen=311, decmplen=487, crc=C8B54313
NOTE: It is assumed that all files in each archive have the same password.
If that is not the case, the hash may be uncrackable. To avoid this, use
option -o to pick a file at a time.
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> Get-Content zip_hash.txt
encrypted.zip/contacts.txt:$zip2$*0*3*0*179a05e256d6d91802f24317653d0649*afd0*118*8a3c10a3bd349ced29ce6eb1740a69c9fe7124
66430547c11ccca9b430d8c2427f505a5fd4c2350728d41ae944dd801f901e44079f9b600e1bbce48c7d14e85d4081ce8bb954fc8da64d94c1213f7d
51e8afb5ef9fa0060595d4903fb490e18801db9918db020c33c8ac11c73659ce0dda4d3957a4b21980848acade80233c6212b3b8e07f7c9d38fee4b5
0a57a0d6fdded0423e5d2a68f79578f67d79054cf768ec1f49b3b519729118a0804d17c11131582315dfe7227d0ef63064e4612f7972b30a25c4b9672
3852a8de469b72789bf42027af477cb9263782e86eda7b50965f3e9d09d46eed50f520dfc817704c8aec2ead1b42794bd7ce3edef1a35befdae4f657d181
497da3924f33128c97c347c1df963870dca714eaba6c0242ce6ff6ee91*$/zip2$:contacts.txt:encrypted.zip:C:\Users\HP\OneDrive\
Pictures\SS Daniel\evidencefile\encrypted.zip
encrypted.zip:$pkzip2$2*1*1*0*63*24*c8b5*b548*81f5550c98bebe7a3078e60e0204d68db08d32b3463289d79f82eb9af9ccb41084c9ad8*2
*0*134*1fe*70a40ba0*0*35*63*134*70a4*b4fd*179a05e256d6d91802f24317653d0649afd08a3c10a3bd349ced29ce6eb1740a69c9fe71246643
0547c11ccca9b430d8c2427f505a5fd4c2350728d41ae944dd801f901e44079f9b600e1bbce48c7d14e85d4081ce8bb954fc8da64d94c1213f7d51e0
afb5ef9fa0060595d4903fb490e18801db9918db020c33c8ac11c73659ce0dda4d3957a4b21980848acade80233c6212b3b8e07f7c9d38fee4b50a57
a0d6fdded0423e5d2a68f79578f67d79054cf768ec1f49b3b519729118a0804d17c11131582315dfe7227d0ef63064e4612f7972b30a25c4b96723852
a8de469b72789bf42027af477cb9263782e86eda7b50965f3e9d09d46eed50f520dfc817704c8aec2ead1b42794bd7ce3edef1a35befdae4f657d181
497da3924f33128c97c347c1df963870dca714eaba6c0242ce6ff6ee91*$/pkzip2$:encrypted.zip:contacts.txt, safehouse locations.tx
t:C:\Users\HP\OneDrive\Pictures\SS Daniel\evidencefile\encrypted.zip
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> Test-Path rockyou.txt
False
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> Test-Path "C:\Users\HP\Downloads\rockyou
.txt"
True
PS C:\Users\HP\Downloads\john-1.9.0-jumbo-1-win64\john-1.9.0-jumbo-1-win64\run> |

```

Figure 6.3.1j — Confirmation of the recovered password via john --show, displaying encrypted.zip/contacts.txt:123456 ("1 password hash cracked, 0 left").

These figures document the password recovery process, including hash extraction, dictionary attack, and verification of the recovered password. The recovered password (123456) is consistent with the password obtained independently using pyzipper, as described in Section 6.3.2, confirming the result through two independent recovery methods.

6.3.2 Encryption and File Cracking

This subsection presents the findings obtained from the examination of encrypted files and recovered artifacts. All findings are presented objectively without interpretation. Correlation with other evidence is discussed in Chapter 8.

File Extension Mismatch Detection

All recovered files were checked using the file command and verified with xxd when necessary. Only client_list showed a mismatch: it had no extension but its signature (50 4B 03 04 / “PK..”) identified it as a Microsoft Excel 2007+ (.xlsx) file, indicating the extension had been intentionally removed. All other files matched their stated extensions.

File	Stated Extension	file Output / Signature	Mismatch
client_list	(none)	Microsoft Excel 2007+ (hex 50 4B 03 04 / "PK..")	Yes — extension removed
Shipment List.pdf	.pdf	PDF document, version 1.7	No
HiddenMessage.bmp	.bmp	PC bitmap, Windows 3.x, 474x266x24	No
safehouse location.bmp	.bmp	PC bitmap, Windows 3.x, 474x316x24	No
safehouse location.jpg	.jpg	JPEG image data, JFIF 1.01	No
encrypted.zip	.zip	Zip archive, AES Encrypted	No

usb_backup_files.zip	.zip	Zip archive, deflate	No
----------------------	------	----------------------	----

Table 6.3.2 — file Extension Mismatch Results

```

berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ ls -la "$SEVID/Emails"
total 680
-rwxrwxrwx 1 berlianavjk berlianavjk 4096 Jul 16 12:56 '$I30'
-rwxrwxrwx 1 berlianavjk berlianavjk 4096 Jul 16 12:57 '$I30'
-rwxrwxrwx 1 berlianavjk berlianavjk 10549 Mar 11 2025 'fake reviews.docx'
-rwxrwxrwx 1 berlianavjk berlianavjk 283 Mar 11 2025 'login details.txt'
-rwxrwxrwx 1 berlianavjk berlianavjk 327 Mar 11 2025 'onion links.txt'
-rwxrwxrwx 1 berlianavjk berlianavjk 0 Jul 16 12:56 'fake reviews.docx'
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ ls -la "$SEVID/KingsoftData"
total 0
-rwxrwxrwx 1 berlianavjk berlianavjk 96 Mar 11 2025 '$I30'
-rwxrwxrwx 1 berlianavjk berlianavjk 4096 Jul 16 12:56 '$I30'
-rwxrwxrwx 1 berlianavjk berlianavjk 4096 Jul 16 12:57 '$I30'
-rwxrwxrwx 1 berlianavjk berlianavjk 617299 Mar 14 2025 'Email 1.eml'
-rwxrwxrwx 1 berlianavjk berlianavjk 1592 Mar 14 2025 'Email 2.eml'
-rwxrwxrwx 1 berlianavjk berlianavjk 62621 Mar 14 2025 'Email 3.eml'
-rwxrwxrwx 1 berlianavjk berlianavjk 1213 Mar 14 2025 'Email 4.eml'
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ ls -la "$SEVID/New folder"
total 0
-rwxrwxrwx 1 berlianavjk berlianavjk 96 Mar 14 2025 '$I30'
-rwxrwxrwx 1 berlianavjk berlianavjk 4096 Jul 16 12:56 '$I30'
-rwxrwxrwx 1 berlianavjk berlianavjk 4096 Jul 16 12:57 '$I30'
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ file "$SEVID/Hidden"/*
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/$I30: data
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/$I30: data
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/HiddenMessage.bmp: PC bitmap, Windows 3.x format, 474 x 266 x 24, image si
ze 378784, cbSize 378838, bits offset 54
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/Shipment List.pdf: PDF document, version 1.7, 1 page(s)
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/client_list: Microsoft Excel 2007+
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/encrypted.zip: Zip archive data, at least v2.0 to extract, compression
method: AES Encrypted
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/safehouse location.bmp: PC bitmap, Windows 3.x format, 474 x 316 x 24, image si
ze 449984, cbSize 450038, bits offset 54
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/safehouse location.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI),
density 0x0, segment length 16, Exif Standard: [TIFF image data, big-endian, direntries=2], baseline, precision 8, 474x316, components 3
/mnt/c/Users/HP/Downloads/Berlian/Documents/Hidden/usb_backup_files.zip: Zip archive data, at least v2.0 to extract, compression
method: deflate

```

Figure 6.3.2a — file command output confirming client_list as a disguised Excel file

```

berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ xxd "$SEVID/Hidden/client_list" | head -n 3
00000000: 504b 0300 0000 0000 0000 0000 0000 0000  PK.....M.E..
00000010: 0000 0000 0000 0000 0000 0000 0000 0000  .....d
00000020: 6358 726f 7073 2f50 4b03 0414 0000 0000  cProps/PK.....
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ file "$SEVID/Hidden"/* "$SEVID/Dark Web/" "$SEVID/Emails"/* >> ~/kan42_workspace/file_type_results.txt
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$

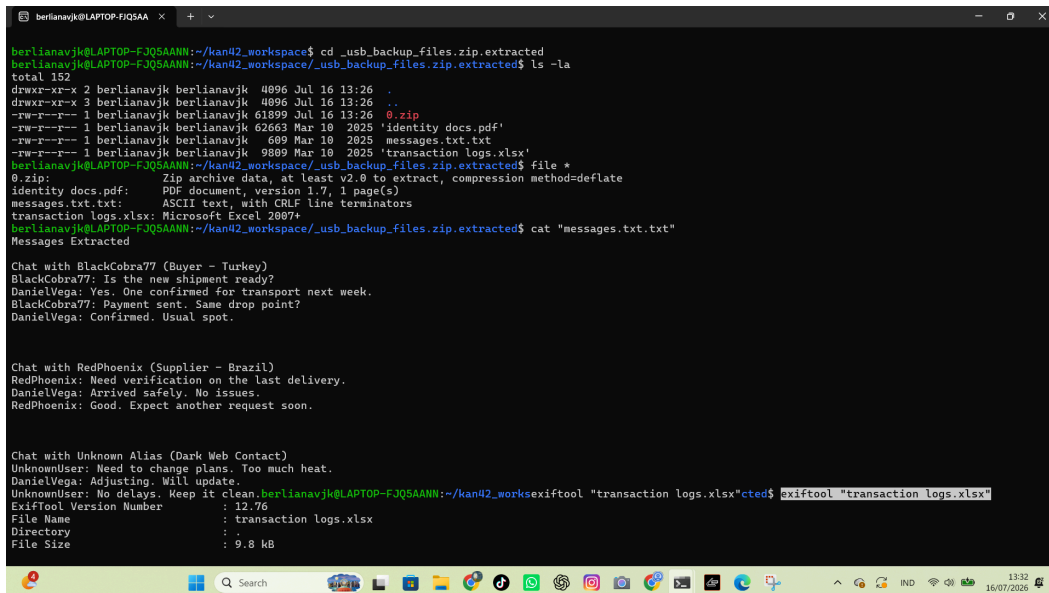
```

Figure 6.3.2b — xxd output confirming the Microsoft Excel OOXML signature (50 4B 03 04) of client_list.

Binwalk Analysis on Suspicious Files

HiddenMessage.bmp, safehouse location.bmp, and safehouse location.jpg showed only their native image headers, normal entropy values, and no embedded or appended data. No GPS

metadata was found in the images. `usb_backup_files.zip` was successfully extracted and contained `identity docs.pdf`, `messages.txt.txt`, and `transaction logs.xlsx`, all of which were verified with the `file` command. After assigning the `.xlsx` extension, `client_list` was confirmed to contain buyer names and payment amounts.



```

berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ cd _usb_backup_files.zip.extracted
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace/_usb_backup_files.zip.extracted$ ls -la
total 152
drwxr-xr-x 2 berlianavjk berlianavjk 4096 Jul 16 13:26 .
drwxr-xr-x 3 berlianavjk berlianavjk 4096 Jul 16 13:26 ..
-rw-r--r-- 1 berlianavjk berlianavjk 61899 Jul 16 13:26 0.zip
-rw-r--r-- 1 berlianavjk berlianavjk 62663 Mar 10 2825 'identity docs.pdf'
-rw-r--r-- 1 berlianavjk berlianavjk 689 Mar 10 2825 messages.txt.txt
-rw-r--r-- 1 berlianavjk berlianavjk 9889 Mar 10 2825 'transaction logs.xlsx'
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace/_usb_backup_files.zip.extracted$ file *
0.zip: Zip archive data, at least v2.0 to extract, compression method=deflate
identity docs.pdf: PDF document, version 1.7, 1 page(s)
messages.txt.txt: ASCII text, with CRLF line terminators
transaction logs.xlsx: Microsoft Excel 2007+
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace/_usb_backup_files.zip.extracted$ cat "messages.txt.txt"
Messages Extracted

Chat with BlackCobra77 (Buyer - Turkey)
BlackCobra77: Is the new shipment ready?
DanielVega: Yes. One confirmed for transport next week.
BlackCobra77: Payment sent. Same drop point?
DanielVega: Confirmed. Usual spot.

Chat with RedPhoenix (Supplier - Brazil)
RedPhoenix: Need verification on the last delivery.
DanielVega: Arrived safely. No issues.
RedPhoenix: Good. Expect another request soon.

Chat with Unknown Alias (Dark Web Contact)
UnknownUser: Need to change plans. Too much heat.
DanielVega: Adjusting. Will update.
UnknownUser: No delays. Keep it clean.
berlianavjk@LAPTOP-FJQ5AANN:~/kan42_workspace$ exiftool "transaction logs.xlsx"
Exiftool Version Number      : 12.76
File Name                   : transaction logs.xlsx
Directory                   : .
File Size                   : 9.8 kB

```

Figure 6.3.2c — Extracted contents of `usb_backup_files.zip` verified with the `file` command

Crack VeraCrypt/Encrypted Volumes :

No VeraCrypt container or encrypted volume was found in the provided evidence. The only encrypted artifact was `encrypted.zip`, identified as WinZip AES (method 99) using `zipinfo`. Initial password attempts with `unzip` failed due to AES incompatibility, while `7z` also failed against a case-derived wordlist. A dictionary attack using `pyzipper` and the `rockyou.txt` wordlist successfully recovered the password `123456`. The archive was extracted with `7z`, revealing `contacts.txt` and `safehouse_locations.txt`. The original evidence file remained unchanged throughout the process.

```

berlianyx@LAPTOP-FJQ5AA: ~
$ unzip -l /usr/share/doc/unzip/examples/zipinfo.v -v "HIDDEN/encrypted.zip"
Archive: /usr/share/doc/unzip/examples/zipinfo.v
There is no zipfile comment.

End-of-central-directory record:

Zip archive file size:      1811 (0000000000000313)h
Actual end-of-cdir record offset: 989 (0000000000001029)h
Expected end-of-cdir record offset: 869 (0000000000001009)h
(based on the length of the central directory and its expected offset)

This zipfile constitutes the sole disk of a single-part archive; its
central directory contains 2 entries.
The central directory is 722 (000000000000028a) bytes long,
and its (expected) offset is bytes from the beginning of the zipfile
is 706 (0000000000000286)h.

Central directory entry #1:

offset of local header from start of archive: 0
                                         (0000000000000000) bytes
file system or operating system of origin: 0
                                         M-DOS, OS/2 or NT FAT
version of encoding software:           2.1
minimum file system compatibility required: M-DOS, OS/2 or NT FAT
minimum software version required to extract: 2.0
compression method:                     unknown (99)
file security status:                    encrypted
external file header:                   288
file last modified as (DOS date/time):  2023 Mar 10 22:39:58
5-bit CRC value (hex):                  8a46a6a
compressed size:                        300 bytes
length of filename:                     12 characters
length of extra field:                   0 bytes
length of file comment:                  0 characters
data marker on which file begins:        0x1f
apparent file type:                     binary
non-RDOS external file attributes:       none hex
M-DOS file attributes (20 hex):          arc

The central directory extra field contains:
- 4 conflict with 10 bytes (unknown 0020) and 23 data bytes. The first
  20 are: 00 00 00 00 01 00 18 00 77 51 50 71 43 91 d8 01 39 b1 b0 71.
- 4 conflict with 10 bytes (unknown) and 7 data bytes:
  01 00 41 40 03 08 00.

There is no file comment.

Central directory entry #2:

There are an extra -28 bytes preceding this file.

safehouse locations list

offset of local header from start of archive: 377
                                         (000000000000017b) bytes
file system or operating system of origin: 0
                                         M-DOS, OS/2 or NT FAT
version of encoding software:           2.1
minimum file system compatibility required: M-DOS, OS/2 or NT FAT
minimum software version required to extract: 2.0
compression method:                     unknown (99)
file security status:                    encrypted

```

Figure 6.3.2d — zipinfo -v output identifying encrypted.zip as a WinZip AES encrypted archive.

[illegible]

Figure 6.3.2e — pyzipper dictionary attack using the rockyou.txt wordlist, successfully recovering the password for encrypted.zip.

Analysis of this evidence is discussed in Chapter 8.

6.4 Steganography Findings

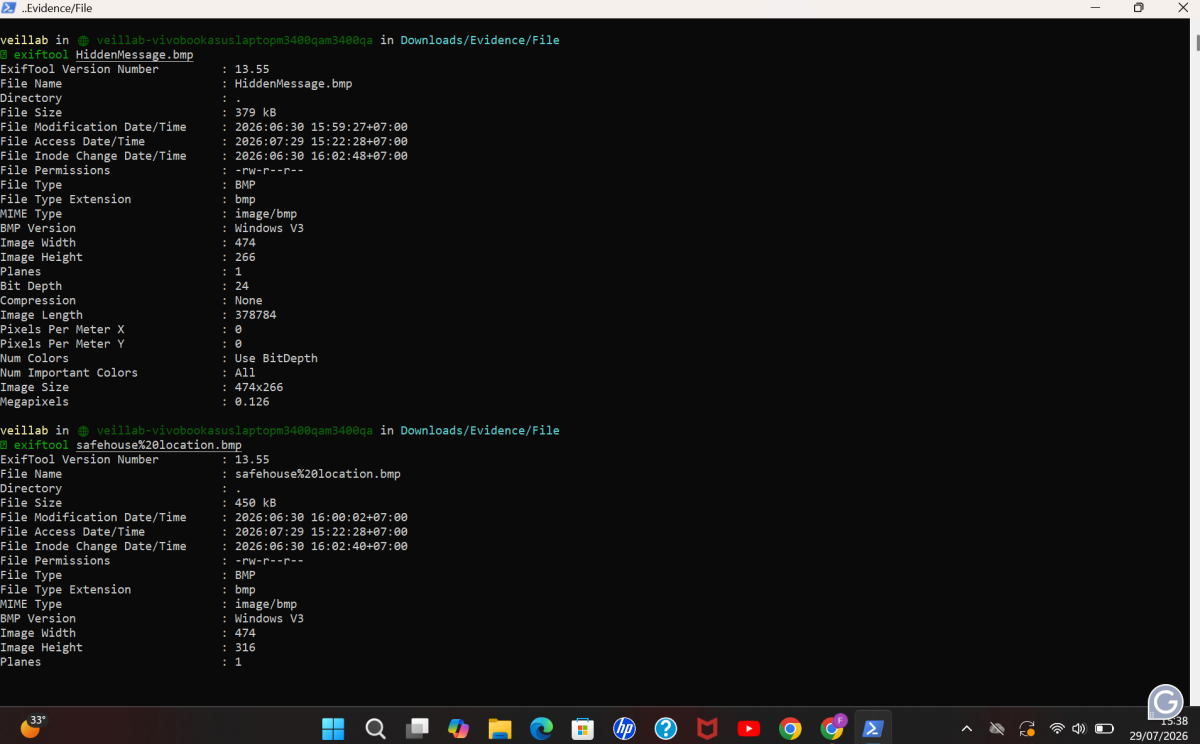
This section presents the findings obtained from the steganographic examination of image files recovered during the network analysis. The examination focused on identifying hidden data, embedded files, metadata, and other concealed information within the recovered image files. All findings are presented objectively without interpretation.

6.4.1 Carrier Files Examined

Both files were selected for steganographic examination because they were image files recovered during the investigation and were considered relevant to the forensic examination.

6.4.2 Metadata Analysis

ExifTool was used to examine the metadata of both image files. No unusual or evidential metadata was identified in either HiddenMessage.bmp or safehouse location.bmp.



```

veillab in @ veillab-vivobookasuslaptopm3400qam3400qam in Downloads/Evidence/File
$ exiftool HiddenMessage.bmp
ExifTool Version Number      : 13.55
File Name                    : HiddenMessage.bmp
Directory                    : .
File Size                    : 379 kB
File Modification Date/Time   : 2026:06:30 15:59:27+07:00
File Access Date/Time        : 2026:07:29 15:22:28+07:00
File Inode Change Date/Time   : 2026:06:30 16:02:48+07:00
File Permissions              : -rw-r--r--
File Type                    : BMP
File Type Extension           : bmp
MIME Type                    : image/bmp
BMP Version                  : Windows V3
Image Width                  : 474
Image Height                 : 266
Planes                       : 1
Bit Depth                    : 24
Compression                  : None
Image Length                 : 378784
Pixels Per Meter X           : 0
Pixels Per Meter Y           : 0
Num Colors                   : Use BitDepth
Num Important Colors          : All
Image Size                   : 474x266
Megapixels                   : 0.126

veillab in @ veillab-vivobookasuslaptopm3400qam3400qam in Downloads/Evidence/File
$ exiftool safehouse%20location.bmp
ExifTool Version Number      : 13.55
File Name                    : safehouse%20location.bmp
Directory                    : .
File Size                    : 450 kB
File Modification Date/Time   : 2026:06:30 16:00:02+07:00
File Access Date/Time        : 2026:07:29 15:22:28+07:00
File Inode Change Date/Time   : 2026:06:30 16:02:40+07:00
File Permissions              : -rw-r--r--
File Type                    : BMP
File Type Extension           : bmp
MIME Type                    : image/bmp
BMP Version                  : Windows V3
Image Width                  : 474
Image Height                 : 316
Planes                       : 1

```

Figure 6.4.1 — ExifTool metadata output for HiddenMessage.bmp and safehouse location.bmp.

6.4.3 String and Hex Analysis

Both image files were examined using strings and xxd to identify readable text and inspect the raw hexadecimal structure of the files. No plaintext payloads or suspicious appended data were identified.

```

veillab in @ veillab-wivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$ strings HiddenMessage.bmp | head -20
, .!!!1$3"#7!$9
049R
+9.=M"/E
/L!$@+
sd1$ &
J-<X$[f=BX2-06"N; P;%M8"H2 B+
8<
CC
068c
0H-1W>|
X*sE
VR-[. /
8"7
1^ nW
, .!!!1$3"$6!%8
049R
,8.=M"/E
/L!$@+
sd1$ &
J-<X$[f=BX3-05"N; IP;%M8"H2 B+
8<
CC

veillab in @ veillab-wivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$ xxd -l 128 HiddenMessage.bmp
00000000: 424d d6c7 0500 0000 0000 3600 0000 2800  BM.....6...(.
00000010: 0000 da01 0000 0a01 0000 0100 1800 0000  .....>C.....
00000020: 0000 a0c7 0500 0000 0000 0000 0000 0000  .....>C.....
00000030: 0000 0000 0000 363b 420e 0912 130e 1700  .....>C.....
00000040: 030c 1712 1b12 0d16 1611 1e17 121b 1e15  .....>C.....
00000050: 1e16 111a 130e 1715 1019 1914 1d1b 161f  .....>C.....
00000060: 1a15 1e17 111c 1b14 211d 1625 2019 2820  .....>C.....
00000070: 1928 1f15 251d 1323 1d13 231d 1323 2015  .....>C.....

```

Figure 6.4.2 — GNU strings and xxd output showing printable strings and the initial hexadecimal structure of HiddenMessage.bmp.

6.4.4 Embedded File Detection

Binwalk was used to scan both BMP image files for embedded or appended file signatures. No additional signatures were identified in HiddenMessage.bmp. During the examination of safehouse location.bmp, Binwalk reported a JBOOT STAG header at offset 0x5DA3D. Manual hexadecimal inspection of the reported offset did not identify a valid embedded file or recoverable payload. Accordingly, the reported JBOOT STAG signature was assessed as a false positive, and no embedded files were confirmed.

```

veillab in ~veillab-vivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$ binwalk HiddenMessage.bmp

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
0            0x0            PC bitmap, Windows 3.x format,, 474 x 266 x 24

veillab in ~veillab-vivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$ binwalk "safehouse%20location.bmp"

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
0            0x0            PC bitmap, Windows 3.x format,, 474 x 316 x 24
383549       0x5DA3D        JBOOT STAG header, image id: 4, timestamp 0x45474228, image size: 995054672 bytes, image JBOOT checksum: 0x383C, header JBOOT checksum: 0x100C

veillab in ~veillab-vivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$

```

Figure 6.4.3 — Binwalk scan identifying a reported JBOOT STAG signature in *safehouse location.bmp* prior to manual verification.

6.4.5 Steganography Analysis

zsteg was used to examine both BMP image files for potential least significant bit (LSB) steganography and hidden data patterns. The tool reported multiple potential OpenPGP Public Key and OpenPGP Secret Key signatures within both images. Manual verification of the reported locations using hexadecimal inspection did not identify valid OpenPGP key material or recoverable hidden payloads. Based on the combined results of zsteg and manual verification, no confirmed steganographic payload was identified within either image file.

```

veillab in @ veillab-vivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$ zsteg HiddenMessage.bmp
imagedata .. text: ", , 1111131\#7159"
b4,lsb,bY .. text: "A4$9e$8D%2%2Dbv(b"
b1,r,msb,xY .. file: OpenPGP Secret Key
b1,b,msb,xY .. file: OpenPGP Secret Key
b2,g,lsb,xY .. text: "UaUUUUUUU^g"
b2,b,lsb,xY .. text: "Z8>ch5B\r"
b3,b,msb,xY .. file: OpenPGP Public Key
b4,r,lsb,xY .. text: "fffffwfDUD34x"
b4,g,lsb,xY .. text: "\3DDUT\("\("\\"UUU34DUDEVf("
b4,b,lsb,xY .. text: "\("\("\\"ffff"
b4,b,msb,xY .. text: "DDDDffff"
b4,rgb,lsb,xY .. text: "1T&D$85#R5$Brh&"

veillab in @ veillab-vivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$ zsteg "safehouse%20location.bmp"
imagedata .. text: "qpl\WYUP[UNKE>"
b4,msb,bY .. file: OpenPGP Public Key
b1,g,msb,xY .. file: OpenPGP Secret Key
b1,b,msb,xY .. file: OpenPGP Secret Key
b1,rgb,msb,xY .. file: OpenPGP Public Key
b2,rgb,lsb,xY .. file: OpenPGP Secret Key
b4,g,lsb,xY .. text: "#333Vol"FTahus"
b4,rgb,msb,xY .. file: OpenPGP Secret Key

veillab in @ veillab-vivobookasuslaptopm3400qam3400qa in Downloads/Evidence/File
$

```

Figure 6.4.4 — zsteg output reporting potential OpenPGP Public Key and OpenPGP Secret Key signatures prior to manual verification.

Analysis of this evidence is discussed in Chapter 8.

6.5 Document Findings

This section presents the findings obtained from the forensic examination of documents recovered from the evidence. The examination focused on identifying document metadata, document contents, and other artifacts relevant to the investigation. All findings are presented objectively without interpretation.

6.5.1 Recovered Documents

The forensic examination recovered several documents and related artifacts from the disk image, including PDF documents, spreadsheet files, email messages, and Windows shortcut (.lnk) files. Examples of recovered documents include Shipment List.pdf, identity docs.pdf, INVOICE.pdf, and transaction logs.xlsx, together with email artifacts and recent document shortcuts. These recovered items may provide information regarding previous user activities and are preserved as digital evidence.

In addition to the documents already listed in Table 6.5.1, the following recovered items provide additional factual content:

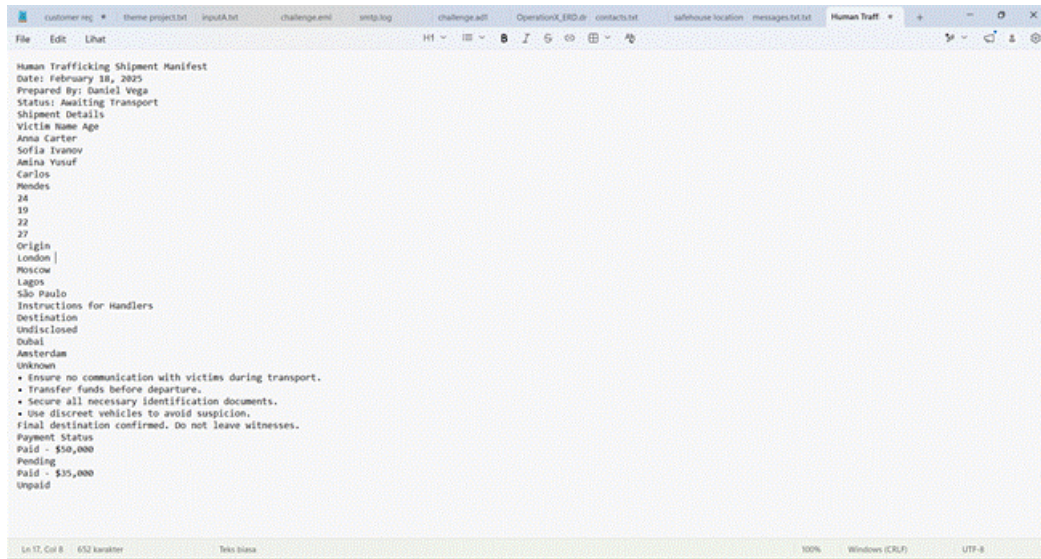


Figure 6.5.2 — Content of the recovered Human Trafficking Shipment Manifest, listing named individuals, origin/destination cities, handler instructions, and payment status.

Recovered File	Descripted
Shipment List.pdf	Shipment manifest recovered from the disk image.
identity docs.pdf	Identity-related document recovered during examination.
INVOICE.pdf	Financial document recovered from the evidence.
transaction logs.xlsx	Transaction records recovered from the evidence

Table 6.5.1 Summary of Recovered Documents

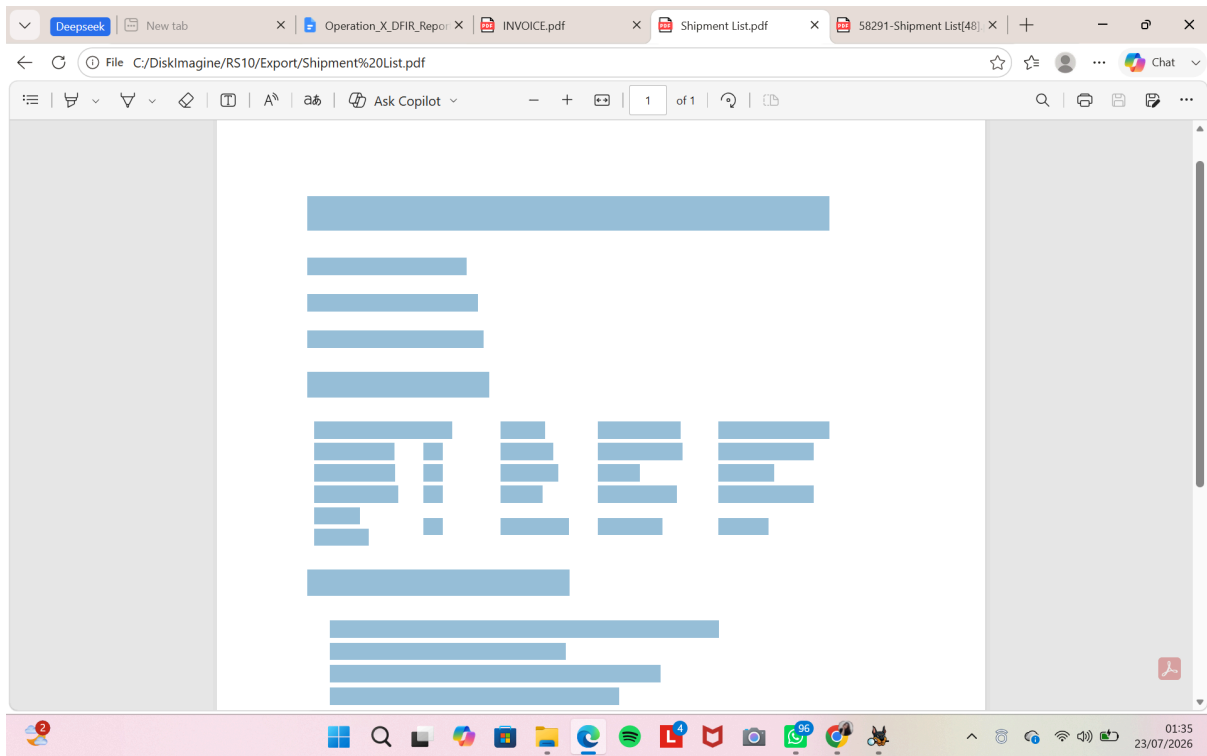


Figure 6.5.1 — Recovered documents and related artifacts identified during forensic examination (Shipment List.pdf).

Human Trafficking Shipment Manifest

Date: February 18, 2025

Prepared By: Daniel Vega **Status:**

Awaiting Transport

Shipment Details

Victim Name	Age	Origin	Destination	Payment Status
Anna Carter	24	London	Undisclosed	Paid - \$50,000
Sofia Ivanov	19	Moscow	Dubai	Pending
Amina Yusuf	22	Lagos	Amsterdam	Paid - \$35,000
Carlos Mendes	27	São Paulo	Unknown	Unpaid

Instructions for Handlers

- Ensure no communication with victims during transport.
- Transfer funds before departure.
- Secure all necessary identification documents.
- Use discreet vehicles to avoid suspicion.

Final destination confirmed. Do not leave witnesses.

Figure 6.5.1 — Recovered documents and related artifacts identified during forensic examination (*Shipment List.pdf*).

Analysis of this evidence is discussed in Chapter 8.

7 TIMELINE RECONSTRUCTION

This chapter presents a chronological reconstruction of events identified during the forensic examination. The timeline consolidates timestamps recovered from the forensic disk image, network traffic captures, recovered documents, email artifacts, and other digital evidence to reconstruct the sequence of activities relevant to the investigation. All timeline entries are derived from the examination results presented in Chapter 6.

Date and Time	Evidence Source	Description	Related Section
2025-03-07 17:33:06	Daniel_Laptop.E01	Tor Browser Portable installer identified from downloaded artifacts	6.1

Date and Time	Evidence Source	Description	Related Section
2025-03-11 00:22:19	Daniel_Laptop.E01	Nightclub.jpg recovered from the Downloads directory.	6.1
2025-03-11 00:22:19	encrypted.zip	Encrypted ZIP archive identified within Hidden folder.	6.3
2025-03-11 01:06:26	Daniel_Laptop.E01	transaction logs.xlsx modified within usb_backup_files.zip archive	6.1
2025-03-11 01:14:50	Daniel_Laptop.E01	identity docs.pdf modified within the same archive.	6.1
2025-03-11 01:16:32	encrypted.zip	Password-protected ZIP archive successfully accessed after password recovery, allowing forensic examination of its contents.	6.3
2025-03-11 09:23:40	Daniel_Laptop.E01	Email "Everything is in place" recovered; decoded message confirmed payment acknowledgement.	6.1
2025-03-13 00:55:01	Daniel_Laptop.E01	Browser history showed a search for OpenStego followed by the installer download.	6.1
2025-03-13 10:49:23	Daniel_Network.pcap	TLS Client Hello traffic observed during encrypted communication.	6.2
2025-03-13 11:08:45	Daniel_Network.pcap	HTTP browsing session to internal web server (192.168.40.135:8080) identified.	6.2
2025-03-13 11:14:32	Daniel_Network.pcap	encrypted.zip downloaded via an HTTP GET request.	6.2

Date and Time	Evidence Source	Description	Related Section
2025-03-13 11:32:13	Daniel_Network.pcap	FTP RETR command used to retrieve INVOICE.pdf.	6.2
2025-03-13 11:32:27	Daniel_Laptop.E01	Tor Browser and OpenStego execution recorded within a one-second interval.	6.1
2025-03-22 14:50:27	Daniel_Laptop.E01	Found a few Ciphertext in a each email that already decoded with Cryptii in Caesar Cipher	6.1.3
2025-03-12 10:10:14	Daniel_Laptop.E01	Subject: Urgent Cleanup Needed email from dan_vega_11041990@outlook.com to : fixer@darkops.xyz;	6.1.3
2004-11-27 10:41:44	Daniel_Laptop.E01	msg_43.txt Subject: Banned auto__mail.python.bat in mail from you	6.1.3
2001-04-07 00:23:06	Daniel_Laptop.E01	Msg25.txt Subject: Returned mail: Too mant hops 19 (17 max): from < linuxuser-admin@www.linux.org.uk > via [199.164.235.226], to <scoffman@wellpart...	6.1.3
2019-12-07 09:17:28	Daniel_Laptop.E01	Noted and followed up several installed program during investigation	6.1
2025-03-07 17:33:06	Daniel_Laptop.E0	Identified several downloaded jpg,png and application in Web Download	6.1

Date and Time	Evidence Source	Description	Related Section
2025-03-13 13:50:29	Daniel_Laptop.E01	Identified many document that suspicious inn Recent Document	6.1
2025-03-06 13:25:00	Daniel_Laptop.E01	Running Program during the Human trafficking and murder investigation	6.1
2025-03-08 21:25:09	Daniel_Laptop.E01	Remote monitoring/management software found installed, indicating potential remote access to the system.	6.1
2025-03-13 00:59:59	Daniel_Laptop.E01	File or folder suspected of being encrypted, identified during the examination of system artifacts.	6.1
2025-03-14 21:11:59	Daniel_Laptop.E01	Encryption application found installed on the system, indicating possible use of encryption to conceal data.	6.1
2023-03-19 09:14:13	Daniel_Laptop.E01	User-owned content suspected of being relevant to the investigation, found during examination of the user directory.	6.1
2025-03-13 19:18:35	Daniel_Laptop.E01	Evidence of an external USB device previously connected to the system, indicating potential data transfer to removable storage media.	6.1
2019-03-13 19:18:24	Daniel_Laptop.E01	Additional (child device) detail of the detected USB device, further confirming the identification of the external storage device used.	6.1

Table 7.1 — Master Investigation Timeline

8 ANALYSIS

This chapter presents the interpretation of the forensic findings obtained during the examination. Unlike Chapter 6, which reports factual findings, this chapter correlates evidence from multiple sources to explain the significance of the recovered artifacts in relation to the investigation.

8.1 Temporal Analysis

The reconstructed timeline presented in Chapter 7 shows that the recovered evidence is interconnected rather than representing isolated events. On 7 March 2025, Tor Browser Portable was identified as having been downloaded and installed, indicating the availability of an anonymised browsing tool before other relevant activities occurred. Four days later, on 11 March 2025, several files, including *Nightclub.jpg*, *transaction logs.xlsx*, and *identity docs.pdf*, were accessed or modified within a relatively short period. Later on the same day, the email titled "*Everything is in place*" was recovered, with its decoded content confirming that a payment had been received. The close timing of these activities suggests that they were part of the same operational session rather than unrelated user actions.

One of the most notable findings in the timeline is the near-simultaneous execution of **Tor Browser Portable** and **OpenStego** on 13 March 2025 at 11:32:27–11:32:28 WIB. The minimal time difference between the execution of both applications suggests that they were likely used during the same activity rather than independently. On the same day, the **Shipment List.pdf** attachment was also recovered from the email evidence. Although the steganography examination did not identify any hidden payload within the analysed BMP files, the combined presence of Tor Browser, OpenStego, recovered documents, and network activity indicates an attempt to utilise privacy and concealment-related tools during the investigation period.

8.2 Relational Analysis

This section analyzes the relationships between evidence obtained from different sources to identify connections between digital artifacts, recovered documents, network communications, and other entities relevant to the investigation.

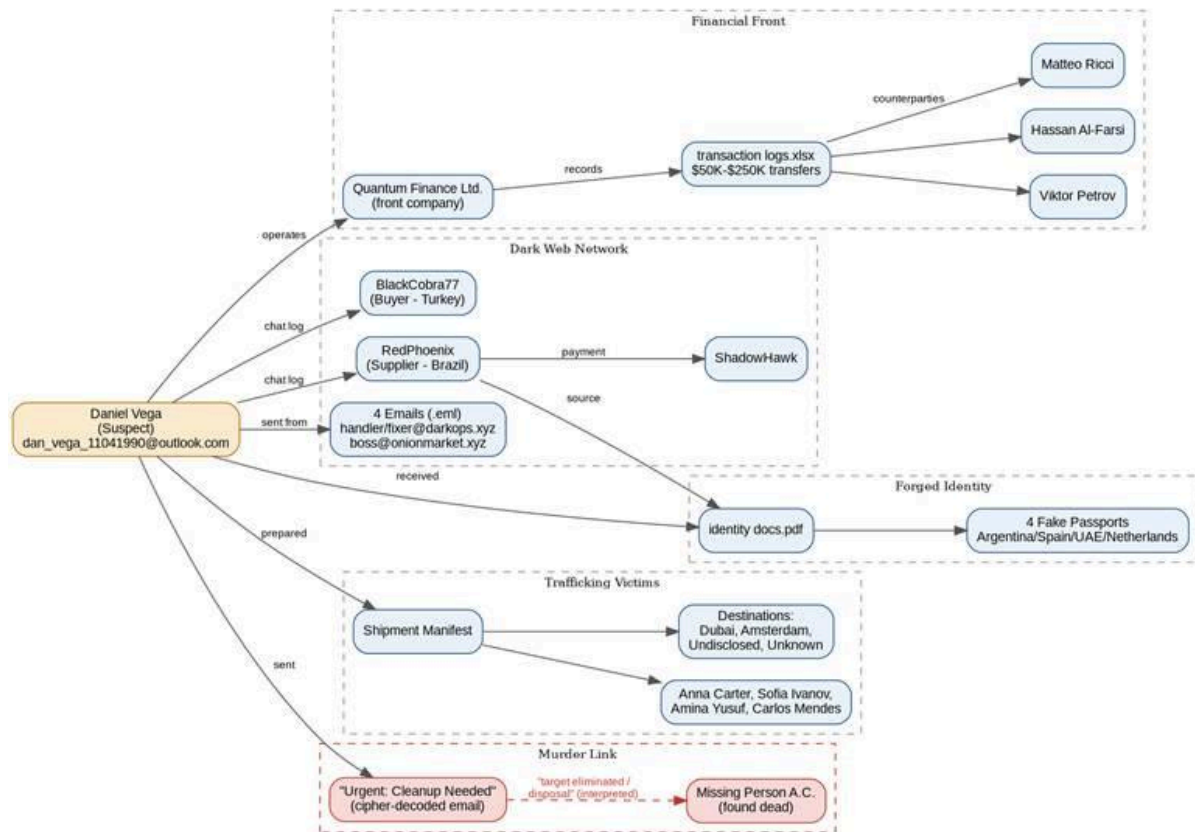


Figure 8.2.1 — Evidence relationship diagram connecting the suspect, financial front, dark-web network, forged identities, trafficking victims, and the murder-linked email.

Figure 8.2.1 illustrates how the recovered evidence is interconnected rather than existing as isolated artefacts. The relationships identified during the forensic examination can be grouped into five major evidence clusters: financial activities, communication network, forged identity documents, shipment records, and recovered encoded communications. Each cluster contributes contextual information that strengthens the overall reconstruction of the investigated activities.

The first cluster represents the financial records recovered from *transaction logs.xlsx*. The spreadsheet references **Quantum Finance Ltd.** together with several counterparties, including **Matteo Ricci**, **Hassan Al-Farsi**, and **Viktor Petrov**, indicating documented financial transactions associated with the investigation.

The second cluster focuses on recovered email communications involving aliases such as **BlackCobra77**, **RedPhoenix**, and **ShadowHawk**. These communications demonstrate interactions between multiple parties and provide supporting context for document exchange and operational coordination.

The third cluster consists of the recovered *identity docs.pdf* and associated counterfeit passport records. These artefacts indicate the preparation or exchange of forged identity documents involving multiple nationalities.

The fourth cluster relates to the recovered *Shipment List*, which records multiple individuals together with their intended destinations. When analysed alongside the recovered financial records and communications, this document provides additional contextual links between different evidence sources.

The final cluster concerns the decoded email entitled "*Urgent: Cleanup Needed*". This communication establishes a relationship between the recovered email evidence and the reported murder case involving Missing Person A.C. While the email alone cannot establish criminal responsibility, it provides contextual evidence when interpreted together with the other recovered artefacts.

8.3 Functional Analysis

The forensic findings from the disk image and network capture indicate a sequence of activities that can be interpreted as four functional stages. These stages describe how the recovered digital evidence is related throughout the investigation.

1. Preparation

The forensic examination identified several files related to the investigation, including the shipment list, identity documents, financial records, and other supporting files. Some of these files were stored within hidden directories and protected using an encrypted ZIP archive, indicating that the data was organised before further activities were performed.

2. Concealment

The examination identified the presence of Tor Browser and OpenStego on the examined system. Timeline analysis also showed that both applications were accessed within a very short time interval. In addition, an encrypted ZIP archive was recovered and successfully decrypted during the investigation. These findings indicate that encryption and privacy-related tools were available and actively used during the period covered by the investigation.

3. Transmission

Network traffic analysis identified HTTP and FTP communication involving several recovered files, including **HiddenMessage.bmp**, **safehouse location.bmp**, **encrypted.zip**, **Shipment List.pdf**, and **INVOICE.pdf**. The network capture also recorded encrypted TLS sessions during the same investigation period. These findings indicate that multiple files were transmitted across the network while encrypted communication was also observed.

4. Coordination and confirmation

Recovered email messages and decoded Caesar cipher text provide additional context for the activities identified during the forensic examination. Messages such as **"Everything is in place," "Safehouse Ready,"** and **"Urgent: Cleanup Needed"** indicate communication between involved parties during the investigated period. The decoded email stating **"Everything is in place"** also confirms that payment had been received, providing supporting evidence that complements the recovered documents and network artifacts.

Overall, the forensic findings indicate a consistent sequence of activities involving document preparation, data protection, network transmission, and communication between related parties. The combination of disk artifacts, recovered documents, network traffic, encrypted archives, and decoded messages provides a coherent reconstruction of the events identified during the investigation.

9 CONCLUSIONS

This chapter summarizes the results of the digital forensic examination by addressing the investigation objectives and describing the limitations encountered during the examination process.

9.1 Answers to Investigation Objectives

Objective 1: Identify digital evidence related to the suspect's activities

Status : Confirmed

Digital evidence associated with the suspect's account (dan_vega_11041990@outlook.com) was successfully recovered from the forensic image. The recovered evidence includes trafficking-related documents, forged identity

records, financial records, browser artifacts, and installation artifacts associated with Tor Browser and OpenStego.

Objective 2: Analyze communications and transferred files associated with the investigation

Status : Confirmed

Four recovered email messages and multiple HTTP and FTP file transfers were successfully analyzed and correlated with system artifacts recovered from the forensic image. The decoded email messages and network activity collectively support the transfer of investigation-related files between communicating systems.

Objective 3: Reconstruct the sequence of events based on recovered digital evidence

Status: Highly Likely

A chronological sequence of events was successfully reconstructed in Chapter 7 by correlating timestamps recovered from the forensic disk image with supporting network evidence. The recovered timestamps are internally consistent and describe a logical sequence of user activities throughout the investigation period. The confidence level is assessed as **Highly Likely** rather than **Confirmed** because the network capture timestamps could not be fully verified against an independent system clock.

Objective 4: Produce a forensic report supporting the investigation

Status: Confirmed

This report documents the forensic examination methodology, findings, timeline reconstruction, and analysis in accordance with NIST SP 800-86. It serves as the consolidated forensic report produced by the examination team to support the investigation.

9.2 Limitations of Examination

9.2.1 Disk Image Acquisition Limitations

- No reference hash values were provided at the time the evidence was received. Therefore, evidence integrity could only be verified by generating a baseline SHA-256 hash rather than comparing it with a previously known reference hash.

- The forensic image was examined as a static, read-only source. Live system memory (RAM) was not available during the examination; therefore, artifacts that existed only in volatile memory could not be recovered.

9.2.2 Deleted File and Artifact Recovery Limitations

- Deleted file recovery was limited to artifacts that remained recoverable through file system metadata or were identified within Autopsy's **Deleted Files** view. Files that had been overwritten at the storage level could not be recovered.
- Some shortcut (.lnk) artifacts referenced file paths whose original target files were no longer available. As a result, the existence of the shortcut could be confirmed, but the contents of the referenced files could not always be independently verified.

9.2.3 Network and Steganography Limitations

- No hidden content was recovered from **HiddenMessage.bmp** or **safehouse location.bmp** using ExifTool, Strings, XXD, Binwalk, or Steghide. The examination was limited to the steganography tools available in the analysis environment and does not exclude the possibility that other specialized techniques could reveal additional hidden content.
- The presence of Tor Browser is supported by installation artifacts recovered from the forensic image. However, the associated network traffic could not be conclusively identified as Tor traffic because the captured communications remained encrypted.
- No cryptocurrency transaction data was observed in cleartext within **Daniel_Network.pcap**. If cryptocurrency transactions occurred, they were not directly observable within the available network capture.

9.2.4 Encryption & File Cracking Limitations

- The standard **unzip** utility available in the analysis environment does not support WinZip AE (AES) encryption. As a result, password recovery using **unzip -P** was unsuccessful despite the correct password being available. This limitation was resolved by using 7-Zip and Pyzipper, both of which support WinZip AE encryption.
- No standalone VeraCrypt volume or other full-disk encrypted container was identified within the evidence provided for examination. The only encrypted file encountered during this investigation was **encrypted.zip**.

- Password recovery for **encrypted.zip** relied on dictionary-based attacks using a custom wordlist followed by **rockyou.txt**. The effectiveness of this approach depends on the completeness of the available wordlists.
- The recovered **client_list** file required a working copy with the **.xlsx** extension to be created for examination. The original evidence file remained preserved in its original state without modification.

10 EXAMINATION NOTES / INVESTIGATION LOG

This chapter records the activities performed during the forensic examination in chronological order. The log provides an audit trail of the examination process, including the analyst responsible, the tools used, and the outcome of each examination activity.

10.1 Disk Image and Deleted File Recovery Examination Log

Date & Time	Analyst	Action / Tool Used	Outcome
2026-07-06 13:56	Disk Image	Reviewed Operating System Information – Autopsy	Confirmed OS version, computer name, and installation date on Daniel_Laptop.E01
2026-07-12 23:59	Disk Image	Reviewed Installed Programs artifacts — Autopsy	Identified installed encryption and remote monitoring software on the system
2026-07-15 13:01	Disk Image	Reviewed Run Programs / execution artifacts — Autopsy	Confirmed execution history of encryption tool
2026-07-16 11:47	Disk Image	Reviewed Recent Documents artifact — Autopsy	Confirmed document access timeline consistent with recovered PDF/XLSX files

2026-07 -16 11:50 11:52	Disk Image	Reviewed USB Device attachment and child device artifacts (USBDeviceAttached.png, USBChild.png) — Autopsy	artifacts (USBDeviceAttached.png, USBChild.png) — AutopsyIdentified external USB device connection and associated device details, consistent with potential data transfer
2026-07 -16 11:55	Disk Image	Reviewed encryption-related artifacts (EncryptionProgram.png, EncryptionSuspected.png) — Autopsy	Identified installed encryption software and files/folders suspected to be encrypted
2026-07 -16 04:38	Disk Image	Reviewed Remote Monitoring/Management software artifact — Autopsy	Identified installed remote access/monitoring tool on the system
2026-07 -16 04:37	Disk Image	Reviewed User Content directory artifact (UserContentSuspected.png) — Autopsy	Flagged user-generated files suspected to be relevant to the investigation
2026-07 -08 18:00 2026-07 -12 16:16	Disk Image	Reviewed Browser Search and Download history — Autopsy	Confirmed search query for OpenStego followed by installer download

Table 10 — Examination Log Examination Log (Disk Image Analyst)

10.2 Network and Steganography Examination Log

Date & Time	Analyst	Action / Tool Used	Outcome
2026-06-30 15:07	Network & Steganography	capinfos Daniel_Network.pcap	Verified capture file information (packet count, capture duration, interfaces).
2026-06-30 15:07	Network & Steganography	tshark -q -z io,phs	Identified major protocols present in the capture (HTTP, FTP, DNS, TLS).
2026-06-30 15:08	Network & Steganography	tshark -q -z endpoints,ip	Enumerated communicating IP addresses.
2026-06-30 15:08	Network & Steganography	tshark -q -z conv,tcp	Reviewed TCP conversations between hosts.
2026-07-06 13:05	Network & Steganography	tshark HTTP request filtering	Analysed HTTP request activity and reconstructed communications.
2026-07-06 13:11–13:17	Network & Steganography	Multiple tshark filtering commands	Reviewed DNS queries, FTP sessions, and TLS handshakes for suspicious traffic.
2026-07-07 10:40–10:53	Network & Steganography	Additional tshark analysis	Verified protocol findings and cross-checked previous observations.

Date & Time	Analyst	Action / Tool Used	Outcome
2026-07-19 18:59	Network & Steganography	HTTP traffic analysis (tcp.port==8080)	Identified web communication over TCP port 8080.
2026-07-19 19:01	Network & Steganography	DNS and HTTP correlation	Correlated DNS requests with HTTP activity.
2026-07-19 19:06	Network & Steganography	FTP session analysis	Confirmed FTP file transfer activity.
2026-07-28 12:44	Network & Steganography	HTTP inspection	Reviewed complete HTTP communications.
2026-07-28 12:45	Network & Steganography	FTP inspection	Verified FTP control and data sessions.
2026-07-28 12:46	Network & Steganography	DNS inspection	Examined DNS query records.
2026-07-28 12:47	Network & Steganography	TLS handshake inspection	Reviewed encrypted TLS connections.
2026-07-29 15:06	Network & Steganography	Final HTTP verification (tshark)	Final verification before report preparation.
2026-06-30 16:26	Network & Steganography	exiftool HiddenMessage.bmp	Metadata examined; no suspicious metadata identified.
2026-06-30 16:26	Network & Steganography	exiftool safehouse location.bmp	Metadata examined; no anomalies detected.

Date & Time	Analyst	Action / Tool Used	Outcome
2026-06-30 16:26	Network & Steganography	strings	Extracted printable strings; no hidden plaintext recovered.
2026-07-06 12:56	Network & Steganography	strings	Repeated string extraction for verification
2026-07-06 12:58	Network & Steganography	xxd	Examined file header and hexadecimal structure.
2026-07-07 11:12	Network & Steganography	file	Verified actual file format of BMP evidence.
2026-07-07 11:14	Network & Steganography	ExifTool	Revalidated metadata consistency.
2026-07-07 11:49	Network & Steganography	strings	Reviewed printable contents of BMP images.
2026-07-07 11:57	Network & Steganography	zsteg	Attempted extraction of hidden payload; no recoverable payload found.
2026-07-07 12:03	Network & Steganography	binwalk	Scanned for embedded files and signatures; none detected.
2026-07-08 16:51	Network & Steganography	ExifTool	Final metadata validation.

Date & Time	Analyst	Action / Tool Used	Outcome
2026-07-08 16:52	Network & Steganography	strings	Searched for hidden keywords and credentials; negative result.
2026-07-08 16:53	Network & Steganography	xxd	Verified hexadecimal contents.
2026-07-29 15:22	Network & Steganography	binwalk	Final embedded-file scan; no hidden objects identified.
2026-07-29 15:46	Network & Steganography	zsteg	Final steganography verification; negative result confirmed.

Table 10.1 — Examination Log (Network & Steganography Analyst)

10.3 Encryption and File Cracking Examination Log

Date	Analyst	Action / Tool Used	Outcome
[Day 1]	Encryption & File Cracking	File type verification — file, xxd (Documents root + Hidden folder)	One extension mismatch identified: client_list (Excel, no extension)
[Day 1]	Encryption & File Cracking	Binwalk scan + extraction — HiddenMessage.bmp, safehouse location.bmp/.jpg, usb_backup_files.zip	No embedded signatures in BMP/JPG files; usb_backup_files.zip yielded 3 archived files
[Day 2]	Encryption & File Cracking	Encryption identification — zipinfo -v, unzip -v on encrypted.zip	Identified as WinZip AE (compression method 99)

[Day 3]	Encryption & File Cracking	Dictionary attack — pyzipper against rockyou.txt	Password recovered: 123456
[Day 3]	Encryption & File Cracking	Extraction — 7z x	contacts.txt and safehouse_locations.txt recovered and reviewed

10.4 Email and Document Recovery Examination Log

Date	Analyst	Action / Tool Used	Outcome
[Day 1]	Email & Document Recovery	Reviewed E-Mail Messages module for account dan_vega_11041990@outlook.com Autopsy	Recovered 4 email artifacts (Everything is in Place, Safehouse Ready, Urgent: Cleanup Needed, File Transfer – Urgent)
[Day 1]	Email & Document Recovery	Cross-checked email attachment-save events against Recent Documents Autopsy	Confirmed Shipment List[48].pdf saved as email attachment at 2025-03-13 01:11:24 WIB
[Day 2]	Email & Document Recovery	Reviewed recovered PDF and spreadsheet documents (Shipment List.pdf, identity docs.pdf, INVOICE.pdf, transaction logs.xlsx) Autopsy Data Artifacts	Confirmed document content consistent with trafficking manifest, forged identity records, and financial ledger
[Day 2]	Email & Document Recovery	Decoded Caesar-cipher (shift 3) body text in "Everything is in place[47].eml" — manual decoding / Cryptii	Recovered plaintext message confirming payment and operational instructions

[Day 3]	Email & Document Recovery	Metadata cross-check of recovered documents against .lnk shortcut artifacts (Emails.lnk, Dark Web.lnk) Autopsy	Confirmed consistent creation/access/modification timestamps; no evidence of metadata manipulation
[Day 3]	Email & Document Recovery	Compiled recovered document summary for Section 6.5.1 Autopsy Report export	Finalized Table 6.5.1 (Summary of Recovered Documents)
[Day 3]	Email & Document Recovery	Decoded additional recovered ciphertext messages — manual decoding / Cryptii	Recovered plaintext content confirming further operational/payment details
[Day 3]	Email & Document Recovery	Reviewed Remote Monitoring/Management software artifact — Autopsy	Identified installed remote access/monitoring tool on the system

Table 10.4 — Examination Log (Email & Document Recovery)

11 REFERENCES

This chapter lists the standards, forensic tools, and external references cited throughout this report. All references are provided to support the methodology and findings presented during the digital forensic examination.

11.1 Standards

- National Institute of Standards and Technology. (2006). NIST Special Publication 800-86: Guide to Integrating Forensic Techniques into Incident Response. <https://csrc.nist.gov/publications/detail/sp/800-86/final>
- Association of Chief Police Officers. (2012). Good Practice Guide for Digital Evidence.
- International Organization for Standardization. (2012). ISO/IEC 27037:2012: Guidelines for Identification, Collection, Acquisition and Preservation of Digital Evidence.

11.2 Tool References

- Basis Technology. (n.d.). Autopsy. <https://www.autopsy.com/>
- ExifTool by Phil Harvey. (n.d.). ExifTool. <https://exiftool.org/>
- Exterro. (n.d.). FTK Imager. <https://www.exterro.com/ftk-imager>
- GNU Project. (n.d.). GNU Binutils. <https://www.gnu.org/software/binutils/>
- Hashcat Team. (n.d.). Hashcat. <https://hashcat.net/hashcat/>
- Netresec. (n.d.). NetworkMiner. <https://www.netresec.com/>
- Openwall. (n.d.). John the Ripper. <https://www.openwall.com/john/>
- ReFirm Labs. (n.d.). Binwalk. <https://github.com/ReFirmLabs/binwalk>
- Steghide Project. (n.d.). Steghide. <https://steghide.sourceforge.net/>
- Wireshark Foundation. (n.d.). TShark User's Guide.
<https://www.wireshark.org/docs/man-pages/tshark.html>
- Wireshark Foundation. (n.d.). Wireshark. <https://www.wireshark.org/>

11.3 Additional Tool References

- Darwin, I. (n.d.). file(1): Determine File Type Using Magic Numbers.
<https://www.darwinsys.com/file/>
- Miessler, D. (n.d.). SecLists. <https://github.com/danielmiessler/SecLists>
- Pavlov, I. (n.d.). 7-Zip. <https://www.7-zip.org/>
- pyzipper Contributors. (n.d.). pyzipper. <https://github.com/danifus/pyzipper>

12 APPENDIX

This appendix contains supplementary materials supporting the findings presented throughout this report. The appendices include command histories, raw tool outputs, supplementary screenshots, and technical references that were not included in the main chapters.

Appendix A — Network Traffic Analysis Commands

The following commands were executed during the network traffic investigation using **TShark** to analyse HTTP, FTP, DNS, TLS, and Dark Web (.onion) activities extracted from the packet capture.

```
tshark -r Daniel_Network.pcap -q -z conv,tcp

tshark -r Daniel_Network.pcap -q -z io,phs

tshark -r Daniel_Network.pcap -Y http

tshark -r Daniel_Network.pcap -Y ftp

tshark -r Daniel_Network.pcap -Y ftp-data

tshark -r Daniel_Network.pcap -Y dns

tshark -r Daniel_Network.pcap -Y tls.handshake


tshark -r Daniel_Network.pcap \

-Y "http.request" \

-T fields \

-e frame.time \

-e http.request.method \

-e http.request.uri \

-e http.host


tshark -r Daniel_Network.pcap \

-Y 'ftp.request.command=="RETR"' \

-T fields \
```

```
-e frame.time \  
  
-e ftp.request.command \  
  
-e ftp.request.arg  
  
tshark -r Daniel_Network.pcap \  
  
-Y "dns.qry.name contains onion" \  
  
-T fields \  
  
-e frame.time_relative \  
  
-e ip.src \  
  
-e ip.dst \  
  
-e dns.qry.name  
  
tshark -r Daniel_Network.pcap \  
  
-Y "tls.handshake.type==1" \  
  
-T fields \  
  
-e frame.time \  
  
-e ip.src \  
  
-e ip.dst \  
  
-e tls.handshake.extensions_server_name  
  
tshark -r Daniel_Network.pcap \  
  
--export-objects http,http_objects_full
```

Appendix B — File, Steganography, and Archive Analysis Commands

The following commands were used to inspect file metadata, verify file integrity, analyse bitmap files for hidden content, and examine encrypted archives.

File Type Identification

```
file *.pdf
```

Metadata Analysis

```
exiftool HiddenMessage.bmp  
exiftool "safehouse location.bmp"
```

String Extraction

```
strings HiddenMessage.bmp
```

Hex Inspection

```
xxd -l 128 HiddenMessage.bmp
```

Embedded Data Analysis

```
binwalk HiddenMessage.bmp  
binwalk "safehouse location.bmp"
```

Archive Inspection

```
zipinfo encrypted.zip  
zipinfo -v encrypted.zip  
unzip -l encrypted.zip
```

Integrity Verification

```
sha256sum Daniel_Network.pcap
```

Appendix C — Supplementary Screenshots

This appendix contains supplementary screenshots referenced throughout the report but not included in the main chapters.

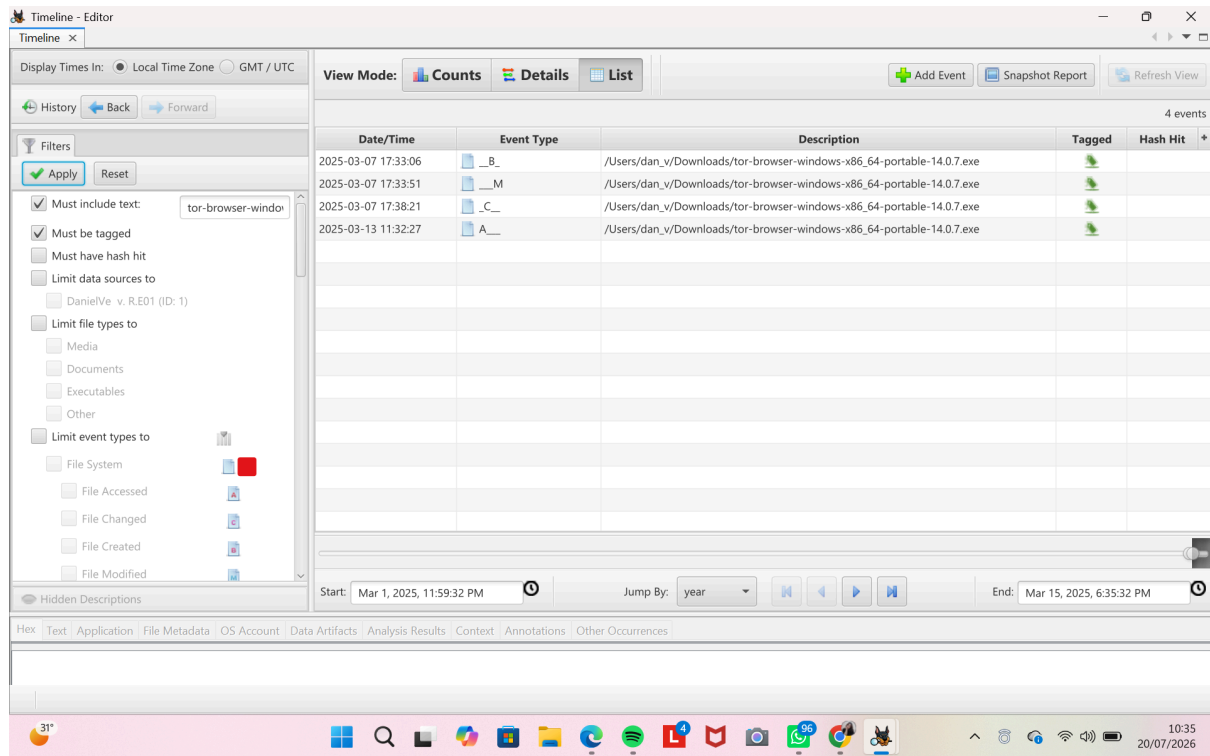


Figure C.1 - Tor Browser access timeline

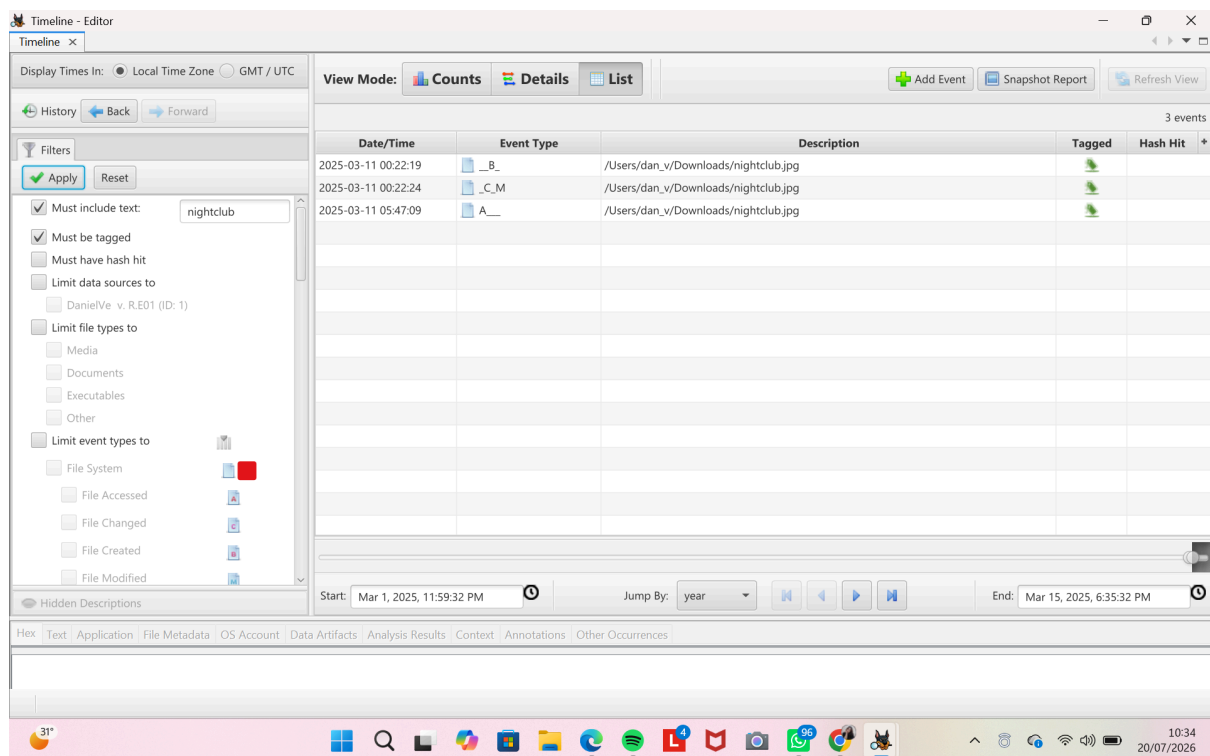
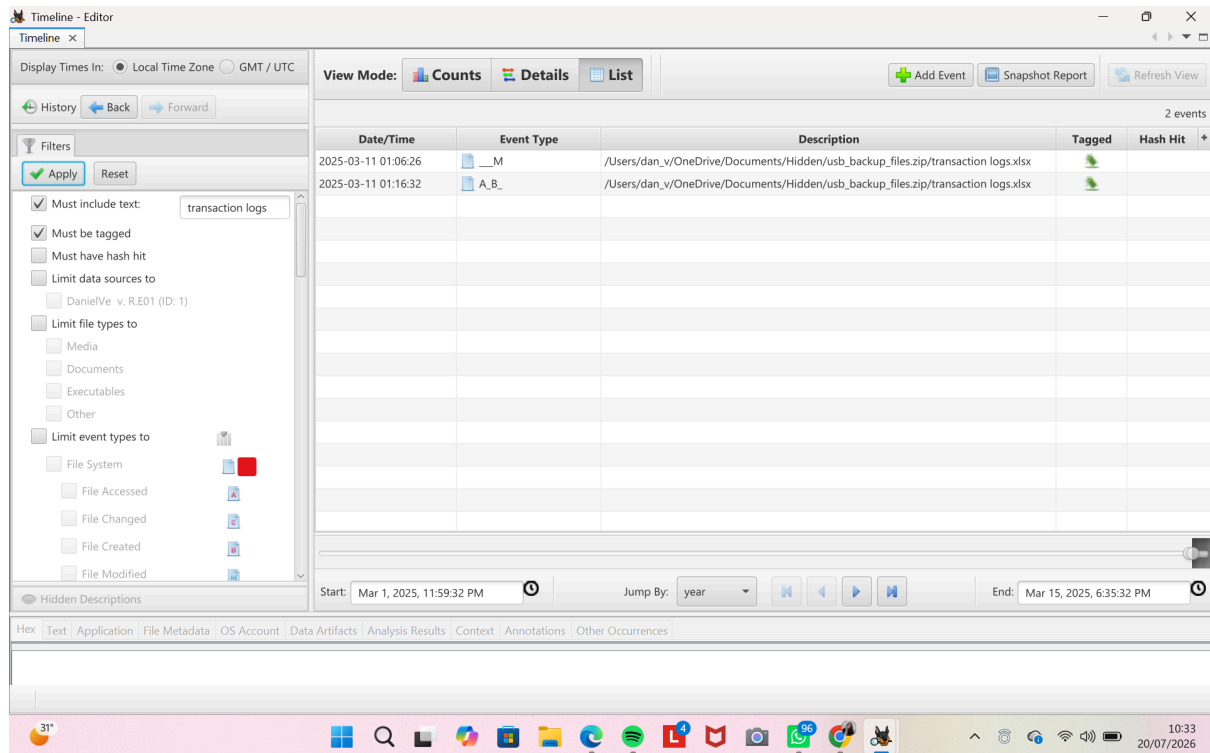
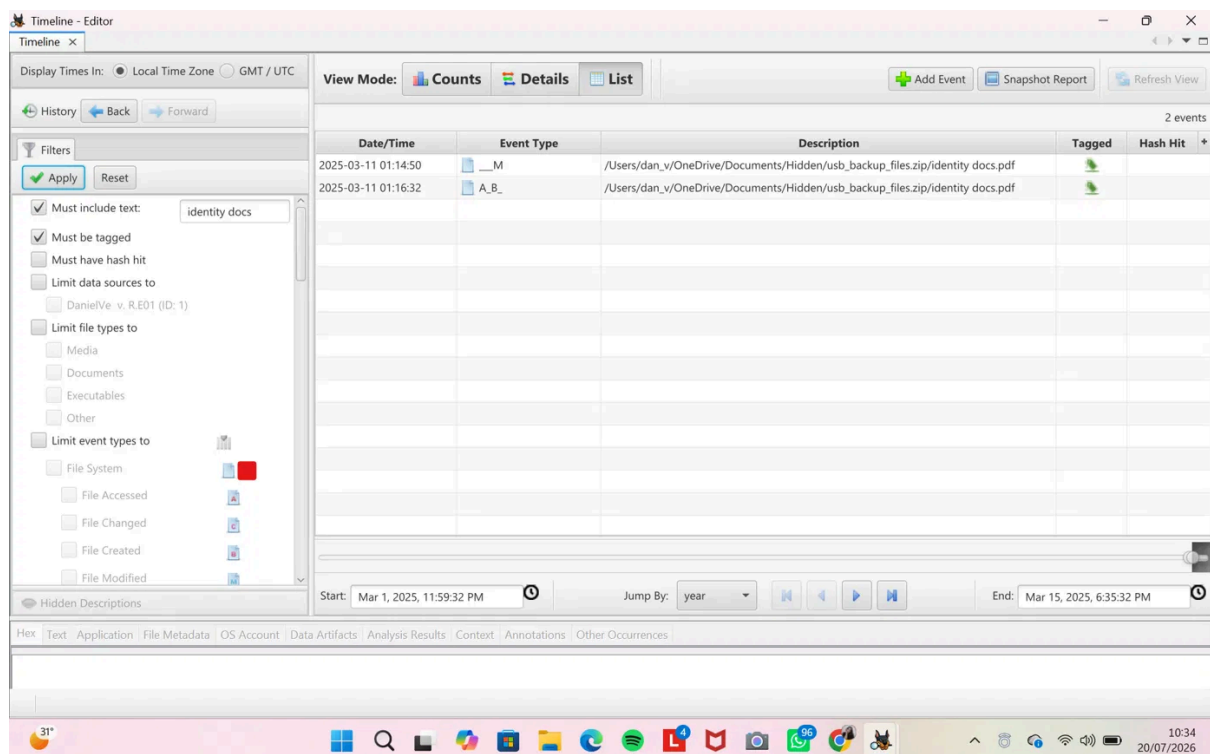


Figure C.2 - nightclub.jpg file timeline

*Figure C.3 - transaction logs.xlsx timeline**Figure C.4 - identity docs.pdf timeline*

The screenshot shows the Timeline - Editor interface. The left sidebar contains filters, and the main pane displays a single event in List view.

Date/Time	Event Type	Description	Tagged	Hash Hit
2025-03-13 01:11:24	ACBM	/Users/dan_v/AppData/Local/Packages/micr ... s/50/7/Attachments/Shipment List[48].pdf		

Filters applied: Must include text: shipment list. Other filters are unchecked.

Timeline range: Start: Mar 1, 2025, 11:59:32 PM; End: Mar 15, 2025, 6:35:32 PM. Jump By: year.

Figure C.5 - Shipment List attachment event

The screenshot shows the Timeline - Editor interface with three events displayed in List view.

Date/Time	Event Type	Description	Tagged	Hash Hit
2025-03-11 09:23:40	Email Sent	Sent from: dan_vega_11041990@outlook.com ... sodqqhg. Qr lvxhv. Sdbphqw frqiluphg. Dzdvlwqj ilqdo lqvwuxfwlrqv. Zlao surfhqg dv glvfxvvhq.		
2025-03-11 09:23:40	Email Received	Message from: dan_vega_11041990@outlook.com ... sodqqhg. Qr lvxhv. Sdbphqw frqiluphg. Dzdvlwqj ilqdo lqvwuxfwlrqv. Zlao surfhqg dv glvfxvvhq.		
2025-03-13 01:11:24	ACBM	/Users/dan_v/AppData/Local/Packages/micro ... ttachments/Everything is in place[47].eml		

Filters applied: Must include text: everything is in place. Other filters are unchecked.

Timeline range: Start: Mar 1, 2025, 11:59:32 PM; End: Mar 15, 2025, 6:35:32 PM. Jump By: year.

Figure C.6 - Email Everything is in place timeline

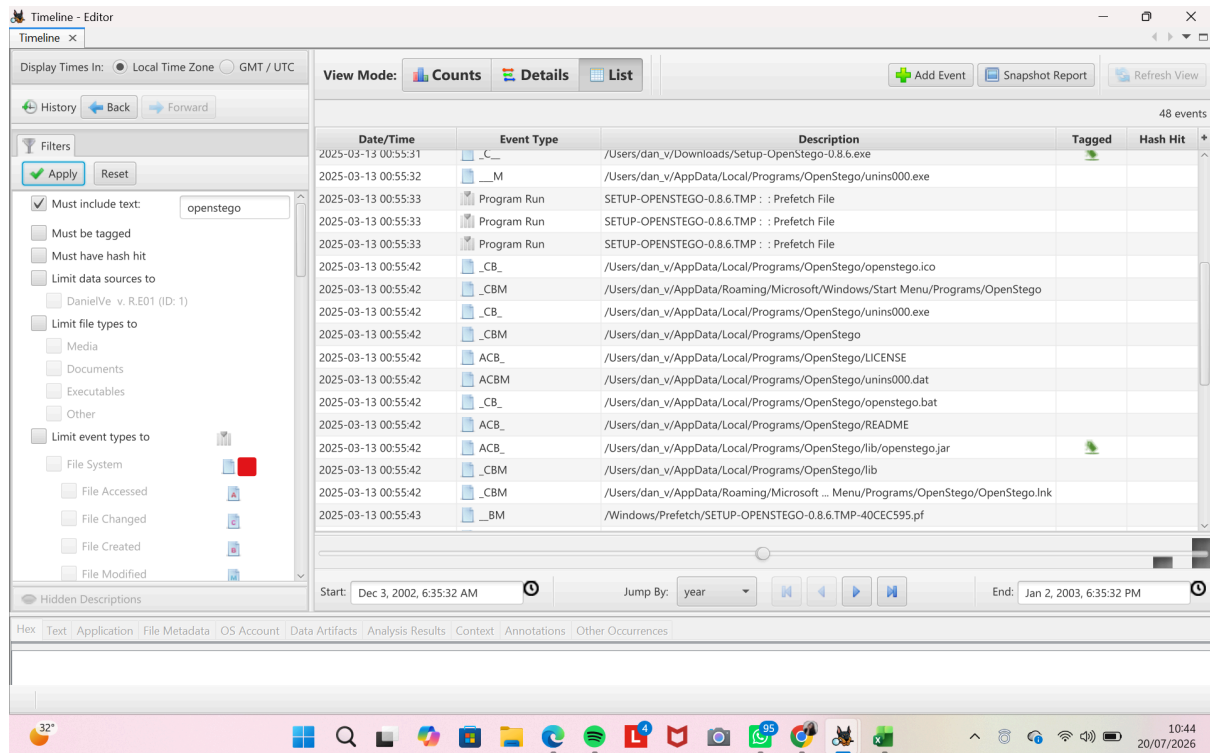


Figure C.7 - OpenStego installation detail

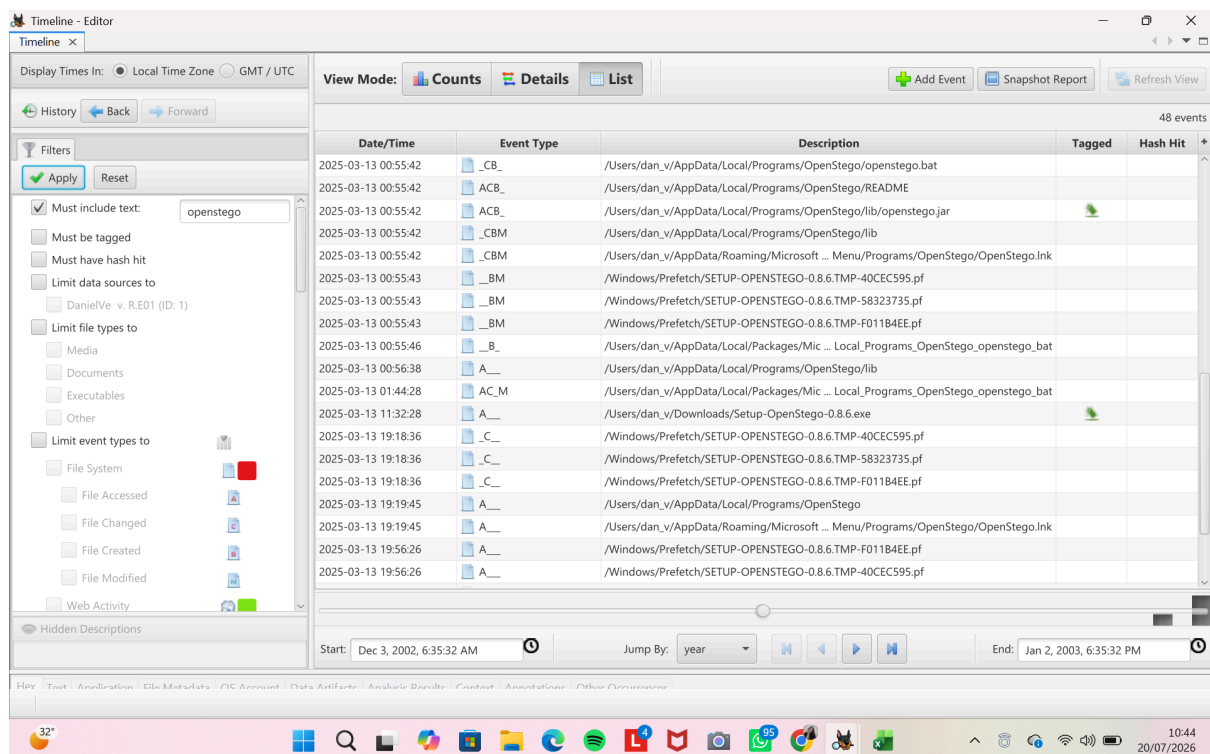


Figure C.8 - OpenStego installation detail

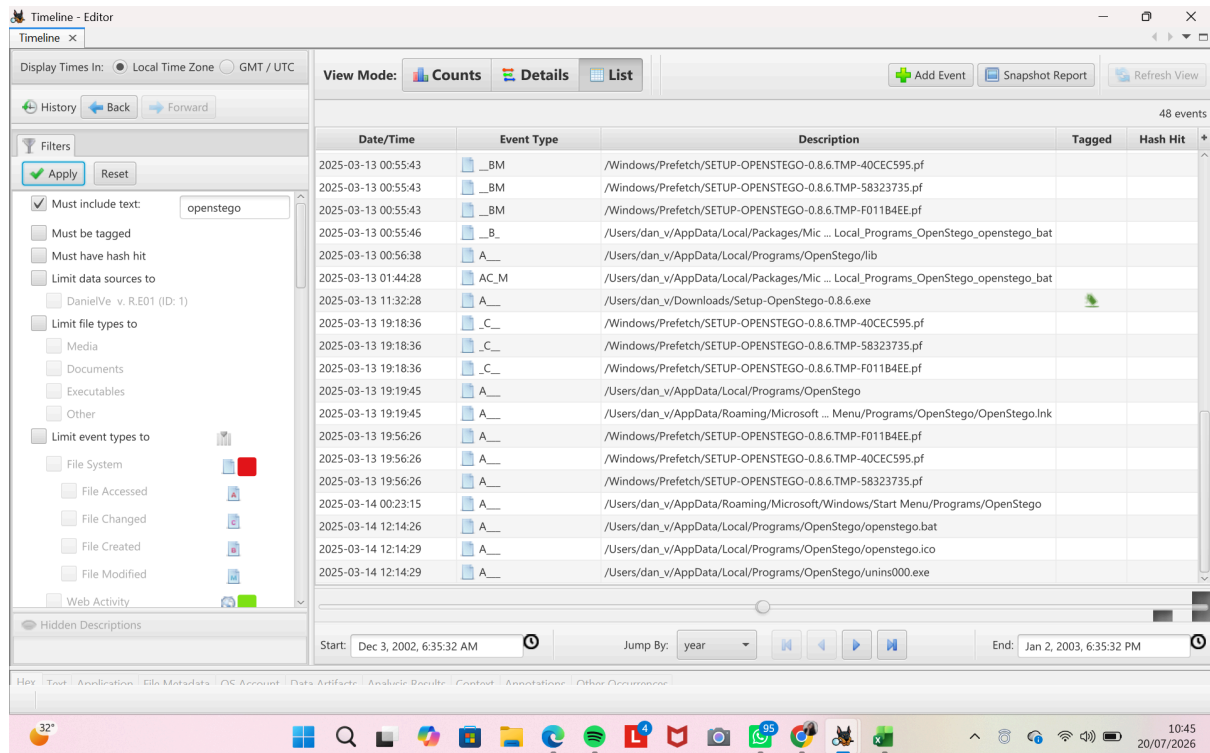


Figure C.9 - OpenStego installation detail

Appendix D — Glossary

The following glossary defines technical terms used throughout this report

- **PCAP** a saved recording of network traffic, viewable in tools like Wireshark.
- **TCP Stream** the full back-and-forth of a single network conversation, reassembled from individual packets.
- **Tor** a network that routes traffic through multiple relays to hide who is talking to whom.
- **Steganography** hiding data inside another file, such as an image, so its presence isn't obvious.
- **Hash** a fixed-length fingerprint of a file or password, used to verify integrity or attempt password recovery without altering the original file.
- **Brute-force** trying many possible passwords systematically until the correct one is found.

- E01 the EnCase disk image format, a common forensic format for full disk/drive images.